Phenix Refinement

5 documentation pages
Generated from local Phenix documentation

Contents

X-ray Structure Refinement

xray-refinement.html

Why

In crystallographic structure determination (X-ray or neutron), once an atomic model has been built into an electron density map, or obtained by molecular replacement, you need to optimize it to best fit the experimental data while also preserving good agreement with prior chemical knowledge.

For crystallographic data, this refinement process is usually done in reciprocal space. Using a least-squares or maximum-likelihood target, the model parameters are changed so the model-derived structure factors match the amplitudes or intensities of the experimental structure factors. For example, these model parameters can include (a) atomic parameters, such as coordinates, atomic displacement parameters, occupancies, and scattering factors, and (b) non-atomic parameters that describe contributions from bulk solvent, twinning, and crystal anisotropy.

How

In Phenix, the primary program for refining atomic models is phenix.refine. This applies optimization algorithms (minimization or simulated annealing) to change the model parameters (typically coordinates and atomic displacement parameters) to improve the fit of the model to the data. At the same time, it applies restraints to prior chemical information (such as bond lengths, angles, and torsions) to maintain good stereochemistry. This customizable program allows multiple refinement strategies to be combined and applied to any selected part of the model in a single run.

How to use the phenix.refine GUI: Click here

Phenix reference manual for phenix.refine

Common issues

- Frequently asked questions for structure refinement

Related programs

- phenix.rosetta_refine: This program combines crystallographic structure refinement algorithms from Phenix with physically realistic potentials for model optimization from Rosetta. The tool is useful at low resolution, where it combines a wide radius of convergence across distinct local minima with realistic geometry.
- phenix.ensemble_refinement: This program combines crystallographic refinement with molecular dynamics to produce ensemble models fitted to diffraction data. These ensemble models can contain ~50-500 individual structures, and can simultaneously account for anisotopic and anharmonic atomic distributions.

phenix.geometry_minimization: regularize model geometry

geometry_minimization.html

Description

The program idealizes model geometry according to standard geometry restraints. In addition it can use extra restraints such as: Ramachandran plot, secondary structure, reference to starting position, C-beta deviations restraints, amino-acid residue side-chain rotamer restraints, other user-defined (custom) restraints.

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- Ramachandran plot,
- secondary structure,
- reference to starting position,
- C-beta deviations restraints,
- amino-acid residue side-chain rotamer restraints,
- other user-defined (custom) restraints.

Command line usage examples:

```
% phenix.geometry_minimization model.pdb
```

GUI

Graphical user interface is available

List of all available keywords

silent = False write_geo_file = True file_name = None show_states = False restraints = None restraints_directory = None output_file_name_prefix = None directory = None job_title = None Job title in PHENIX GUI, not used on command linefix_rotamer_outliers = True Remove outliersallow_allowed_rotamers = True More strict fixing outliersstop_for_unknowns = True selection = all Atom selection string: selected atoms are subject to movepdb_interpretationsort_atoms = True use_ncs_to_build_restraints = True show_restraints_histograms = True flip_symmetric_amino_acids = True superpose_ideal_ligand = *None all SF4 F3S DVT disable_uc_volume_vs_n_atoms_check = False allow_polymer_cross_special_position = False correct_hydrogens = True c_beta_restraints = True use_neutron_distances = False Use neutron X-H distances (which are longer than X-ray ones)disulfide_bond_exclusions_selection_string = None exclusion_distance_cutoff = 3 If SG of CYS forming SS bond is closer than this distance to an atom that it may coordinate then this SG is excluded from SS bond.link_distance_cutoff = 3 Length of link between the linked residuesdisulfide_distance_cutoff = 3 add_angle_and_dihedral_restraints_for_disulfides = True dihedral_function_type = *determined_by_sign_of_periodicity all_sinusoidal all_harmonic chir_volume_esd = 0.2 max_reasonable_bond_distance = 50.0 nonbonded_distance_cutoff = None default_vdw_distance = 1 min_vdw_distance = 1 nonbonded_buffer = 1 **EXPERIMENTAL, developers only**nonbonded_weight = None Weighting of nonbonded restraints term. By default, this will be set to 16 if explicit hydrogens are used (this was the default in earlier versions of Phenix), or 100 if hydrogens are missing.const_shrink_donor_acceptor = 0 **EXPERIMENTAL, developers only**vdw_1_4_factor = 0.8

min_distance_sym_equiv = 0.5 custom_nonbonded_symmetry_exclusions = None translate_cns_dna_rna_residue_names = None proceed_with_excessive_length_bonds = False restraints_librarycdl = True Use Conformation Dependent Library (CDL) for geometry restraintsinclude_modified_amino_acid_in_cdl = False mcl = True Use Metal Coordination Library (MCL) for tetrahedral Zn++ and iron-sulfur clusters SF4, FES, F3S, ...cis_pro_eh99 = True omega_cdl = False Use Omega Conformation Dependent Library (omega-CDL) for geometry restraintscdl_svl = False rdl = False rdl_selection = *all TRP hpdl = False cdl_nucleotidesenable = False Use RestraintsLib for DNA and RNA for geometry restraintsesd = *phenix csd factor = 2.0 user_suppliedpath = None Root directory of user supplied restraintsaction = *pre post Choice to look here before GeoStd, the default, or aftersecondary_structuress_by_chain = True Only applies if search_method = from_ca. Find secondary structure only within individual chains. Alternative is to allow H-bonds between chains. Can be much slower with ss_by_chain=False. If your model is complete use ss_by_chain=True. If your model is many fragments, use ss_by_chain=False.from_ca_conservative = False various parameters changed to make from_ca method more conservative, hopefully to closer resemble ksdssp.max_rmsd = 1 Only applies if search_method = from_ca. Maximum rmsd to consider two chains with identical sequences as the same for ss identificationuse_representative_chains = True Only applies if search_method = from_ca. Use a representative of all chains with the same sequence. Alternative is to examine each chain individually. Can be much slower with use_representative_of_chain=False if there are many symmetry copies. Ignored unless ss_by_chain is True.max_representative_chains = 100 Only applies if search_method = from_ca. Maximum number of representative chainsenabled = False Turn on secondary structure restraints (main switch)proteinenabled = True Turn on secondary structure restraints for proteinsearch_method = *ksdssp from_ca cablam Particular method to search protein secondary structure.distance_ideal_n_o = 2.9 Target length for N-O hydrogen bonddistance_cut_n_o = 3.5 Hydrogen ...

- silent = False
- write_geo_file = True
- file_name = None
- show_states = False
- restraints = None
- restraints_directory = None
- output_file_name_prefix = None
- directory = None
- job_title = None Job title in PHENIX GUI, not used on command line
- fix_rotamer_outliers = True Remove outliers
- allow_allowed_rotamers = True More strict fixing outliers
- stop_for_unknowns = True
- selection = all Atom selection string: selected atoms are subject to move
- pdb_interpretationsort_atoms = True use_ncs_to_build_restraints = True show_restraints_histograms = True flip_symmetric_amino_acids = True superpose_ideal_ligand = *None all SF4 F3S DVT disable_uc_volume_vs_n_atoms_check = False allow_polymer_cross_special_position = False correct_hydrogens = True c_beta_restraints = True use_neutron_distances = False Use neutron X-H distances (which are longer than X-ray ones)disulfide_bond_exclusions_selection_string = None exclusion_distance_cutoff = 3 If SG of CYS forming SS bond is closer than this distance to an atom that it may coordinate then this SG is excluded from SS bond.link_distance_cutoff = 3 Length of link between the linked residuesdisulfide_distance_cutoff = 3 add_angle_and_dihedral_restraints_for_disulfides =

True dihedral_function_type = *determined_by_sign_of_periodicity all_sinusoidal all_harmonic chir_volume_esd = 0.2 max_reasonable_bond_distance = 50.0 nonbonded_distance_cutoff = None default_vdw_distance = 1 min_vdw_distance = 1 nonbonded_buffer = 1 **EXPERIMENTAL, developers only**nonbonded_weight = None Weighting of nonbonded restraints term. By default, this will be set to 16 if explicit hydrogens are used (this was the default in earlier versions of Phenix), or 100 if hydrogens are missing.const_shrink_donor_acceptor = 0 **EXPERIMENTAL, developers only**vdw_1_4_factor = 0.8 min_distance_sym_equiv = 0.5 custom_nonbonded_symmetry_exclusions = None translate_cns_dna_rna_residue_names = None proceed_with_excessive_length_bonds = False restraints_librarycdl = True Use Conformation Dependent Library (CDL) for geometry restraintsinclude_modified_amino_acid_in_cdl = False mcl = True Use Metal Coordination Library (MCL) for tetrahedral Zn++ and iron-sulfur clusters SF4, FES, F3S, ...cis_pro_eh99 = True omega_cdl = False Use Omega Conformation Dependent Library (omega-CDL) for geometry restraintscdl_svl = False rdl = False rdl_selection = *all TRP hpdl = False cdl_nucleotidesenable = False Use RestraintsLib for DNA and RNA for geometry restraintsesd = *phenix csd factor = 2.0 user_suppliedpath = None Root directory of user supplied restraintsaction = *pre post Choice to look here before GeoStd, the default, or aftersecondary_structuress_by_chain = True Only applies if search_method = from_ca. Find secondary structure only within individual chains. Alternative is to allow H-bonds between chains. Can be much slower with ss_by_chain=False. If your model is complete use ss_by_chain=True. If your model is many fragments, use ss_by_chain=False.from_ca_conservative = False various parameters changed to make from_ca method more conservative, hopefully to closer resemble ksdssp.max_rmsd = 1 Only applies if search_method = from_ca. Maximum rmsd to consider two chains with identical sequences as the same for ss identificationuse_representative_chains = True Only applies if search_method = from_ca. Use a representative of all chains with the same sequence. Alternative is to examine each chain individually. Can be much slower with use_representative_of_chain=False if there are many symmetry copies. Ignored unless ss_by_chain is True.max_representative_chains = 100 Only applies if search_method = from_ca. Maximum number of representative chainsenabled = False Turn on secondary structure restraints (main switch)proteinenabled = True Turn on secondary structure restraints for proteinsearch_method = *ksdssp from_ca cablam Particular method to search protein secondary structure.distance_ideal_n_o = 2.9 Target length for N-O hydrogen bonddistance_cut_n_o = 3.5 Hydrogen bond with length exceeding this value will not be establishedremove_outliers = True If true, h-bonds exceeding distance_cut_n_o length will not be establishedrestrain_hbond_angles = True helixserial_number = None helix_identifier = None enabled = True Restrain this particular helixselection = None helix_type = *alpha pi 3_10 unknown Type of helix, defaults to alpha. Only alpha, pi, and 3_10 helices are used for hydrogen-bond res...

- sort_atoms = True
- use_ncs_to_build_restraints = True
- show_restraints_histograms = True
- flip_symmetric_amino_acids = True
- superpose_ideal_ligand = *None all SF4 F3S DVT
- disable_uc_volume_vs_n_atoms_check = False
- allow_polymer_cross_special_position = False
- correct_hydrogens = True
- c_beta_restraints = True
- use_neutron_distances = False Use neutron X-H distances (which are longer than X-ray ones)
- disulfide_bond_exclusions_selection_string = None

- exclusion_distance_cutoff = 3 If SG of CYS forming SS bond is closer than this distance to an atom that it may coordinate then this SG is excluded from SS bond.
- link_distance_cutoff = 3 Length of link between the linked residues
- disulfide_distance_cutoff = 3
- add_angle_and_dihedral_restraints_for_disulfides = True
- dihedral_function_type = *determined_by_sign_of_periodicity all_sinusoidal all_harmonic
- chir_volume_esd = 0.2
- max_reasonable_bond_distance = 50.0
- nonbonded_distance_cutoff = None
- default_vdw_distance = 1
- min_vdw_distance = 1
- nonbonded_buffer = 1 **EXPERIMENTAL, developers only**
- nonbonded_weight = None Weighting of nonbonded restraints term. By default, this will be set to 16 if explicit hydrogens are used (this was the default in earlier versions of Phenix), or 100 if hydrogens are missing.
- const_shrink_donor_acceptor = 0 **EXPERIMENTAL, developers only**
- vdw_1_4_factor = 0.8
- min_distance_sym_equiv = 0.5
- custom_nonbonded_symmetry_exclusions = None
- translate_cns_dna_rna_residue_names = None
- proceed_with_excessive_length_bonds = False
- restraints_librarycdl = True Use Conformation Dependent Library (CDL) for geometry restraintsinclude_modified_amino_acid_in_cdl = False mcl = True Use Metal Coordination Library (MCL) for tetrahedral Zn++ and iron-sulfur clusters SF4, FES, F3S, ...cis_pro_eh99 = True omega_cdl = False Use Omega Conformation Dependent Library (omega-CDL) for geometry restraintscdl_svl = False rdl = False rdl_selection = *all TRP hpdl = False cdl_nucleotidesenable = False Use RestraintsLib for DNA and RNA for geometry restraintsesd = *phenix csd factor = 2.0 user_suppliedpath = None Root directory of user supplied restraintsaction = *pre post Choice to look here before GeoStd, the default, or after
- cdl = True Use Conformation Dependent Library (CDL) for geometry restraints
- include_modified_amino_acid_in_cdl = False
- mcl = True Use Metal Coordination Library (MCL) for tetrahedral Zn++ and iron-sulfur clusters SF4, FES, F3S, ...
- cis_pro_eh99 = True
- omega_cdl = False Use Omega Conformation Dependent Library (omega-CDL) for geometry restraints
- cdl_svl = False
- rdl = False
- rdl_selection = *all TRP
- hpdl = False
- cdl_nucleotidesenable = False Use RestraintsLib for DNA and RNA for geometry restraintsesd = *phenix csd factor = 2.0

- enable = False Use RestraintsLib for DNA and RNA for geometry restraints
- esd = *phenix csd
- factor = 2.0
- user_suppliedpath = None Root directory of user supplied restraintsaction = *pre post Choice to look here before GeoStd, the default, or after
- path = None Root directory of user supplied restraints
- action = *pre post Choice to look here before GeoStd, the default, or after
- secondary_structuress_by_chain = True Only applies if search_method = from_ca. Find secondary structure only within individual chains. Alternative is to allow H-bonds between chains. Can be much slower with ss_by_chain=False. If your model is complete use ss_by_chain=True. If your model is many fragments, use ss_by_chain=False.from_ca_conservative = False various parameters changed to make from_ca method more conservative, hopefully to closer resemble ksdssp.max_rmsd = 1 Only applies if search_method = from_ca. Maximum rmsd to consider two chains with identical sequences as the same for ss identificationuse_representative_chains = True Only applies if search_method = from_ca. Use a representative of all chains with the same sequence. Alternative is to examine each chain individually. Can be much slower with use_representative_of_chain=False if there are many symmetry copies. Ignored unless ss_by_chain is True.max_representative_chains = 100 Only applies if search_method = from_ca. Maximum number of representative chainsenabled = False Turn on secondary structure restraints (main switch)proteinenabled = True Turn on secondary structure restraints for proteinsearch_method = *ksdssp from_ca cablam Particular method to search protein secondary structure.distance_ideal_n_o = 2.9 Target length for N-O hydrogen bonddistance_cut_n_o = 3.5 Hydrogen bond with length exceeding this value will not be establishedremove_outliers = True If true, h-bonds exceeding distance_cut_n_o length will not be establishedrestrain_hbond_angles = True helixserial_number = None helix_identifier = None enabled = True Restrain this particular helixselection = None helix_type = *alpha pi 3_10 unknown Type of helix, defaults to alpha. Only alpha, pi, and 3_10 helices are used for hydrogen-bond restraints.sigma = 0.05 slack = 0 top_out = False angle_sigma_scale = 1 Multiply sigmas for h-bond angles by this value. Original sigmas range from 5 to 10.angle_sigma_set = None Use this parameter to set sigmas for h-bond angles to a particular valuehbonddonor = None acceptor = None sheetenabled = True Restrain this particular sheetfirst_strand = None sheet_id = None sigma = 0.05 slack = 0 top_out = False angle_sigma_scale = 1 Multiply sigmas for h-bond angles by this value. Original sigmas range from 5 to 10.angle_sigma_set = None Use this parameter to set sigmas for h-bond angles to a particular valuestrandselection = None sense = parallel antiparallel *unknown bond_start_current = None bond_start_previous = None hbonddonor = None acceptor = None nucleic_acidenabled = True Turn on secondary structure restraints for nucleic acidshbond_distance_cutoff = 3.4 Hydrogen bonds with length exceeding this limit will not be establishedscale_bonds_sigma = 1. All sigmas for h-bond length will be multiplied by this number. The smaller number is tighter restraints.base_pairenabled = True Restraint this particular base-pairbase1 = None Selection string selecting at least one atom in the desired residuebase2 = None Selection string selecting at least one atom in the desired residuesaenger_class = 0 Type of base-pairingrestrain_planarity = False Apply planarity restraint to this base-pairplanarity_sigma = 0.176 restrain_hbonds = True Restrain hydrogen bonds length for this base-pairrestrain_hb_angles = True Restrain angles around hydrogen bonds for this base-pairrestrain_parallelity = True Apply parallelity restraint to this base-pairparallelity_target = 0 parallelity_sigma = 0.0335 stacking_pairenabled = True Restraint this particular base-pairbase1 = None Selection string selecting at least one atom in the desired residuebase2 = None Selection string selecting at least one atom in the desired residueangle = 0 sigma = 0.027

- ss_by_chain = True Only applies if search_method = from_ca. Find secondary structure only within individual chains. Alternative is to allow H-bonds between chains. Can be much slower with ss_by_chain=False. If your model is complete use ss_by_chain=True. If your model is many fragments, use ss_by_chain=False.
- from_ca_conservative = False various parameters changed to make from_ca method more conservative, hopefully to closer resemble ksdssp.
- max_rmsd = 1 Only applies if search_method = from_ca. Maximum rmsd to consider two chains with identical sequences as the same for ss identification
- use_representative_chains = True Only applies if search_method = from_ca. Use a representative of all chains with the same sequence. Alternative is to examine each chain individually. Can be much slower with use_representative_of_chain=False if there are many symmetry copies. Ignored unless ss_by_chain is True.
- max_representative_chains = 100 Only applies if search_method = from_ca. Maximum number of representative chains
- enabled = False Turn on secondary structure restraints (main switch)
- proteinenabled = True Turn on secondary structure restraints for proteinsearch_method = *ksdssp from_ca cablam Particular method to search protein secondary structure.distance_ideal_n_o = 2.9 Target length for N-O hydrogen bonddistance_cut_n_o = 3.5 Hydrogen bond with length exceeding this value will not be establishedremove_outliers = True If true, h-bonds exceeding distance_cut_n_o length will not be establishedrestrain_hbond_angles = True helixserial_number = None helix_identifier = None enabled = True Restrain this particular helixselection = None helix_type = *alpha pi 3_10 unknown Type of helix, defaults to alpha. Only alpha, pi, and 3_10 helices are used for hydrogen-bond restraints.sigma = 0.05 slack = 0 top_out = False angle_sigma_scale = 1 Multiply sigmas for h-bond angles by this value. Original sigmas range from 5 to 10.angle_sigma_set = None Use this parameter to set sigmas for h-bond angles to a particular valuehbonddonor = None acceptor = None sheetenabled = True Restrain this particular sheetfirst_strand = None sheet_id = None sigma = 0.05 slack = 0 top_out = False angle_sigma_scale = 1 Multiply sigmas for h-bond angles by this value. Original sigmas range from 5 to 10.angle_sigma_set = None Use this parameter to set sigmas for h-bond angles to a particular valuestrandselection = None sense = parallel antiparallel *unknown bond_start_current = None bond_start_previous = None hbonddonor = None acceptor = None
- enabled = True Turn on secondary structure restraints for protein
- search_method = *ksdssp from_ca cablam Particular method to search protein secondary structure.
- distance_ideal_n_o = 2.9 Target length for N-O hydrogen bond
- distance_cut_n_o = 3.5 Hydrogen bond with length exceeding this value will not be established
- remove_outliers = True If true, h-bonds exceeding distance_cut_n_o length will not be established
- restrain_hbond_angles = True
- helixserial_number = None helix_identifier = None enabled = True Restrain this particular helixselection = None helix_type = *alpha pi 3_10 unknown Type of helix, defaults to alpha. Only alpha, pi, and 3_10 helices are used for hydrogen-bond restraints.sigma = 0.05 slack = 0 top_out = False angle_sigma_scale = 1 Multiply sigmas for h-bond angles by this value. Original sigmas range from 5 to 10.angle_sigma_set = None Use this parameter to set sigmas for h-bond angles to a particular valuehbonddonor = None acceptor = None
- serial_number = None
- helix_identifier = None

- enabled = True Restrain this particular helix
- selection = None
- helix_type = *alpha pi 3_10 unknown Type of helix, defaults to alpha. Only alpha, pi, and 3_10 helices are used for hydrogen-bond restraints.
- sigma = 0.05
- slack = 0
- top_out = False
- angle_sigma_scale = 1 Multiply sigmas for h-bond angles by this value. Original sigmas range from 5 to 10.
- angle_sigma_set = None Use this parameter to set sigmas for h-bond angles to a particular value
- hbonddonor = None acceptor = None
- donor = None
- acceptor = None
- sheetenabled = True Restrain this particular sheetfirst_strand = None sheet_id = None sigma = 0.05 slack = 0 top_out = False angle_sigma_scale = 1 Multiply sigmas for h-bond angles by this value. Original sigmas range from 5 to 10.angle_sigma_set = None Use this parameter to set sigmas for h-bond angles to a particular valuestrandselection = None sense = parallel antiparallel *unknown bond_start_current = None bond_start_previous = None hbonddonor = None acceptor = None
- enabled = True Restrain this particular sheet
- first_strand = None
- sheet_id = None
- sigma = 0.05
- slack = 0
- top_out = False
- angle_sigma_scale = 1 Multiply sigmas for h-bond angles by this value. Original sigmas range from 5 to 10.
- angle_sigma_set = None Use this parameter to set sigmas for h-bond angles to a particular value
- strandselection = None sense = parallel antiparallel *unknown bond_start_current = None bond_start_previous = None
- selection = None
- sense = parallel antiparallel *unknown
- bond_start_current = None
- bond_start_previous = None
- hbonddonor = None acceptor = None
- donor = None
- acceptor = None
- nucleic_acidenabled = True Turn on secondary structure restraints for nucleic acidshbond_distance_cutoff = 3.4 Hydrogen bonds with length exceeding this limit will not be establishedscale_bonds_sigma = 1. All sigmas for h-bond length will be multiplied by this number. The smaller number is tighter restraints.base_pairenabled = True Restraint this particular base-pairbase1 =

None Selection string selecting at least one atom in the desired residuebase2 = None Selection string selecting at least one atom in the desired residuesaenger_class = 0 Type of base-pairingrestrain_planarity = False Apply planarity restraint to this base-pairplanarity_sigma = 0.176 restrain_hbonds = True Restrain hydrogen bonds length for this base-pairrestrain_hb_angles = True Restrain angles around hydrogen bonds for this base-pairrestrain_parallelity = True Apply parallelity restraint to this base-pairparallelity_target = 0 parallelity_sigma = 0.0335 stacking_pairenabled = True Restraint this particular base-pairbase1 = None Selection string selecting at least one atom in the desired residuebase2 = None Selection string selecting at least one atom in the desired residueangle = 0 sigma = 0.027

- enabled = True Turn on secondary structure restraints for nucleic acids
- hbond_distance_cutoff = 3.4 Hydrogen bonds with length exceeding this limit will not be established
- scale_bonds_sigma = 1. All sigmas for h-bond length will be multiplied by this number. The smaller number is tighter restraints.
- base_pairenabled = True Restraint this particular base-pairbase1 = None Selection string selecting at least one atom in the desired residuebase2 = None Selection string selecting at least one atom in the desired residuesaenger_class = 0 Type of base-pairingrestrain_planarity = False Apply planarity restraint to this base-pairplanarity_sigma = 0.176 restrain_hbonds = True Restrain hydrogen bonds length for this base-pairrestrain_hb_angles = True Restrain angles around hydrogen bonds for this base-pairrestrain_parallelity = True Apply parallelity restraint to this base-pairparallelity_target = 0 parallelity_sigma = 0.0335
- enabled = True Restraint this particular base-pair
- base1 = None Selection string selecting at least one atom in the desired residue
- base2 = None Selection string selecting at least one atom in the desired residue
- saenger_class = 0 Type of base-pairing
- restrain_planarity = False Apply planarity restraint to this base-pair
- planarity_sigma = 0.176
- restrain_hbonds = True Restrain hydrogen bonds length for this base-pair
- restrain_hb_angles = True Restrain angles around hydrogen bonds for this base-pair
- restrain_parallelity = True Apply parallelity restraint to this base-pair
- parallelity_target = 0
- parallelity_sigma = 0.0335
- stacking_pairenabled = True Restraint this particular base-pairbase1 = None Selection string selecting at least one atom in the desired residuebase2 = None Selection string selecting at least one atom in the desired residueangle = 0 sigma = 0.027
- enabled = True Restraint this particular base-pair
- base1 = None Selection string selecting at least one atom in the desired residue
- base2 = None Selection string selecting at least one atom in the desired residue
- angle = 0
- sigma = 0.027
- reference_coordinate_restraintsRestrains coordinates in Cartesian space to stay near their starting positions. This is intended for use in generating simulated annealing omit maps, to prevent refined atoms from collapsing in on the region missing atoms. For conserving geometry quality at low resolution, the

more flexible reference model restraints should be used instead.enabled = False exclude_outliers = True selection = all sigma = 0.2 limit = 1.0 top_out = False reference_is_average_alt_confs = False Sets the reference coordinate to the average of two alt. confs. Ignores selection option.

• enabled = False

• exclude_outliers = True

• selection = all

• sigma = 0.2

• limit = 1.0

• top_out = False

• reference_is_average_alt_confs = False Sets the reference coordinate to the average of two alt. confs. Ignores selection option.

• automatic_restraintsside_chain_parallelity = False

• side_chain_parallelity = False

• automatic_linkinglink_all = False If True, bond restraints will be generated for any appropriate ligand-protein or ligand-nucleic acid covalent bonds. This includes sugars, amino acid modifications, and other prosthetic groups.link_none = False link_metals = Auto link_residues = True link_amino_acid_rna_dna = False link_carbohydrates = True link_ligands = True link_small_molecules = False metal_coordination_cutoff = 3.0 amino_acid_bond_cutoff = 1.9 inter_residue_bond_cutoff = 2.2 buffer_for_second_row_elements = 0.5 carbohydrate_bond_cutoff = 1.99 ligand_bond_cutoff = 1.99 small_molecule_bond_cutoff = 1.98 exclude_hydrogens_from_bonding_decisions = False

• link_all = False If True, bond restraints will be generated for any appropriate ligand-protein or ligand-nucleic acid covalent bonds. This includes sugars, amino acid modifications, and other prosthetic groups.

• link_none = False

• link_metals = Auto

• link_residues = True

• link_amino_acid_rna_dna = False

• link_carbohydrates = True

• link_ligands = True

• link_small_molecules = False

• metal_coordination_cutoff = 3.0

• amino_acid_bond_cutoff = 1.9

• inter_residue_bond_cutoff = 2.2

• buffer_for_second_row_elements = 0.5

• carbohydrate_bond_cutoff = 1.99

• ligand_bond_cutoff = 1.99

• small_molecule_bond_cutoff = 1.98

• exclude_hydrogens_from_bonding_decisions = False

• include_in_automatic_linkingselection_1 = None selection_2 = None bond_cutoff = 4.5

• selection_1 = None

- selection_2 = None
- bond_cutoff = 4.5
- exclude_from_automatic_linkingselection_1 = None selection_2 = None
- selection_1 = None
- selection_2 = None
- apply_cis_trans_specificationcis_trans_mod = cis *trans residue_selection = None Residues containing C-alpha atom of omega dihedral
- cis_trans_mod = cis *trans
- residue_selection = None Residues containing C-alpha atom of omega dihedral
- apply_cif_restraintsrestraints_file_name = None residue_selection = None
- restraints_file_name = None
- residue_selection = None
- apply_cif_modificationdata_mod = None residue_selection = None
- data_mod = None
- residue_selection = None
- apply_cif_linkdata_link = None residue_selection_1 = None residue_selection_2 = None
- data_link = None
- residue_selection_1 = None
- residue_selection_2 = None
- peptide_linkramachandran_restraints = False !!! OBSOLETED. Kept for backward compatibility only !!! Restrains peptide backbone to fall within allowed regions of Ramachandran plot. Although it does not eliminate outliers, it can significantly improve the percent favored and percent outliers at low resolution. Probably not useful (and maybe even harmful) at resolutions much higher than 3.5A.cis_threshold = 45 apply_all_trans = False discard_omega = False discard_psi_phi = True apply_peptide_plane = False omega_esd_override_value = None rama_weight = 1.0 scale_allowed = 1.0 rama_potential = *oldfield emsley rama_selection = None Selection of part of the model for which Ramachandran restraints will be set up.restrain_rama_outliers = True Apply restraints to Ramachandran outliersrestrain_rama_allowed = True Apply restraints to residues in allowed region on Ramachandran plotrestrain_allowed_outliers_with_emsley = False In case of restrain_rama_outliers=True and/or restrain_rama_allowed=True still restrain these residues with emsley. Make sense only in case of using oldfield potential.oldfieldesd = 10.0 weight_scale = 1.0 dist_weight_max = 10.0 weight = None plot_cutoff = 0.027
- ramachandran_restraints = False !!! OBSOLETED. Kept for backward compatibility only !!! Restrains peptide backbone to fall within allowed regions of Ramachandran plot. Although it does not eliminate outliers, it can significantly improve the percent favored and percent outliers at low resolution. Probably not useful (and maybe even harmful) at resolutions much higher than 3.5A.
- cis_threshold = 45
- apply_all_trans = False
- discard_omega = False
- discard_psi_phi = True
- apply_peptide_plane = False

- omega_esd_override_value = None
- rama_weight = 1.0
- scale_allowed = 1.0
- rama_potential = *oldfield emsley
- rama_selection = None Selection of part of the model for which Ramachandran restraints will be set up.
- restrain_rama_outliers = True Apply restraints to Ramachandran outliers
- restrain_rama_allowed = True Apply restraints to residues in allowed region on Ramachandran plot
- restrain_allowed_outliers_with_emsley = False In case of restrain_rama_outliers=True and/or restrain_rama_allowed=True still restrain these residues with emsley. Make sense only in case of using oldfield potential.
- oldfieldesd = 10.0 weight_scale = 1.0 dist_weight_max = 10.0 weight = None plot_cutoff = 0.027
- esd = 10.0
- weight_scale = 1.0
- dist_weight_max = 10.0
- weight = None
- plot_cutoff = 0.027
- ramachandran_plot_restraintsenabled = False favored = *oldfield emsley emsley8k phi_psi_2 allowed = *oldfield emsley emsley8k phi_psi_2 outlier = *oldfield emsley emsley8k phi_psi_2 selection = None Selection of part of the model for which Ramachandran restraints will be set up.inject_emsley8k_into_oldfield_favored = True Backdoor to disable temporary dirty hack to use botholdfieldweight = 0. Direct weight value. If 0 the weight will be calculated as following: (w, op.esd, op.dist_weight_max, 2.0, op.weight_scale) 1 / esd^2 * max(2.0, min(current_distance_to_allowed, dist_weight_max)) * weight_scale max(2.0, current_distance_to_allowed) 1 / esd^2 * weight_scale * max(distance_to_allowed_cutoff, current_distance_to_allowed) weight_scale(=0.01) * max(distance_weight_min(=2.), min(distance_weight_max(=10.), current_distance_to_allowed))weight_scale = 0.01 distance_weight_min = 2.0 minimum coefficient when scaling depending on how far the residue is from allowed region.distance_weight_max = 10.0 maximum coefficient when scaling depending on how far the residue is from allowed region.plot_cutoff = 0.027 emsleyweight = 1.0 scale_allowed = 1.0 emsley8kweight_favored = 5.0 weight_allowed = 10.0 weight_outlier = 10.0 phi_psi_2favored_strategy = *closest highest_probability random weighted_random allowed_strategy = *closest highest_probability random weighted_random outlier_strategy = *closest highest_probability random weighted_random
- enabled = False
- favored = *oldfield emsley emsley8k phi_psi_2
- allowed = *oldfield emsley emsley8k phi_psi_2
- outlier = *oldfield emsley emsley8k phi_psi_2
- selection = None Selection of part of the model for which Ramachandran restraints will be set up.
- inject_emsley8k_into_oldfield_favored = True Backdoor to disable temporary dirty hack to use both
- oldfieldweight = 0. Direct weight value. If 0 the weight will be calculated as following: (w, op.esd, op.dist_weight_max, 2.0, op.weight_scale) 1 / esd^2 * max(2.0, min(current_distance_to_allowed, dist_weight_max)) * weight_scale max(2.0, current_distance_to_allowed) 1 / esd^2 * weight_scale * max(distance_to_allowed_cutoff, current_distance_to_allowed) weight_scale(=0.01) *

max(distance_weight_min(=2.), min(distance_weight_max(=10.), current_distance_to_allowed))weight_scale = 0.01 distance_weight_min = 2.0 minimum coefficient when scaling depending on how far the residue is from allowed region.distance_weight_max = 10.0 maximum coefficient when scaling depending on how far the residue is from allowed region.plot_cutoff = 0.027

- weight = 0. Direct weight value. If 0 the weight will be calculated as following: (w, op.esd, op.dist_weight_max, 2.0, op.weight_scale) 1 / esd^2 * max(2.0, min(current_distance_to_allowed, dist_weight_max)) * weight_scale max(2.0, current_distance_to_allowed) 1 / esd^2 * weight_scale * max(distance_to_allowed_cutoff, current_distance_to_allowed) weight_scale(=0.01) * max(distance_weight_min(=2.), min(distance_weight_max(=10.), current_distance_to_allowed))
- weight_scale = 0.01
- distance_weight_min = 2.0 minimum coefficient when scaling depending on how far the residue is from allowed region.
- distance_weight_max = 10.0 maximum coefficient when scaling depending on how far the residue is from allowed region.
- plot_cutoff = 0.027
- emsleyweight = 1.0 scale_allowed = 1.0
- weight = 1.0
- scale_allowed = 1.0
- emsley8kweight_favored = 5.0 weight_allowed = 10.0 weight_outlier = 10.0
- weight_favored = 5.0
- weight_allowed = 10.0
- weight_outlier = 10.0
- phi_psi_2favored_strategy = *closest highest_probability random weighted_random allowed_strategy = *closest highest_probability random weighted_random outlier_strategy = *closest highest_probability random weighted_random
- favored_strategy = *closest highest_probability random weighted_random
- allowed_strategy = *closest highest_probability random weighted_random
- outlier_strategy = *closest highest_probability random weighted_random
- rna_sugar_pucker_analysisbond_min_distance = 1.2 bond_max_distance = 1.8 epsilon_range_min = 155.0 epsilon_range_max = 310.0 delta_range_2p_min = 129.0 delta_range_2p_max = 162.0 delta_range_3p_min = 65.0 delta_range_3p_max = 104.0 p_distance_c1p_outbound_line_2p_max = 2.9 o3p_distance_c1p_outbound_line_2p_max = 2.4 bond_detection_distance_tolerance = 0.5 enable = True
- bond_min_distance = 1.2
- bond_max_distance = 1.8
- epsilon_range_min = 155.0
- epsilon_range_max = 310.0
- delta_range_2p_min = 129.0
- delta_range_2p_max = 162.0
- delta_range_3p_min = 65.0
- delta_range_3p_max = 104.0

- p_distance_c1p_outbound_line_2p_max = 2.9
- o3p_distance_c1p_outbound_line_2p_max = 2.4
- bond_detection_distance_tolerance = 0.5
- enable = True
- show_histogram_slotsbond_lengths = 5 nonbonded_interaction_distances = 5 bond_angle_deviations_from_ideal = 5 dihedral_angle_deviations_from_ideal = 5 chiral_volume_deviations_from_ideal = 5
- bond_lengths = 5
- nonbonded_interaction_distances = 5
- bond_angle_deviations_from_ideal = 5
- dihedral_angle_deviations_from_ideal = 5
- chiral_volume_deviations_from_ideal = 5
- show_max_itemsnot_linked = 5 bond_restraints_sorted_by_residual = 5 nonbonded_interactions_sorted_by_model_distance = 5 bond_angle_restraints_sorted_by_residual = 5 dihedral_angle_restraints_sorted_by_residual = 3 chirality_restraints_sorted_by_residual = 3 planarity_restraints_sorted_by_residual = 3 residues_with_excluded_nonbonded_symmetry_interactions = 12 fatal_problem_max_lines = 10
- not_linked = 5
- bond_restraints_sorted_by_residual = 5
- nonbonded_interactions_sorted_by_model_distance = 5
- bond_angle_restraints_sorted_by_residual = 5
- dihedral_angle_restraints_sorted_by_residual = 3
- chirality_restraints_sorted_by_residual = 3
- planarity_restraints_sorted_by_residual = 3
- residues_with_excluded_nonbonded_symmetry_interactions = 12
- fatal_problem_max_lines = 10
- ncs_groupThe definition of one NCS group. Note, that almost always in refinement programs they will be checked and filtered if needed.reference = None Residue selection string for the complete master NCS copyselection = None Residue selection string for each NCS copy location in ASU
- reference = None Residue selection string for the complete master NCS copy
- selection = None Residue selection string for each NCS copy location in ASU
- ncs_searchSet of parameters for NCS search procedure. Some of them also used for filtering user-supplied ncs_group.enabled = False Enable NCS restraints or constraints in refinement (in some cases may be switched on inside refinement program).exclude_selection = "element H or element D or water" Atoms selected by this selection will be excluded from the model before any NCS search and/or filtering procedures. There is no way atoms defined by this selection will be in NCS.chain_similarity_threshold = 0.85 Threshold for sequence similarity between matching chains. A smaller value may cause more chains to be grouped together and can lower the number of common residueschain_max_rmsd = 2. limit of rms difference between chains to be considered as copiesresidue_match_radius = 4.0 Maximum allowed distance difference between pairs of matching atoms of two residuestry_shortcuts = False Try very quick check to speed up the search when chains are

identical. If failed, regular search will be performed automatically.minimum_number_of_atoms_in_copy = 3 Do not create ncs groups where master and copies would contain less than specified amount of atomsvalidate_user_supplied_groups = True Enable validation of user-supplied ncs_group. Need to exercise a lot of caution turning this off. This option is for developers only.

- enabled = False Enable NCS restraints or constraints in refinement (in some cases may be switched on inside refinement program).
- exclude_selection = "element H or element D or water" Atoms selected by this selection will be excluded from the model before any NCS search and/or filtering procedures. There is no way atoms defined by this selection will be in NCS.
- chain_similarity_threshold = 0.85 Threshold for sequence similarity between matching chains. A smaller value may cause more chains to be grouped together and can lower the number of common residues
- chain_max_rmsd = 2. limit of rms difference between chains to be considered as copies
- residue_match_radius = 4.0 Maximum allowed distance difference between pairs of matching atoms of two residues
- try_shortcuts = False Try very quick check to speed up the search when chains are identical. If failed, regular search will be performed automatically.
- minimum_number_of_atoms_in_copy = 3 Do not create ncs groups where master and copies would contain less than specified amount of atoms
- validate_user_supplied_groups = True Enable validation of user-supplied ncs_group. Need to exercise a lot of caution turning this off. This option is for developers only.
- clash_guardnonbonded_distance_threshold = 0.5 max_number_of_distances_below_threshold = 100 max_fraction_of_distances_below_threshold = 0.1
- nonbonded_distance_threshold = 0.5
- max_number_of_distances_below_threshold = 100
- max_fraction_of_distances_below_threshold = 0.1
- geometry_restraintseditsexcessive_bond_distance_limit = 10 bondaction = *add delete change atom_selection_1 = None atom_selection_2 = None symmetry_operation = None The bond is between atom_1 and symmetry_operation * atom_2, with atom_1 and atom_2 given in fractional coordinates. Example: symmetry_operation = -x-1,-y,zdistance_ideal = None sigma = None slack = None limit = -1.0 top_out = False angleaction = *add delete change atom_selection_1 = None atom_selection_2 = None atom_selection_3 = None angle_ideal = None sigma = None dihedralaction = *add delete change atom_selection_1 = None atom_selection_2 = None atom_selection_3 = None atom_selection_4 = None angle_ideal = None alt_angle_ideals = None sigma = None periodicity = 1 planarityaction = *add delete change atom_selection = None sigma = None parallelityaction = *add delete change atom_selection_1 = None atom_selection_2 = None sigma = 0.027 target_angle_deg = 0 removeangles = None dihedrals = None chiralities = None planarities = None parallelities = None
- editsexcessive_bond_distance_limit = 10 bondaction = *add delete change atom_selection_1 = None atom_selection_2 = None symmetry_operation = None The bond is between atom_1 and symmetry_operation * atom_2, with atom_1 and atom_2 given in fractional coordinates. Example: symmetry_operation = -x-1,-y,zdistance_ideal = None sigma = None slack = None limit = -1.0 top_out = False angleaction = *add delete change atom_selection_1 = None atom_selection_2 = None atom_selection_3 = None angle_ideal = None sigma = None dihedralaction = *add delete change atom_selection_1 = None atom_selection_2 = None atom_selection_3 = None atom_selection_4 = None angle_ideal = None alt_angle_ideals = None sigma = None periodicity = 1 planarityaction = *add delete

change atom_selection = None sigma = None parallelityaction = *add delete change atom_selection_1 = None atom_selection_2 = None sigma = 0.027 target_angle_deg = 0

- excessive_bond_distance_limit = 10
- bondaction = *add delete change atom_selection_1 = None atom_selection_2 = None symmetry_operation = None The bond is between atom_1 and symmetry_operation * atom_2, with atom_1 and atom_2 given in fractional coordinates. Example: symmetry_operation = -x-1,-y,zdistance_ideal = None sigma = None slack = None limit = -1.0 top_out = False
- action = *add delete change
- atom_selection_1 = None
- atom_selection_2 = None
- symmetry_operation = None The bond is between atom_1 and symmetry_operation * atom_2, with atom_1 and atom_2 given in fractional coordinates. Example: symmetry_operation = -x-1,-y,z
- distance_ideal = None
- sigma = None
- slack = None
- limit = -1.0
- top_out = False
- angleaction = *add delete change atom_selection_1 = None atom_selection_2 = None atom_selection_3 = None angle_ideal = None sigma = None
- action = *add delete change
- atom_selection_1 = None
- atom_selection_2 = None
- atom_selection_3 = None
- angle_ideal = None
- sigma = None
- dihedralaction = *add delete change atom_selection_1 = None atom_selection_2 = None atom_selection_3 = None atom_selection_4 = None angle_ideal = None alt_angle_ideals = None sigma = None periodicity = 1
- action = *add delete change
- atom_selection_1 = None
- atom_selection_2 = None
- atom_selection_3 = None
- atom_selection_4 = None
- angle_ideal = None
- alt_angle_ideals = None
- sigma = None
- periodicity = 1
- planarityaction = *add delete change atom_selection = None sigma = None
- action = *add delete change

- atom_selection = None
- sigma = None
- parallelityaction = *add delete change atom_selection_1 = None atom_selection_2 = None sigma = 0.027 target_angle_deg = 0
- action = *add delete change
- atom_selection_1 = None
- atom_selection_2 = None
- sigma = 0.027
- target_angle_deg = 0
- removeangles = None dihedrals = None chiralities = None planarities = None parallelities = None
- angles = None
- dihedrals = None
- chiralities = None
- planarities = None
- parallelities = None
- reference_modelenabled = False Restrains the dihedral angles to a high-resolution reference structure to reduce overfitting at low resolution. You will need to specify a reference PDB file (in the input list in the main window) to use this option.file = None use_starting_model_as_reference = False sigma = 1.0 limit = 15.0 hydrogens = False Include dihedrals with hydrogen atomsmain_chain = True Include dihedrals formed by main chain atomsside_chain = True Include dihedrals formed by side chain atomsfix_outliers = True Try to fix rotamer outliers in refined modelstrict_rotamer_matching = False Make sure that rotamers in refinement model matches those in reference model even when they are not outliersauto_shutoff_for_ncs = False Do not apply to parts of structure covered by NCS restraintssecondary_structure_only = False Only apply reference model restraints to secondary structure elements (helices and sheets)reference_groupreference = None selection = None file_name = None this is to used internally to disambiguate cases where multiple reference models contain the same chain ID. This normally does not need to be set by the usersearch_optionsSet of parameters for NCS search procedure. Some of them also used for filtering user-supplied ncs_group.exclude_selection = "element H or element D or water" Atoms selected by this selection will be excluded from the model before any NCS search and/or filtering procedures. There is no way atoms defined by this selection will be in NCS.chain_similarity_threshold = 0.85 Threshold for sequence similarity between matching chains. A smaller value may cause more chains to be grouped together and can lower the number of common residueschain_max_rmsd = 100. limit of rms difference between chains to be considered as copiesresidue_match_radius = 1000 Maximum allowed distance difference between pairs of matching atoms of two residuestry_shortcuts = False Try very quick check to speed up the search when chains are identical. If failed, regular search will be performed automatically.minimum_number_of_atoms_in_copy = 3 Do not create ncs groups where master and copies would contain less than specified amount of atomsvalidate_user_supplied_groups = True Enable validation of user-supplied ncs_group. Need to exercise a lot of caution turning this off. This option is for developers only.
- enabled = False Restrains the dihedral angles to a high-resolution reference structure to reduce overfitting at low resolution. You will need to specify a reference PDB file (in the input list in the main window) to use this option.
- file = None

- use_starting_model_as_reference = False
- sigma = 1.0
- limit = 15.0
- hydrogens = False Include dihedrals with hydrogen atoms
- main_chain = True Include dihedrals formed by main chain atoms
- side_chain = True Include dihedrals formed by side chain atoms
- fix_outliers = True Try to fix rotamer outliers in refined model
- strict_rotamer_matching = False Make sure that rotamers in refinement model matches those in reference model even when they are not outliers
- auto_shutoff_for_ncs = False Do not apply to parts of structure covered by NCS restraints
- secondary_structure_only = False Only apply reference model restraints to secondary structure elements (helices and sheets)
- reference_groupreference = None selection = None file_name = None this is to used internally to disambiguate cases where multiple reference models contain the same chain ID. This normally does not need to be set by the user
- reference = None
- selection = None
- file_name = None this is to used internally to disambiguate cases where multiple reference models contain the same chain ID. This normally does not need to be set by the user
- search_optionsSet of parameters for NCS search procedure. Some of them also used for filtering user-supplied ncs_group.exclude_selection = "element H or element D or water" Atoms selected by this selection will be excluded from the model before any NCS search and/or filtering procedures. There is no way atoms defined by this selection will be in NCS.chain_similarity_threshold = 0.85 Threshold for sequence similarity between matching chains. A smaller value may cause more chains to be grouped together and can lower the number of common residueschain_max_rmsd = 100. limit of rms difference between chains to be considered as copiesresidue_match_radius = 1000 Maximum allowed distance difference between pairs of matching atoms of two residuestry_shortcuts = False Try very quick check to speed up the search when chains are identical. If failed, regular search will be performed automatically.minimum_number_of_atoms_in_copy = 3 Do not create ncs groups where master and copies would contain less than specified amount of atomsvalidate_user_supplied_groups = True Enable validation of user-supplied ncs_group. Need to exercise a lot of caution turning this off. This option is for developers only.
- exclude_selection = "element H or element D or water" Atoms selected by this selection will be excluded from the model before any NCS search and/or filtering procedures. There is no way atoms defined by this selection will be in NCS.
- chain_similarity_threshold = 0.85 Threshold for sequence similarity between matching chains. A smaller value may cause more chains to be grouped together and can lower the number of common residues
- chain_max_rmsd = 100. limit of rms difference between chains to be considered as copies
- residue_match_radius = 1000 Maximum allowed distance difference between pairs of matching atoms of two residues
- try_shortcuts = False Try very quick check to speed up the search when chains are identical. If failed, regular search will be performed automatically.

- minimum_number_of_atoms_in_copy = 3 Do not create ncs groups where master and copies would contain less than specified amount of atoms
- validate_user_supplied_groups = True Enable validation of user-supplied ncs_group. Need to exercise a lot of caution turning this off. This option is for developers only.
- minimizationGeometry minimization parametersmax_iterations = 500 Maximun number of minimization iterationsmacro_cycles = 5 Number of minimization macro-cyclesalternate_nonbonded_off_on = False rmsd_bonds_termination_cutoff = 0 stop after reaching specified cutoff valuermsd_angles_termination_cutoff = 0 stop after reaching specified cutoff valuegrms_termination_cutoff = 0 stop after reaching specified cutoff valuecorrect_special_position_tolerance = 1.0 riding_h = True Use riding model for HmoveDefine what to include into refinement targetbond = True nonbonded = True angle = True dihedral = True chirality = True planarity = True parallelity = True
- max_iterations = 500 Maximun number of minimization iterations
- macro_cycles = 5 Number of minimization macro-cycles
- alternate_nonbonded_off_on = False
- rmsd_bonds_termination_cutoff = 0 stop after reaching specified cutoff value
- rmsd_angles_termination_cutoff = 0 stop after reaching specified cutoff value
- grms_termination_cutoff = 0 stop after reaching specified cutoff value
- correct_special_position_tolerance = 1.0
- riding_h = True Use riding model for H
- moveDefine what to include into refinement targetbond = True nonbonded = True angle = True dihedral = True chirality = True planarity = True parallelity = True
- bond = True
- nonbonded = True
- angle = True
- dihedral = True
- chirality = True
- planarity = True
- parallelity = True
- amberParameters for using Amber in refinement.use_amber = False Use Amber for all the gradients in refinementtopology_file_name = None A topology file needed by Amber. Can be generated using phenix.AmberPrep.coordinate_file_name = None A coordinate file needed by Amber. Can be generated using phenix.AmberPrep.order_file_name = None A file that maps amber atom numbers to phenix atom numbers.wxc_factor = 1. restraint_wt = 0. restraintmask = '' reference_file_name = '' bellymask = '' If given, turn on belly in sanderqmmask = '' If given, turn on QM/MM with the given maskqmcharge = 0 Charge of the QM/MM regionnetcdf_trajectory_file_name = '' If given, turn on writing netcdf trajectoryprint_amber_energies = False Print details of Amber energies during refinement
- use_amber = False Use Amber for all the gradients in refinement
- topology_file_name = None A topology file needed by Amber. Can be generated using phenix.AmberPrep.
- coordinate_file_name = None A coordinate file needed by Amber. Can be generated using phenix.AmberPrep.

- order_file_name = None A file that maps amber atom numbers to phenix atom numbers.
- wxc_factor = 1.
- restraint_wt = 0.
- restraintmask = ''
- reference_file_name = ''
- bellymask = '' If given, turn on belly in sander
- qmmask = '' If given, turn on QM/MM with the given mask
- qmcharge = 0 Charge of the QM/MM region
- netcdf_trajectory_file_name = '' If given, turn on writing netcdf trajectory
- print_amber_energies = False Print details of Amber energies during refinement
- qiQMworking_directory = None qm_restraintsselection = None selection for core of atoms to calculate new restraints via a QM geometry minimisationrun_in_macro_cycles = *first_only first_and_last all last_only test the steps of the refinement that the restraints generation is runbuffer = 3.5 distance to include entire residues into the enviroment of the corecalculate = *in_situ_opt starting_energy final_energy starting_strain final_strain starting_bound final_bound starting_binding final_binding starting_higher_single_point final_higher_single_point Choose QM calculations to runwrite_files = *restraints pdb_core pdb_buffer pdb_final_core pdb_final_buffer which ligand or cluster files to writeprotein_optimisation_freeze = *all None main_chain main_chain_to_beta main_chain_to_delta torsions the parts of protein residues that are frozen when an amino acid is the main selectionremove_water = False restraints_filename = Auto restraints filename is based on model name if not specifiedcleanup = all *most None ignore_x_h_distance_protein = False skip check on transfer of proton during QM optimisationignore_lack_of_h_on_ligand = False skip check on protonation of ligand for entities such as MgF3capping_groups = True include_nearest_neighbours_in_optimisation = False include the side chains of protein in the QM optimisationinclude_inter_residue_restraints = False do_not_update_restraints = False For testing and maybe getting strain energy of standard restraintsbuffer_selection = None use this instead of distance from selectionspecific_atom_chargesatom_selection = None charge = None specific_atom_multiplicitiesatom_selection = None multiplicity = None freeze_specific_atomsatom_selection = None packageprogram = *mopac test charge = Auto multiplicity = Auto method = Auto basis_set = Auto solvent_model = None nproc = 1 read_output_to_skip_opt_if_available = False ignore_input_differences = False view_output = None
- working_directory = None
- qm_restraintsselection = None selection for core of atoms to calculate new restraints via a QM geometry minimisationrun_in_macro_cycles = *first_only first_and_last all last_only test the steps of the refinement that the restraints generation is runbuffer = 3.5 distance to include entire residues into the enviroment of the corecalculate = *in_situ_opt starting_energy final_energy starting_strain final_strain starting_bound final_bound starting_binding final_binding starting_higher_single_point final_higher_single_point Choose QM calculations to runwrite_files = *restraints pdb_core pdb_buffer pdb_final_core pdb_final_buffer which ligand or cluster files to writeprotein_optimisation_freeze = *all None main_chain main_chain_to_beta main_chain_to_delta torsions the parts of protein residues that are frozen when an amino acid is the main selectionremove_water = False restraints_filename = Auto restraints filename is based on model name if not specifiedcleanup = all *most None ignore_x_h_distance_protein = False skip check on transfer of proton during QM optimisationignore_lack_of_h_on_ligand = False skip check on protonation of ligand for entities such as MgF3capping_groups = True include_nearest_neighbours_in_optimisation = False include the side

chains of protein in the QM optimisationinclude_inter_residue_restraints = False do_not_update_restraints = False For testing and maybe getting strain energy of standard restraintsbuffer_selection = None use this instead of distance from selectionspecific_atom_chargesatom_selection = None charge = None specific_atom_multiplicitiesatom_selection = None multiplicity = None freeze_specific_atomsatom_selection = None packageprogram = *mopac test charge = Auto multiplicity = Auto method = Auto basis_set = Auto solvent_model = None nproc = 1 read_output_to_skip_opt_if_available = False ignore_input_differences = False view_output = None

- selection = None selection for core of atoms to calculate new restraints via a QM geometry minimisation
- run_in_macro_cycles = *first_only first_and_last all last_only test the steps of the refinement that the restraints generation is run
- buffer = 3.5 distance to include entire residues into the enviroment of the core
- calculate = *in_situ_opt starting_energy final_energy starting_strain final_strain starting_bound final_bound starting_binding final_binding starting_higher_single_point final_higher_single_point Choose QM calculations to run
- write_files = *restraints pdb_core pdb_buffer pdb_final_core pdb_final_buffer which ligand or cluster files to write
- protein_optimisation_freeze = *all None main_chain main_chain_to_beta main_chain_to_delta torsions the parts of protein residues that are frozen when an amino acid is the main selection
- remove_water = False
- restraints_filename = Auto restraints filename is based on model name if not specified
- cleanup = all *most None
- ignore_x_h_distance_protein = False skip check on transfer of proton during QM optimisation
- ignore_lack_of_h_on_ligand = False skip check on protonation of ligand for entities such as MgF3
- capping_groups = True
- include_nearest_neighbours_in_optimisation = False include the side chains of protein in the QM optimisation
- include_inter_residue_restraints = False
- do_not_update_restraints = False For testing and maybe getting strain energy of standard restraints
- buffer_selection = None use this instead of distance from selection
- specific_atom_chargesatom_selection = None charge = None
- atom_selection = None
- charge = None
- specific_atom_multiplicitiesatom_selection = None multiplicity = None
- atom_selection = None
- multiplicity = None
- freeze_specific_atomsatom_selection = None
- atom_selection = None
- packageprogram = *mopac test charge = Auto multiplicity = Auto method = Auto basis_set = Auto solvent_model = None nproc = 1 read_output_to_skip_opt_if_available = False ignore_input_differences = False view_output = None

- program = *mopac test
- charge = Auto
- multiplicity = Auto
- method = Auto
- basis_set = Auto
- solvent_model = None
- nproc = 1
- read_output_to_skip_opt_if_available = False
- ignore_input_differences = False
- view_output = None

Placing CA atoms in a map with refine_ca_model

refine_ca_model.html

Author(s)

- refine_ca_model: Tom Terwilliger

Purpose

The routine refine_ca_model is a tool for adjusting CA/CB positions in a preliminary model to increase its suitability for generating an all-atom model.

Usage

How refine_ca_model works:

The refine_ca_model tool takes a set of CA positions, and optionally, a set of CB positions. It adjusts the CA/CB positions to put the atoms in density (if possible) and also to place the CB atoms in plausible positions relative to the CA positions and also to make the CA-CA distances as close to 3.8 A as possible.

Using refine_ca_model:

The tool refine_ca_model is usually run automatically as part of trace_and_build. However you can run it yourself with a CA/CB model, a map, and a resolution.

Input map file: The map file should cover the model you supply.

Resolution: Specify the resolution of your map (usually the resolution defined by your half-dataset Fourier shell correlation

CA/CB model: Supply a model that at least has CA positions. If you supply CB positions the new CB positions will be restrained towards the ones you supply.

Examples

Standard run of refine_ca_model:

You can use refine_ca_model to adjust a CA/CB model based on a cryo-EM map:

```
phenix.refine_ca_model my_map.mrc resolution=2.8 ca_cb_model.pdb
```

Possible Problems

Specific limitations and problems:

Literature

Additional information

List of all available keywords

job_title = None Job title in PHENIX GUI, not used on command lineinput_filesmap_file = None File with CCP4-style map. May have origin in any location.model_file = None Input PDB file with CA positions to be adjusted. Other atoms ignoredoutput_filestarget_output_format = *None pdb mmcif Desired output format (if possible). Choices are None (try to use input format), pdb, mmcif. If output model does not fit in pdb format, mmcif will be used. Default is pdb.pdb_out = refined_model.pdb Model with refined CA and CB positions (and optional full model)temp_dir = None Temporary directorycrystal_inforesolution = None High-resolution limit for map analysis.scattering_table = n_gaussian wk1995 it1992 *electron neutron Choice of scattering table for structure factor calculations. Standard for X-ray is n_gaussian, for cryoEM is electron.chain_type = *PROTEIN Chain type. Must be PROTEINca_ca_distance = 3.8 CA-CA distancesolvent_content = None Solvent fraction of the cell. If this is density cut out from a bigger cell, you can specify the fraction of the volume of this cell that is taken up by the macromolecule. Normally set automatically. Values go from 0 to 1. Used to scale density.solvent_content_iterations = 3 Iterations of solvent fraction estimationcb_up_from_ca = 1.36 CB atom is cb_up_from_ca in direction of CA atom from mid-point between previous and next CA atoms.cb_right_from_ca = 0.68 CB atom is cb_right_from_ca in direction of (CA atom from mid-point between previous and next CA atoms) cross direction of (previous CA atom to next CA atom).cb_scale = 0.80 Scale on distance for CA-CB. Use 1/cb_scale for densityorigin_cart = None Origin (cartesian coordinates, overrides value based on map)refinerefine_group_length = 10 Length of segments to refinerefine_in_groups = False Refine in groups, not all at oncerefine_start = None first residue to refinerefine_end = None Last residue to refineresidual_start = None first residue to include in residualresidual_end = None Last residue to include in residualmax_iterations = 10 Iterations in refinementweightsweight_ca_ca_dist = 1. weight on CA-CA distanceweight_cb = 5. weight on CB initial positionsweight_proximity_to_initial_position = 0.5 weight on proximity to input CA positionweight_density = -1.0 weight on density at CA atoms, between CA atoms, and CB positionsweight_cb_density = -5. weight on density at CA atoms, between CA atoms, and CB positionscontrolnproc = 1 Number of processors to userandom_seed = 171731 Random seedresolve_size = None Size of resolve to use. ignore_symmetry_conflicts = False You can ignore the symmetry information (CRYST1) from coordinate files. This may be necessary if your model has been placed in a box with box_map for example.verbose = False Verbose outputskip_optimization = False Skip optimizationmax_dirs = 1000 Maximum number of directories (refine_ca_model_xxx)guiGUI-specific parameter required for output directoryoutput_dir = None

- job_title = None Job title in PHENIX GUI, not used on command line
- input_filesmap_file = None File with CCP4-style map. May have origin in any location.model_file = None Input PDB file with CA positions to be adjusted. Other atoms ignored
- map_file = None File with CCP4-style map. May have origin in any location.
- model_file = None Input PDB file with CA positions to be adjusted. Other atoms ignored
- output_filestarget_output_format = *None pdb mmcif Desired output format (if possible). Choices are None (try to use input format), pdb, mmcif. If output model does not fit in pdb format, mmcif will be used. Default is pdb.pdb_out = refined_model.pdb Model with refined CA and CB positions (and optional full model)temp_dir = None Temporary directory
- target_output_format = *None pdb mmcif Desired output format (if possible). Choices are None (try to use input format), pdb, mmcif. If output model does not fit in pdb format, mmcif will be used. Default is pdb.
- pdb_out = refined_model.pdb Model with refined CA and CB positions (and optional full model)
- temp_dir = None Temporary directory

- crystal_inforesolution = None High-resolution limit for map analysis.scattering_table = n_gaussian wk1995 it1992 *electron neutron Choice of scattering table for structure factor calculations. Standard for X-ray is n_gaussian, for cryoEM is electron.chain_type = *PROTEIN Chain type. Must be PROTEINca_ca_distance = 3.8 CA-CA distancesolvent_content = None Solvent fraction of the cell. If this is density cut out from a bigger cell, you can specify the fraction of the volume of this cell that is taken up by the macromolecule. Normally set automatically. Values go from 0 to 1. Used to scale density.solvent_content_iterations = 3 Iterations of solvent fraction estimationcb_up_from_ca = 1.36 CB atom is cb_up_from_ca in direction of CA atom from mid-point between previous and next CA atoms.cb_right_from_ca = 0.68 CB atom is cb_right_from_ca in direction of (CA atom from mid-point between previous and next CA atoms) cross direction of (previous CA atom to next CA atom).cb_scale = 0.80 Scale on distance for CA-CB. Use 1/cb_scale for densityorigin_cart = None Origin (cartesian coordinates, overrides value based on map)
- resolution = None High-resolution limit for map analysis.
- scattering_table = n_gaussian wk1995 it1992 *electron neutron Choice of scattering table for structure factor calculations. Standard for X-ray is n_gaussian, for cryoEM is electron.
- chain_type = *PROTEIN Chain type. Must be PROTEIN
- ca_ca_distance = 3.8 CA-CA distance
- solvent_content = None Solvent fraction of the cell. If this is density cut out from a bigger cell, you can specify the fraction of the volume of this cell that is taken up by the macromolecule. Normally set automatically. Values go from 0 to 1. Used to scale density.
- solvent_content_iterations = 3 Iterations of solvent fraction estimation
- cb_up_from_ca = 1.36 CB atom is cb_up_from_ca in direction of CA atom from mid-point between previous and next CA atoms.
- cb_right_from_ca = 0.68 CB atom is cb_right_from_ca in direction of (CA atom from mid-point between previous and next CA atoms) cross direction of (previous CA atom to next CA atom).
- cb_scale = 0.80 Scale on distance for CA-CB. Use 1/cb_scale for density
- origin_cart = None Origin (cartesian coordinates, overrides value based on map)
- refinerefine_group_length = 10 Length of segments to refinerefine_in_groups = False Refine in groups, not all at oncerefine_start = None first residue to refinerefine_end = None Last residue to refineresidual_start = None first residue to include in residualresidual_end = None Last residue to include in residualmax_iterations = 10 Iterations in refinement
- refine_group_length = 10 Length of segments to refine
- refine_in_groups = False Refine in groups, not all at once
- refine_start = None first residue to refine
- refine_end = None Last residue to refine
- residual_start = None first residue to include in residual
- residual_end = None Last residue to include in residual
- max_iterations = 10 Iterations in refinement
- weightsweight_ca_ca_dist = 1. weight on CA-CA distanceweight_cb = 5. weight on CB initial positionsweight_proximity_to_initial_position = 0.5 weight on proximity to input CA positionweight_density = -1.0 weight on density at CA atoms, between CA atoms, and CB positionsweight_cb_density = -5. weight on density at CA atoms, between CA atoms, and CB positions

- weight_ca_ca_dist = 1. weight on CA-CA distance
- weight_cb = 5. weight on CB initial positions
- weight_proximity_to_initial_position = 0.5 weight on proximity to input CA position
- weight_density = -1.0 weight on density at CA atoms, between CA atoms, and CB positions
- weight_cb_density = -5. weight on density at CA atoms, between CA atoms, and CB positions
- controlnproc = 1 Number of processors to userandom_seed = 171731 Random seedresolve_size = None Size of resolve to use. ignore_symmetry_conflicts = False You can ignore the symmetry information (CRYST1) from coordinate files. This may be necessary if your model has been placed in a box with box_map for example.verbose = False Verbose outputskip_optimization = False Skip optimizationmax_dirs = 1000 Maximum number of directories (refine_ca_model_xxx)
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- random_seed = 171731 Random seed
- resolve_size = None Size of resolve to use.
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- verbose = False Verbose output
- skip_optimization = False Skip optimization
- max_dirs = 1000 Maximum number of directories (refine_ca_model_xxx)
- guiGUI-specific parameter required for output directoryoutput_dir = None
- output_dir = None

The phenix.refine graphical interface

refine_gui.html

Contents

GUI overview

The graphical interface for phenix.refine runs the unmodified command-line version; default settings are unchanged. All parameters should be configurable through the GUI, although some of these (such as NCS restraint groups) are handled in a non-standard way. This document describes the behavior of the GUI only; see the phenix.refine documentation for details on specific parameters and methods. There is also a list of frequently asked questions about phenix.refine.

At present the GUI has the following capabilities in addition to the features included in phenix.refine:

automatic extraction of parameters from PDB and reflection files detection of unknown ligands, and generation of restraint files graphical mirroring and picking of any atom selection project-specific default settings and restraints NCS restraint group detection and editing tool TLS group detection and editing tool real-time visualization of current model and maps during refinement model validation using programs from the MolProbity server graphical comparison of structure quality measures printable plots of final statistics

- automatic extraction of parameters from PDB and reflection files
- detection of unknown ligands, and generation of restraint files
- graphical mirroring and picking of any atom selection
- project-specific default settings and restraints

- NCS restraint group detection and editing tool
- TLS group detection and editing tool
- real-time visualization of current model and maps during refinement
- model validation using programs from the MolProbity server
- graphical comparison of structure quality measures
- printable plots of final statistics

Video Tutorial

The tutorial video is available on the Phenix YouTube channel and covers the following topics:

- Basic overview
- How to run phenix.refine via the GUI
- Example of refinement with default parameters using the data from this tutorial

Configuration

Because phenix.refine currently has nearly 500 distinct parameters, many of which rarely need changing, and the window controls are mostly created automatically, advanced settings are hidden by default. The "user level" may be set within each window, or set globally in the Preferences. The most important parameters (input and output files, strategy, additional procedures such as solvent updating and simulated annealing) are shown in the main window.

Input files

The first tab contains settings for input files, which can be added by clicking the "+" button below the file list, or by dragging files into the list control and releasing the mouse button. Up to five different reflection files may be used (X-ray data, X-ray R-free flags, neutron data, neutron R-free flags, and experimental phases), but in most cases all data will be in a single file. Any number of PDB files for the starting model may be added; these will be combined internally. You may also specify a reference model for additional restraints (see below). Other supported file formats are CIF (restraint) files for non-standard ligands, and phenix.refine parameter files.

The GUI will attempt to guess the purpose of PDB and X-ray files, but you may change this by right-clicking on the "Data type" field associated with the file and selecting an option from a drop-down menu, or select the file and click the button labeled "Modify file data type".

If your reflection and/or PDB files have the appropriate information on crystal symmetry and data labels, these parameters will be extracted automatically. Appropriate column labels for reflection files will also be added to the drop-down menus; by default, anomalous intensities (I+/I-) will be used if present, but you may also use merged data and/or amplitudes. The R-free flags are optional, since phenix.refine will add them automatically if none are present, but once flags have been used in refinement (including the automated building in AutoSol or AutoBuild) the same set should always be used.

Refinement settings

The second tab contains options for the refinement protocol. Many of these controls will open additional dialog windows in response to mouse clicks or file selection. In many cases, you can open these windows on demand by right-clicking on the appropriate control (for instance, right-clicking the "Rigid body" strategy button will show the atom selection controls for defining rigid groups). All other parameters can be found in the Settings menu.

Most of the refinement options presented on this tab essentially fall into three categories:

Model parameterization. This determines which properties of the model will be refined, and how they are grouped. Options include coordinate refinement (individually against reciprocal-space data or real-space maps; or as rigid-body groups against reciprocal-space data); B-factor or ADP (atomic displacement parameter) refinement, with several options for grouping and isotropic versus anisotropic; atomic occupancies (i.e. the fraction of the time the atom is present in the refined position in the crystal); and anomalous groups. Optimization target. This defines the function to be minimized; by default this is at least the agreement of the model amplitudes with the experimentally measured amplitudes. For almost all macromolecular structures restraints on covalent geometry and the B-factors of nearby atoms will also be necessary, and a variety of customizations of the geometry restraints are possible. Optimization method. The standard optimization method used in phenix.refine and similar programs is gradient-driven minimization, but other options are available. These often have a significantly wider radius of convergence, but at the cost of increased runtime.

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The default options have been chosen to work well for partially refined structures at moderate resolutions, which comprise the majority of structures in the Protein Data Bank. However, at significantly higher or lower resolution, or for structures at the early or final stages of refinement, additional options may be necessary or beneficial.

Parameterization (refinement strategy)

The first set of controls determines the parameterization of the model and the attributes to be refined. By default, the Individual Sites, Individual ADPs (B-factors), and Occupancies strategies are selected, since these are usually appropriate at a wide range of resolutions and refinement stages. All strategies have associated atom selections which control what parts of the model are refined. A brief summary follows; note that the resolution limits specified are approximate only, and in practice may vary widely depending on model quality, data quality, and observation-to-parameters ratio.

XYZ coordinates is standard Cartesian refinement of model geometry and agreement with X-ray data. It is almost always appropriate, but at low-resolution (typically 3.0A or worse), additional restraints (NCS, reference model, etc.) are often necessary to prevent overfitting. Real-space also refines the atomic-positions, but uses the electron density map as target instead of F(obs). It will be run both locally (to fit individual residues with poor density or rotamer outliers) and globally. This is generally most helpful early in refinement for moderate- to high-resolution structures; at worse than 3Å the effectiveness falls off rapidly. Rigid-body refines the position of large, user-defined regions of the model without changing geometry. It is typically used immediately after molecular replacement, when the R-factors are usually very high, and at very low resolutions (4.5A or worse). If both rigid-body and individual sites refinement are checked (recommended), the rigid-body step will be done first, in the first cycle of refinement. Individual B-factors is restrained atomic B-factor refinement, and is usually appropriate except at very low resolution (there is some debate about what the cutoff should be, but we generally find the vast majority of structures in the PDB refine well with this option). Whether atoms are treated as isotropic or anisotropic is defined separately (see below). Group B-factors refines B-factors for multiple atoms at a time, most often single residues either as a whole or divided into mainchain and sidechain. (Larger user-defined groups are also allowed.) This is usually only necessary at very low resolution. Note that there are no restraints to keep the B-factors for adjacent groups similar! TLS parameters refines anisotropic displacements for large groups of atoms, typically individual domains or chains. Because this introduces only ten parameters for each group, and relatively few groups are used, it is appropriate at almost any resolution unless individual anisotropic B-factors are refined. (See note about B-factors below.) It is important to provide accurate atom selections here; the TLSMD server may be helpful in determining these automatically. Occupancies refines atomic occupancies. This is appropriate at high resolution (usually 1.7A or better) where alternate conformations are often present, and in situations where a ligand, ion, or solvent atom is not bound in every unit cell. By default, it will only be performed for atoms with partial occupancies, so it is safe to leave this on even if you have no partial-ocupancy atoms. Groupings are determined automatically if possible, but you may specify them manually. (This is especially important if you have complex relationships betweeen alternate conformations in different parts of the model.) Anomalous groups refines the anomalous coefficients f' and f'' for anomalously scattering atoms such as Se or heavy metals; it is only appropriate if you have anomalous data (I+/I- or F+/F-). This usually has minimal impact on R-factors, but can sometimes remove difference map peaks, and may be useful in conjunction with an anomalous residual or LLG map (which shows the location of unmodeled anomalous scatterers).

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• Anomalous groups refines the anomalous coefficients f' and f'' for anomalously scattering atoms such as Se or heavy metals; it is only appropriate if you have anomalous data (I+/I- or F+/F-). This usually has minimal impact on R-factors, but can sometimes remove difference map peaks, and may be useful in conjunction with an anomalous residual or LLG map (which shows the location of unmodeled anomalous scatterers).

Targets and restraints

The refinement protocol can be further modified by manipulating various refinement targets, including restraints (structural and otherwise):

Target function describes the calculation of model-based X-ray terms; "ML" means "Maximum likelihood" and is usually appropriate; "MLHL" use used when experimental phase restraints are added; "ML-SAD" is not suitable for general use. The least-squares (LS) target is only useful for twinned refinement or very small datasets (it will be used automatically if necessary). Weight optimization uses a grid search to determine the best relative weight for the X-ray target function and the geometry or B-factor restraints. At atomic resolution this is rarely necessary, but it will be very helpful at low resolution, and we recommend always enabling these options for at least the final round of refinement before the model is deposited. If you find that the structure is being overfit or the geometry is especially poor, optimizing the X-ray/stereochemistry weight will often fix the problem. NCS restraints preserve non-crystallographic symmetry between identical copies of a molecule. If NCS is present in the crystal, these restraints are usually appropriate at intermediate resolution (2.0-2.7A or worse) and essential at lower resolutions. NCS restraints are available in either torsion-angle or Cartesian parameterization: the former allows the molecules to be flexible while preserving local conformation, while the latter essentially treats NCS-related group as rigid relative to each other. This may be beneficial in some cases, but the torsion-angle restraints are the safest and easiest to use. Phenix will automatically identify appropriate NCS groups for you, and if you are using the torsion-angle version it is very rare that you will need to assign groups manually. Both implementations also allow you to restrain the B-factors of NCS groups to be similar, but this is not always useful or even valid. Automatic covalent bonds supplement the standard covalent geometry restraints for special cases such as modified amino acids, glycoproteins, and metal ions. This is usually a safe option as long as you do not have excessive errors in the input geometry, which may lead to spurious links being created. Note that you can also define custom geometry restraints yourself, and/or use ReadySet to generate them for you. Reference model restraints use the dihedral angles of an existing structure as restraints on the input model conformation, for instance, refining a low-resolution structure of a molecule whose high-resolution structure is available. This is most useful at low resolutions (usually worse than 3.0A). Secondary structure restraints add distance restraints for hydrogen bonds in alpha helices, beta sheets, and Watson-Crick base pairs. These are helpful for maintaining correct geometry at lower resolutions (2.5A or worse). Suitable atom selections will be detected automatically but you may also specify them manually. Use experimental phases uses the experimental phase distributions (Hendrickson-Lattman coefficients) from phasing programs such as AutoSol as part of the refinement target function. This is usually helpful if these phases are available, but only if the same data were used for both phasing and refinement.

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• Use experimental phases uses the experimental phase distributions (Hendrickson-Lattman coefficients) from phasing programs such as AutoSol as part of the refinement target function. This is usually helpful if these phases are available, but only if the same data were used for both phasing and refinement.

Optimization methods and other options

Several other options are available for the types of optimization used; these typically apply globally, except as noted.

Automatically add hydrogens to model runs phenix.ready_set to add hydrogen atoms (or deuteriums, if performing neutron crystallography) where appropriate. This usually only affects the R-factors at high resolution, but can be very helpful for improving geometry at any resolution. We recommend using explicit hydrogens on protein, nucleic acid, and ligand molecules throughout refinement. (We do not reocmmend adding hydrogens to waters unless you have exceptionally high resolution.) Hydrogen atoms will still be defined using the "riding" model unless otherwise requested, so they do not add parameters during refinement. (Note that this option can be left on if you already have hydrogen atoms in place and are refining as "riding"; if you are refining against neutron data and/or allowing hydrogen atoms to refine individually, you should uncheck the box, as it will otherwise replace the existing atoms.) Update waters automatically adds and removes solvent atoms as necessary; this is usually appropriate at any resolution where water molecules are visible in density (usually 2.5A or better). Ion placement extends solvent picking to build elemental ions such as calcium and zinc. To enable this option, simply enter a list of the elements to search for; this will also enable solvent picking as a first step. This option is not recommended if you have data worse than 3.0A resolution, and works best with anomalous data. Simulated annealing uses molecular dynamics with an extra X-ray term as an additional optimization method. It is very helpful for removing phase bias and overcoming energy barriers, and often yields a significant improvement over simple minimization early in refinement. The main disadvantage is speed. Two types of parameterization are available, Cartesian and torsion angles; the latter is more suitable for low resolution. Scattering table determines how atomic scattering is modeled; this is almost always either "n_gaussian" for standard X-ray refinement, or "neutron" (see below). Automatically correct N/Q/H errors uses the program Reduce to flip backwards sidechains, which appear symmetric when explicit hydrogens are not present. This is almost always a good idea and adds very little to overall run time. Twin law enables twinned refinement. Only a single twin domain is supported at present, and the refinement target is least-squares (LS) instead of maximum-likelihood. We recommend caution when using this option, as the R-factors are invariably lower whether or not the crystal was actually twinned, and the detwinned maps will have more model bias. Number of processors determines the parallelization for several optional steps, most importantly the weight optimization where it can reduce the runtime by up to 3x on large multi-core systems. Note that with default settings phenix.refine runs almost entirely in serial, and due to operating system limitations the multiprocessing option is not available on Windows.

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- Update waters automatically adds and removes solvent atoms as necessary; this is usually appropriate at any resolution where water molecules are visible in density (usually 2.5A or better).
- Ion placement extends solvent picking to build elemental ions such as calcium and zinc. To enable this option, simply enter a list of the elements to search for; this will also enable solvent picking as a first step. This option is not recommended if you have data worse than 3.0A resolution, and works best with anomalous data.
- Simulated annealing uses molecular dynamics with an extra X-ray term as an additional optimization method. It is very helpful for removing phase bias and overcoming energy barriers, and often yields a significant improvement over simple minimization early in refinement. The main disadvantage is speed. Two types of parameterization are available, Cartesian and torsion angles; the latter is more suitable for low resolution.
- Scattering table determines how atomic scattering is modeled; this is almost always either "n_gaussian" for standard X-ray refinement, or "neutron" (see below).
- Automatically correct N/Q/H errors uses the program Reduce to flip backwards sidechains, which appear symmetric when explicit hydrogens are not present. This is almost always a good idea and adds very little to overall run time.

- Twin law enables twinned refinement. Only a single twin domain is supported at present, and the refinement target is least-squares (LS) instead of maximum-likelihood. We recommend caution when using this option, as the R-factors are invariably lower whether or not the crystal was actually twinned, and the detwinned maps will have more model bias.
- Number of processors determines the parallelization for several optional steps, most importantly the weight optimization where it can reduce the runtime by up to 3x on large multi-core systems. Note that with default settings phenix.refine runs almost entirely in serial, and due to operating system limitations the multiprocessing option is not available on Windows.

The output tab allows you to change the destination directory and modify the default map coefficients and/or map files written out. By default, an MTZ file containing 2mFo-DFc and mFo-DFc map coefficients will be generated.

Creating restraints

If the PDB file contains ligands or prosthetic groups that are not part of the limited monomer library distributed with PHENIX, a warning message will be displayed when the file is loaded. The necessary restraints must be generated by phenix.elbow in order for refinement to proceed. To start this process, click the ReadySet button in the toolbar or choose "Prepare structure and restraints" from the Utilities menu. This will launch a dialog for running phenix.ready_set, a simple utility for preparing structures for phenix.refine.

Generating the restraints will take between 30 seconds and several minutes, depending on computer speed and the size of the ligand. When ReadySet is finished, another window will be displayed summarizing the results. If a project directory is defined, these can be saved for future runs and will be automatically loaded into the GUI when launched.

If your input model does not contain hydrogen atoms and you would like to use them in refinement, this can also be done via the ReadySet dialog, or by clicking the box labeled "Automatically add hydrogens to model" in the main window.

Note: ReadySet look in the PDB's Chemical Components Database (distributed with PHENIX) for the three-letter code of any unknown ligands it finds, and use the database information to generate restraints. For non-standard ligands, users are advised to run eLBOW with a SMILES string or equivalent source of information. Generating robust geometry restraints from a PDB file alone is problematic, because the coordinates are not a reliable guide to molecular topology, chirality, etc.

Neutron crystallography

phenix.refine can perform refinement using either X-ray data, neutron diffraction data, or both. You can change the type of data contained in a reflections file by right-clicking on the "Data type" field in the file list.

At present, phenix.refine always requires that you define an X-ray dataset; this constraint will be removed in future versions. If you are performing refinement against neutron data alone, you must treat the reflections as if they were X-ray data. All of the data options remain the same, and will be extracted automatically. You must change the scattering table to "neutron" in the "Refinement settings" tabe. If you are doing joint X-ray/neutron refinement, you should leave the scattering table set to the default, and specify one reflections file as neutron data. Parameters for neutron data are not chosen automatically; once you have specified a neutron reflections file, click the "Neutron data. . ." button to select the column labels for the data and test set.

If you need to add hydrogens and deuteriums to your model before running phenix.refine, you can either run ReadySet prior to refinement, or change the options for automatic hydrogen addition. If your model already includes these atoms, you may wish to turn automatic hydrogen addition off to ensure that existing atom types are left in place.

Atom selection

The atom-selection syntax used by PHENIX is described in a separate page. In the GUI, any valid atom selection can be visualized if you have a suitable graphics card and have already loaded a PDB file with valid symmetry information. The graphics window can be opened by clicking the "View/pick" button next to any atom selection field. Depending on the size of your structure, it may take several seconds for PHENIX to determine the atomic connectivity. The current selection, if any, will be highlighted:

The Select atoms button opens a window that allows you to type in selections and immediately visualize the results, including the number of atoms selected. On mice with at least two buttons, clicking on an atom with the right button will open a menu for selecting residue ranges or chains. However, we recommend that you learn the selection syntax, as it is much more flexible than mouse controls. Selections made in the graphics window will be sent automatically to the appropriate control.

Once this window is open, it does not need to be closed; clicking a different "View/pick" button will transfer control of the display. (On Linux, you will first need to open the Actions menu and click Restore parent window to transfer mouse and keyboard controls back to the configuration dialog.)

There is also a unified selection interface for

- Secondary structure annotations (helices, beta sheets, etc.)
- NCS groups
- TLS groups
- Refinement strategy options (occupancies, rigid bodies, etc.)
- Custom geometry restraints (bonds, angles, dihedrals, etc.)

More information can be found here.

Identifying and editing TLS groups

PHENIX includes a tool (phenix.find_tls_groups) for fast identification of appropriate TLS groupings based on isotropic ADPs in a PDB file. This is integrated into the phenix.refine GUI, and can be accessed from the TLS group editor (the "TLS" button on the toolbar, also available on the "Refinement settings" configuration tab). The editor is essentially just an atom selection list control similar to others in PHENIX:

Clicking the "Find TLS" button will launch the identification program, which typically takes a few seconds to a few minutes to run depending on the size of the structure and the number of processors available. All CPU cores will be used by default. You may alternately specify TLS groups manually, or upload a file from the TLSMD server.

Running phenix.refine

Each time phenix.refine is run from the GUI, a new folder is created (in the current output directory) named Refine_X, where X is an integer corresponding to the job ID. This directory will contain all temporary and output files created during the course of the run. Existing folders will not be overwritten. All output files will be displayed in the GUI in the results tab.

Once you are done configuring refinement, switch to the Run tab in the main window and start the process by clicking the "Run" button. The log output will appear at the bottom of the new tab:

Statistics will be continually updated; any that appear to be outliers (e.g. excessively high R-factors) will be highlighted in red. If an error occurs during refinement, a message box will pop up and the process will be halted.

After refinement

Once the refinement is complete, additional tabs will appear. The first of these is a summary page displaying the final statistics, a list of output files, buttons for opening the refined structure in Coot and PyMOL, and links to assorted graphs. Note that the output files may include one ending in "_data.mtz"; if you did not specify R-free flags as part of the input, the new file will contain the automatically generated flags, and should be re-used in future runs.

Additional tabs contain the full validation summary, which is essentially identical to the standalone validation program.

Frequently Asked Questions

Questions not specific to the graphical interface are answered in the phenix.refine FAQ.

I have a parameter file that I want to use; how do I load this in the GUI? You have two options: add it to the list of input files in the "Input data" tab, or select "Load parameter file" from the File menu. If you do the latter, the parameters will be immediately incorporated into the working parameters and the GUI will be redrawn to reflect any changes. The disadvantage of this method is that there is no easy way to revert to the previous settings. Adding the file to the input list will postpone parsing until the program is run, so you will not be able to edit the parameters graphically, but you may remove the file at any time.

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Acknowledgements

phenix.refine was primarily written by Pavel Afonine and Ralf Grosse-Kunstleve, with additional contributions from Peter Zwart, Jeff Headd, Nat Echols, Nigel Moriarty, and Tom Terwilliger. The electron-density isosurface and anisotropic ellipsoid code in the CCTBX (used in the internal graphics viewer) was contributed by Luc Bourhis.

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Structure refinement in PHENIX

refinement.html

phenix.refine - general purpose crystallographic structure refinement program

Contents

- Useful options Changing the number of refinement cycles and minimizer iterations Parallelizing for multi-core systems Creating R-free flags (if not present in the input reflection files) Specify the name for output files Choosing data arrays from input file Reflection output Setting the resolution range for the refinement Bulk solvent correction and anisotropic scaling Default refinement with user specified X-ray target function Modifying the initial model before refinement starts Refinement using FFT or direct structure factor calculation algorithm Ignoring test (free) flags in refinement Using phenix.refine to calculate structure factors Scattering factors Suppressing the output of certain files Random seed Electron density maps Refining with anomalous data (or what phenix.refine does with Fobs+ and Fobs-). Rejecting reflections by sigma Developer's tools Automatic linking CIF modifications and links Definition of custom bonds and angles Making sure atom charge is used in refinement
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- Creating R-free flags (if not present in the input reflection files)
- Specify the name for output files
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- Modifying the initial model before refinement starts
- Refinement using FFT or direct structure factor calculation algorithm
- Ignoring test (free) flags in refinement
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- Scattering factors
- Suppressing the output of certain files
- Random seed
- Electron density maps
- Refining with anomalous data (or what phenix.refine does with Fobs+ and Fobs-).
- Rejecting reflections by sigma
- Developer's tools
- Automatic linking
- CIF modifications and links
- Definition of custom bonds and angles
- Making sure atom charge is used in refinement
- What is a reasonable gap between R and Rfree?
- Interpreting the R values at the end of the phenix refinement log file
- What to do next after running refinement with PhenixRefine
- Depositing refined structure with PDB
- Feedback, more information

- List of all available keywords

Overview

This document describes phenix.refine with the emphasis on the command line use. Graphical User Interface (GUI) is available for phenix.refine (see below). The phenix.refine software is designed for use with crystallographic data such as X-ray diffraction or neutron data. For refinement of structures with cryo-EM data, normally you should use real space refinement with phenix.real_space_refine.

Graphical interface

A complete graphical interface for phenix.refine is available; it includes integration with several refinement-related utilities such as phenix.ready_set, phenix.simple_ncs_from_pdb, phenix.find_tls_groups and more. All of the program details described in this document should apply to the GUI as well.

Citation reference and comprehensive methodology details

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Available features at a glance

- Coordinate refinement:

Restrained or unrestrained individual, in real or reciprocal space Grouped (rigid body) LBFGS minimization, Cartesian or torsion Simulated Annealing Selective removing of stereochemistry restraints Adding custom (user-defined) restraints (bonds, angles, etc) Fixing (not refining) coordinates of any selected part of the structure NCS: global (Cartesian), local (torsion), constraints ("strict" NCS) Restraints specific to low-resolution refinement: secondary structure, reference model, Ramachandran plot restraints

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- Atomic Displacement Parameters (ADP) refinement:

Restrained individual isotropic, anisotropic, mixed Group isotropic (one isotropic B per selected model part). Modes to refine one or two group B per residue (for side and main chains) TLS (Translation-Libration-Screw-rotation model) Comprehensive mode: combined TLS + individual or group ADP

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- Group isotropic (one isotropic B per selected model part). Modes to refine one or two group B per residue (for side and main chains)
- TLS (Translation-Libration-Screw-rotation model)
- Comprehensive mode: combined TLS + individual or group ADP
- Occupancy refinement: individual, group, constrained for alternative conformations
- Anomalous f' and f'' refinement
- Bulk solvent correction (flat model using a mask) and anisotropic scaling
- Refinement target functions: least-squares (ls), maximum-likelihood (ml), phased maximum-likelihood (mlhl)
- FFT and direct summation
- Maps: any pFobs-qFmodel (unweighted: p=q=1 or sigmaA-weighted), anomalous difference. Can be output as Fourier map coefficients (MTZ file) or actual map in CCP4 or X-plor format
- Simple structure factor and map calculation
- Combined automatic ordered solvent building, update and refinement
- Complete model and data statistics: ready to PDB deposition REMARK 3 records
- Detection and using NCS in refinement (NCS restraints or constraints)
- Refinement using X-ray, neutron or both data simultaneously (joint X-ray and neutron refinement)
- Complex refinement strategies in one run
- Refinement at subatomic resolution (approx. < 1.0 A) with IAS model
- Refinement using twinned data
- Local and global real-space refinement of coordinates including rotamer fitting
- Refinement using electron diffraction data

Current limitations

- TLS and individual anisotropic ADP cannot be refined at once for the same group
- Certain refinement strategies are not available for joint X-ray/neutron refinement
- Atoms with anisotropic ADP in NCS groups
- No Simulated Annealing for selected fragments
- Two twin domains only

Remark on using amplitudes (Fobs) vs intensities (Iobs)

Although phenix.refine will read both data types, intensities or amplitudes, internally it uses amplitudes in nearly all calculations. Intensities are converted into amplitudes using French&Wilson method.

phenix.refine organization

A phenix.refine refinement run always consists of three steps:

Reading and processing inputs (model in PDB format, reflections in most known formats, parameters and cif files with stereochemistry definitions for non-standard ligands) Main refinement loop (macro-cycle; see Refinement flowchart below) iteratively performs requested refinement protocols (bulk solvent and scaling, refinement of coordinates and B-factors, solvent update, etc...) Output of PDB file with refined model and ready to PDB deposition REMARK 3 records, MTZ file with Fourier map coefficients and other arrays, refinement log file containing comprehensive summary of refinement steps, model, data and model to data fit statistics.

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Multiple refinement strategies can be combined and applied to any selected part of a model as illustrated below:

Running phenix.refine

phenix.refine can be run from the command line:

```
% phenix.refine &lt;pdb-file(s)&gt; &lt;reflection-file(s)&gt; &lt;monomer-library-file(s)&gt;
 &lt;parameter-keyword(s)&gt; &lt;parameter-file(s)&gt;
```

or from PHENIX GUI.

When you do this a number of things happen:

- The program automatically generates a ".eff" file which contains all of the parameters for the job (for example if you provided lysozyme.pdb the file lysozyme_refine_001.eff will be generated). This is the set of input parameters for this run.
- The program automatically interprets the reflection file(s). If there is an unambiguous choice of data arrays these will be used for the refinement. If there is a choice, you're given a message telling you how

to select the arrays. Several reflection files can be provided, for example: one containing Fobs and another one with R-free flags.

- Once the data arrays are chosen, the program writes all of the data it will be using in the refinement to a new MTZ file, for example, lysozyme_refine_data.mtz. This makes it very easy to keep track of what you actually used in the refinement (instead of having the arrays spread across multiple files).

- At the end of refinement the program generates:

PDB file with the refined model called for example lysozyme_refine_001.pdb. This file contains REMARK records summarizing details about refinement run, as well as model, data and model-to-data fit statistics. Also, it contains REMARK 3 records that can be used for PDB deposition. MTZ file (e.g. lysozyme_refine_001.mtz) that contains: Copy if input data (for example, Iobs, R-free flags) Data actually used in refinement and calculation of reported statistics. These can be different from input data for a number of reasons: a) conversion Iobs to Fobs (if Iobs were input), b) truncation by resolution and/or sigma (if requested by the user), c) automated rejection of reflection-outliers. Total model structure factors (Fmodel). Fmodel includes all scales and bulk-solvent contribution and is defined as Fmodel = k_total * (Fcalc_atoms + k_mask * Fmask) Fourier map coefficients that can be used for example in Coot or XtalView to visualize the maps. They correspond to 2mFobs-DFmodel map calculated using original set of Fobs, 2mFobs-DFmodel "filled" map, where missing Fobs are substituted with some expected values, residual mFobs-DFmodel map, and anomalous difference map (if input data was anomalous: contained Fobs(+) and Fobs(-) separately). log file that contains detailed information about refinement run. geo file is a foot print of all geometry restraints used in refinement, such as bonds, angles, planarity, chirality, dihedral, non-bonded. For each type of restraints current model and ideal (library) values are listed. This file allows to pinpoint all the restraints that an atom in question is involved. Optionally, actual maps can be output in CCP4 binary format or X-plor plain text format. def file: a new defaults file to run the next cycle of refinement, e.g. lysozyme_refine_002.def. This means you can run the next cycle of refinement by typing: % phenix.refine lysozyme_refine_002.def

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```
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```

To get information about command line options type:

```
% phenix.refine --help
```

To have the program generate the default input parameters without running the refinement job (e.g. if you want to modify the parameters prior to running the job):

```
% phenix.refine --dry_run &lt;pdb-file&gt; &lt;reflection-file(s)&gt;
```

If you know the parameter that you want to change you can override it from the command line:

```
% phenix.refine data.hkl model.pdb xray_data.low_resolution=8.0 \
 simulated_annealing.start_temperature=5000
```

Note that you don't have to specify the full parameter name. What you specify on the command line is matched against all known parameters names and the best substring match is used if it is unique. Otherwise a list of possible options is provided.

To rerun a job that was previously run:

```
% phenix.refine --overwrite lysozyme_refine_001.def
```

or:

```
% phenix.refine lysozyme_refine_001.def overwrite=true
```

The order of commands is not important.

The --overwrite (or overwrite=True) option allows the program to overwrite existing files. By default the program will not overwrite existing files - just in case this would remove the results of a refinement job that took a long time to finish.

To see all default parameters:

```
% phenix.refine --show-defaults
```

To see difference between default settings and a parameter file:

```
% phenix.refine --diff-params lysozyme_refine_001.def
```

Giving parameters on the command line or in files

In phenix.refine parameters to control refinement can be given by the user on the command line:

```
% phenix.refine data.hkl model.pdb simulated_annealing=true
```

However, sometimes the number of parameters is large enough to make it difficult to type them all, for example:

```
% phenix.refine data.hkl model.pdb refine.adp.tls="chain A" \
 refine.adp.tls="chain B" main.number_of_macro_cycles=4 \
 xray_data.high_resolution=2.5 wxc_scale=3 wxu_scale=5 \
 output.prefix=my_best_model strategy=tls+individual_sites+individual_adp \
 simulated_annealing.start_temperature=5000
```

The same result can be achieved by using:

```
% phenix.refine data.hkl model.pdb custom_par_1.params
```

where the custom_par_1.params file contains the following lines:

```
refinement.refine.strategy=tls+individual_sites+individual_adp
refinement.refine.adp.tls="chain A"
refinement.refine.adp.tls="chain B"
refinement.main.number_of_macro_cycles=4
refinement.target_weights.wxc_scale=3
refinement.target_weights.wxu_scale=5
refinement.output.prefix=my_best_model
refinement.simulated_annealing.start_temperature=5000
data_manager.fmodel.xray_data.high_resolution=2.5
```

which can also be formatted by grouping the parameters under the relevant scopes (custom_par_2.params):

```
refinement.main {
 number_of_macro_cycles=4
}
data_manager.fmodel.xray_data.high_resolution=2.5
refinement.refine {
 strategy = *individual_sites \
 rigid_body \
 *individual_adp \
 group_adp \
 *tls \
 occupancies \
 group_anomalous \
 none
 adp {
 tls = "chain A"
 tls = "chain B"
 }
}
refinement.target_weights {
 wxc_scale=3
 wxu_scale=5
}
refinement.output.prefix=my_best_model
refinement.simulated_annealing.start_temperature=5000
```

and the refinement run will be:

```
% phenix.refine data.hkl model.pdb custom_par_2.params
```

The easiest way to create a file like the custom_par_2.params file is to generate a template file containing all parameters by using the command phenix.refine --show-defaults and then keep the parameters that you want to use and remove the rest.

Comments in parameter files

Use # for comments:

```
% phenix.refine data.hkl model.pdb comments_in_params_file.params
```

where comments_in_params_file.params file contains the lines:

```
refinement {
 refine {
 #strategy = individual_sites rigid_body individual_adp group_adp tls \
 # occupancies group_anomalous *none
 }
 #main {
 # number_of_macro_cycles = 1
 #}
}
refinement.target_weights.wxc_scale = 1.5
#data_manager.fmodel.xray_data.low_resolution=5.0
```

In this example the only parameter that is used to overwrite the defaults is target_weights.wxc_scale and the rest is commented out.

Refinement scenarios

The refinement of atomic parameters is controlled by the strategy keyword. Those include:

```
- individual_sites (refinement of individual atomic coordinates)
- individual_sites_real_space (same as above performed real space)
- individual_adp (refinement of individual atomic B-factors)
- group_adp (group B-factors refinement)
- group_anomalous (refinement of f' and f" values)
- tls (TLS refinement = refinement of ADP through TLS parameters)
- rigid_body (rigid body refinement)
- occupancies (occupancy refinement: individual, group, group constrained)
- none (bulk solvent and anisotropic scaling only)
```

Below are examples to illustrate the use of the strategy keyword as well as a few others.

Refinement with all default parameters

```
% phenix.refine data.hkl model.pdb
```

This will perform coordinate refinement, restrained ADP refinement and occupancy refinement (if applicable, see Occupancy refinement section for details). Three macrocycles will be executed, each consisting of bulk solvent correction, anisotropic scaling of the data, coordinate refinement (25 iterations of the LBFGS minimizer) and ADP refinement (25 iterations of the LBFGS minimizer).

Refinement of coordinates

Coordinates can be refined using:

individual coordinate refinement using gradient-driven (LBFGS) minimization; individual coordinate refinement in real-space using a combination of gradient-driven (LBFGS) minimization and local torsion-angle grid searches; individual coordinate refinement using simulated annealing (SA refinement); grouped coordinate refinement (rigid body refinement); group-constrained refinement in torsion angle space using SA (often called torsion angle simulated annealing refinement).

- individual coordinate refinement using gradient-driven (LBFGS) minimization;
- individual coordinate refinement in real-space using a combination of gradient-driven (LBFGS) minimization and local torsion-angle grid searches;
- individual coordinate refinement using simulated annealing (SA refinement);
- grouped coordinate refinement (rigid body refinement);
- group-constrained refinement in torsion angle space using SA (often called torsion angle simulated annealing refinement).

The default restrained refinement includes a standard set of stereo-chemical restraints (covalent bonds, angles, dihedrals, planarities, chiralities, non-bonded). The NCS restrains can be added as well. Completely unrestrained refinement is possible.

The total refinement target is defined as:

```
Etotal = wxc_scale * wxc * Exray + wc * Egeom
```

where: Exray is crystallographic refinement target (least-squares, maximum-likelihood, or any other), Egeom is the sum of restraints (including NCS if requested), wc is 1.0 by default and used to turn the restraints off, wxc ~ ratio of gradient norms for geometry and X-ray targets as defined in (Adams et al, 1997, PNAS, Vol. 94, p. 5018), wxc_scale is an empirical scale that is typically between 0 and 10. Egeom can optionally include reference model, secondary structure and Ramachandran plot restraints.

By default coordinates of all atoms are refined. It is possible to refine coordinates of only selected atoms.

Using strategy=rigid_body or strategy=individual_sites will ask phenix.refine to refine only coordinates while other parameters (ADP, occupancies) will be fixed (not refined).

phenix.refine will stop if an atom on special position is included into group of atoms that are subject to rigid body refinement. The solution is to make a new rigid body group selection containing no atoms on special positions.

- Rigid body refinement phenix.refine implementation of rigid body refinement is highly and efficient (large convergence radius, no need to cut high-resolution data). We call this MZ protocol (multiple zones). The essence of MZ protocol is that the refinement starts with a few reflections selected in the lowest resolution zone and proceeds with gradually adding higher resolution reflections. Also, it almost constantly updates the mask and bulk solvent model parameters and this is crucial since the bulk solvent affects the low resolution reflections - exactly those the most important for success of rigid body refinement. The default set of the rigid body parameters is good for most of the cases and is normally not supposed to be changed. Rigid body refinement does not use any restraints. This means that rigid-body groups may bump into each other (overlap) or if covalently bonded atoms belong to different rigid groups then the bond linking these atoms may (and likely will) break as result of rigid body refinement. One rigid body group per chain (default behavior): % phenix.refine data.hkl model.pdb strategy=rigid_body Multiple groups (requires a basic knowledge of the Phenix atom selection syntax): % phenix.refine data.hkl model.pdb strategy=rigid_body \ sites.rigid_body="chain A" sites.rigid_body="chain B" This will refine the chain A and chain B as two rigid bodies. The rest of the model will be kept fixed. If there are many rigid groups to define, typing them in the command line may be a tedious exercise. In this case a better alternative is to create a parameter file rigid_body_selections containing the following lines: refinement.refine.sites { rigid_body = chain A rigid_body = chain B } The command line will then be: % phenix.refine data.hkl model.pdb strategy=rigid_body \ rigid_body_selections.params Files like this can be created, for example, by copy-and-paste from the complete list of parameters (phenix.refine --show-defaults). To switch from MZ protocol to traditional way of doing rigid body refinement (not recommended!): % phenix.refine data.hkl model.pdb strategy=rigid_body \ rigid_body.number_of_zones=1 rigid_body.high_resolution=4.0 For such refinement to be useful one needs to cut the high-resolution data off at some arbitrary point around 3-5 A (depending on model size and data quality). By default rigid body refinement is run only the first macro-cycle. To switch from running rigid body refinement only once at the first macro-cycle to running it every macro-cycle: % phenix.refine data.hkl model.pdb strategy=rigid_body \ rigid_body.mode=every_macro_cycle To change the default number of lowest resolution reflections used to determine the first resolution zone to do rigid body refinement in it (for MZ protocol only): % phenix.refine data.hkl model.pdb strategy=rigid_body \ rigid_body.min_number_of_reflections=250 Decreasing this number may increase the convergence radius of rigid body refinement but small numbers may lead to refinement instability and nonsensical model shifts. To change the number of zones for MZ protocol: % phenix.refine data.hkl model.pdb strategy=rigid_body \ rigid_body.number_of_zones=7 Increasing this number may increase the convergence radius of rigid body refinement at the cost of much longer run time. Rigid body refinement can be combined with individual refinement of coordinates: % phenix.refine data.hkl model.pdb strategy=rigid_body+individual_sites this will perform 3 macro-cycles of individual coordinate refinement and the rigid body refinement will be performed only once at the first macro-cycle. More powerful combination for coordinates refinement is: % phenix.refine data.hkl model.pdb strategy=rigid_body+individual_sites \ simulated_annealing=true this will do the same refinement as above combined with the Simulated Annealing refinement at the second macro-cycle.

Rigid body refinement

phenix.refine implementation of rigid body refinement is highly and efficient (large convergence radius, no need to cut high-resolution data). We call this MZ protocol (multiple zones). The essence of MZ protocol is that the refinement starts with a few reflections selected in the lowest resolution zone and proceeds with gradually adding higher resolution reflections. Also, it almost constantly updates the mask and bulk solvent model parameters and this is crucial since the bulk solvent affects the low resolution reflections - exactly those the most important for success of rigid body refinement. The default set of the rigid body parameters is good for most of the cases and is normally not supposed to be changed.

Rigid body refinement does not use any restraints. This means that rigid-body groups may bump into each other (overlap) or if covalently bonded atoms belong to different rigid groups then the bond linking these atoms may (and likely will) break as result of rigid body refinement.

- One rigid body group per chain (default behavior): % phenix.refine data.hkl model.pdb strategy=rigid_body

One rigid body group per chain (default behavior):

```
% phenix.refine data.hkl model.pdb strategy=rigid_body
```

- Multiple groups (requires a basic knowledge of the Phenix atom selection syntax): % phenix.refine data.hkl model.pdb strategy=rigid_body \ sites.rigid_body="chain A" sites.rigid_body="chain B" This will refine the chain A and chain B as two rigid bodies. The rest of the model will be kept fixed.

Multiple groups (requires a basic knowledge of the Phenix atom selection syntax):

```
% phenix.refine data.hkl model.pdb strategy=rigid_body \
 sites.rigid_body="chain A" sites.rigid_body="chain B"
```

This will refine the chain A and chain B as two rigid bodies. The rest of the model will be kept fixed.

- If there are many rigid groups to define, typing them in the command line may be a tedious exercise. In this case a better alternative is to create a parameter file rigid_body_selections containing the following lines: refinement.refine.sites { rigid_body = chain A rigid_body = chain B } The command line will then be: % phenix.refine data.hkl model.pdb strategy=rigid_body \ rigid_body_selections.params Files like this can be created, for example, by copy-and-paste from the complete list of parameters (phenix.refine --show-defaults).

If there are many rigid groups to define, typing them in the command line may be a tedious exercise. In this case a better alternative is to create a parameter file rigid_body_selections containing the following lines:

```
refinement.refine.sites {
 rigid_body = chain A
 rigid_body = chain B
}
```

The command line will then be:

```
% phenix.refine data.hkl model.pdb strategy=rigid_body \
 rigid_body_selections.params
```

Files like this can be created, for example, by copy-and-paste from the complete list of parameters (phenix.refine --show-defaults).

- To switch from MZ protocol to traditional way of doing rigid body refinement (not recommended!): % phenix.refine data.hkl model.pdb strategy=rigid_body \ rigid_body.number_of_zones=1 rigid_body.high_resolution=4.0 For such refinement to be useful one needs to cut the high-resolution data off at some arbitrary point around 3-5 A (depending on model size and data quality).

To switch from MZ protocol to traditional way of doing rigid body refinement (not recommended!):

```
% phenix.refine data.hkl model.pdb strategy=rigid_body \
 rigid_body.number_of_zones=1 rigid_body.high_resolution=4.0
```

For such refinement to be useful one needs to cut the high-resolution data off at some arbitrary point around 3-5 A (depending on model size and data quality).

- By default rigid body refinement is run only the first macro-cycle. To switch from running rigid body refinement only once at the first macro-cycle to running it every macro-cycle: % phenix.refine data.hkl model.pdb strategy=rigid_body \ rigid_body.mode=every_macro_cycle

By default rigid body refinement is run only the first macro-cycle. To switch from running rigid body refinement only once at the first macro-cycle to running it every macro-cycle:

```
% phenix.refine data.hkl model.pdb strategy=rigid_body \
 rigid_body.mode=every_macro_cycle
```

- To change the default number of lowest resolution reflections used to determine the first resolution zone to do rigid body refinement in it (for MZ protocol only): % phenix.refine data.hkl model.pdb strategy=rigid_body \ rigid_body.min_number_of_reflections=250 Decreasing this number may increase the convergence radius of rigid body refinement but small numbers may lead to refinement instability and nonsensical model shifts.

To change the default number of lowest resolution reflections used to determine the first resolution zone to do rigid body refinement in it (for MZ protocol only):

```
% phenix.refine data.hkl model.pdb strategy=rigid_body \
 rigid_body.min_number_of_reflections=250
```

Decreasing this number may increase the convergence radius of rigid body refinement but small numbers may lead to refinement instability and nonsensical model shifts.

• To change the number of zones for MZ protocol: % phenix.refine data.hkl model.pdb strategy=rigid_body \ rigid_body.number_of_zones=7 Increasing this number may increase the convergence radius of rigid body refinement at the cost of much longer run time.

To change the number of zones for MZ protocol:

```
% phenix.refine data.hkl model.pdb strategy=rigid_body \
 rigid_body.number_of_zones=7
```

Increasing this number may increase the convergence radius of rigid body refinement at the cost of much longer run time.

• Rigid body refinement can be combined with individual refinement of coordinates: % phenix.refine data.hkl model.pdb strategy=rigid_body+individual_sites this will perform 3 macro-cycles of individual coordinate refinement and the rigid body refinement will be performed only once at the first macro-cycle. More powerful combination for coordinates refinement is: % phenix.refine data.hkl model.pdb strategy=rigid_body+individual_sites \ simulated_annealing=true this will do the same refinement as above combined with the Simulated Annealing refinement at the second macro-cycle.

Rigid body refinement can be combined with individual refinement of coordinates:

```
% phenix.refine data.hkl model.pdb strategy=rigid_body+individual_sites
```

this will perform 3 macro-cycles of individual coordinate refinement and the rigid body refinement will be performed only once at the first macro-cycle. More powerful combination for coordinates refinement is:

```
% phenix.refine data.hkl model.pdb strategy=rigid_body+individual_sites \
 simulated_annealing=true
```

this will do the same refinement as above combined with the Simulated Annealing refinement at the second macro-cycle.

• Refinement of individual coordinates Refinement with Simulated Annealing: % phenix.refine data.hkl model.pdb simulated_annealing=true \ strategy=individual_sites This will perform Simulated Annealing refinement and LBFGS minimization for the whole model. SA will run on select macro-cycle(s) only (not every macro-cycle). To change starting SA temperature: % phenix.refine data.hkl model.pdb simulated_annealing=true \ strategy=individual_sites simulated_annealing.start_temperature=10000 There are several options defining of how many times the SA will be performed per refinement run. Run it only the first macro_cycle: % phenix.refine data.hkl model.pdb simulated_annealing=true \ strategy=individual_sites simulated_annealing.mode=first or every macro-cycle: % phenix.refine data.hkl model.pdb simulated_annealing=true \ strategy=individual_sites simulated_annealing.mode=every_macro_cycle or second and before the last macro-cycle: % phenix.refine data.hkl model.pdb simulated_annealing=true \ strategy=individual_sites simulated_annealing.mode=second_and_before_last Other options may exist: check all parameters for details. Refinement with minimization (whole model): % phenix.refine data.hkl model.pdb strategy=individual_sites Refinement with minimization (selected part of model): % phenix.refine data.hkl model.pdb strategy=individual_sites \ sites.individual="chain A" This will refine the coordinates of atoms in chain A only. To perform unrestrained refinement of coordinates (usually at ultra-high resolution): % phenix.refine data.hkl model.pdb strategy=individual_sites wc=0 This assigns the contribution of the geometry restraints target to zero. However, it is still calculated for statistics output. Removing selected geometry restraints In the example below: % phenix.refine data.hkl model.pdb remove_restraints_selections.params where remove_restraints_selections.params contains: refinement { geometry_restraints.remove { angles = chain B dihedrals = name CA chiralities = all planarities = None } } the following restraints will be removed: angle for all atoms in chain B, dihedral for all involving CA atoms, all chirality. All planarity restraints will be preserved.

Refinement of individual coordinates

- Refinement with Simulated Annealing: % phenix.refine data.hkl model.pdb simulated_annealing=true \ strategy=individual_sites This will perform Simulated Annealing refinement and LBFGS minimization for the whole model. SA will run on select macro-cycle(s) only (not every macro-cycle). To change starting SA temperature: % phenix.refine data.hkl model.pdb simulated_annealing=true \ strategy=individual_sites simulated_annealing.start_temperature=10000 There are several options defining of how many times the SA will be performed per refinement run. Run it only the first macro_cycle: % phenix.refine data.hkl model.pdb simulated_annealing=true \ strategy=individual_sites simulated_annealing.mode=first or every macro-cycle: % phenix.refine data.hkl model.pdb simulated_annealing=true \ strategy=individual_sites simulated_annealing.mode=every_macro_cycle or second and before the last macro-cycle: % phenix.refine data.hkl model.pdb simulated_annealing=true \ strategy=individual_sites simulated_annealing.mode=second_and_before_last Other options may exist: check all parameters for details.

Refinement with Simulated Annealing:

```
% phenix.refine data.hkl model.pdb simulated_annealing=true \
 strategy=individual_sites
```

This will perform Simulated Annealing refinement and LBFGS minimization for the whole model. SA will run on select macro-cycle(s) only (not every macro-cycle).

To change starting SA temperature:

```
% phenix.refine data.hkl model.pdb simulated_annealing=true \
 strategy=individual_sites simulated_annealing.start_temperature=10000
```

There are several options defining of how many times the SA will be performed per refinement run. Run it only the first macro_cycle:

```
% phenix.refine data.hkl model.pdb simulated_annealing=true \
 strategy=individual_sites simulated_annealing.mode=first
```

or every macro-cycle:

```
% phenix.refine data.hkl model.pdb simulated_annealing=true \
 strategy=individual_sites simulated_annealing.mode=every_macro_cycle
```

or second and before the last macro-cycle:

```
% phenix.refine data.hkl model.pdb simulated_annealing=true \
strategy=individual_sites simulated_annealing.mode=second_and_before_last
```

Other options may exist: check all parameters for details.

- Refinement with minimization (whole model): % phenix.refine data.hkl model.pdb strategy=individual_sites

Refinement with minimization (whole model):

```
% phenix.refine data.hkl model.pdb strategy=individual_sites
```

- Refinement with minimization (selected part of model): % phenix.refine data.hkl model.pdb strategy=individual_sites \ sites.individual="chain A" This will refine the coordinates of atoms in chain A only.

Refinement with minimization (selected part of model):

```
% phenix.refine data.hkl model.pdb strategy=individual_sites \
sites.individual="chain A"
```

This will refine the coordinates of atoms in chain A only.

- To perform unrestrained refinement of coordinates (usually at ultra-high resolution): % phenix.refine data.hkl model.pdb strategy=individual_sites wc=0 This assigns the contribution of the geometry restraints target to zero. However, it is still calculated for statistics output.

To perform unrestrained refinement of coordinates (usually at ultra-high resolution):

```
% phenix.refine data.hkl model.pdb strategy=individual_sites wc=0
```

This assigns the contribution of the geometry restraints target to zero. However, it is still calculated for statistics output.

- Removing selected geometry restraints In the example below: % phenix.refine data.hkl model.pdb remove_restraints_selections.params where remove_restraints_selections.params contains: refinement { geometry_restraints.remove { angles = chain B dihedrals = name CA chiralities = all planarities = None } } the following restraints will be removed: angle for all atoms in chain B, dihedral for all involving CA atoms, all chirality. All planarity restraints will be preserved.

Removing selected geometry restraints

In the example below:

```
% phenix.refine data.hkl model.pdb remove_restraints_selections.params
```

where remove_restraints_selections.params contains:

```
refinement {
 geometry_restraints.remove {
 angles = chain B
 dihedrals = name CA
 chiralities = all
 planarities = None
 }
}
```

the following restraints will be removed: angle for all atoms in chain B, dihedral for all involving CA atoms, all chirality. All planarity restraints will be preserved.

- Real-space refinement of coordinates Local refinement: amino-acid side-chain rotamer correction to eliminate outliers and find better fit to the map. Enabled by default, and controlled by strategy=individual_sites_real_space keyword. Overall (global) real-space refinement only happens at low resolution if the program decided to do so. Results of global real-space refinement may or may not be accepted by the program, which is based on comparison of R factors before and after refinement.

Real-space refinement of coordinates

Local refinement: amino-acid side-chain rotamer correction to eliminate outliers and find better fit to the map. Enabled by default, and controlled by strategy=individual_sites_real_space keyword. Overall (global) real-space refinement only happens at low resolution if the program decided to do so. Results of global real-space refinement may or may not be accepted by the program, which is based on comparison of R factors before and after refinement.

Refinement of Atomic Displacement Parameters (ADP or B-factors)

An ADP in phenix.refine is defined as a sum of three contributions:

```
Utotal = Ulocal + Utls + Ucryst
```

where Utotal is the total ADP, Ulocal reflects the local atomic vibration (also named as residual B) and Ucryst reflects global lattice vibrations. Ucryst is determined and refined as part of overall anisotropic scaling.

Options for ADP refinement:

individual isotropic, anisotropic or mixed ADP; grouped with one isotropic ADP per selected group; TLS;

- individual isotropic, anisotropic or mixed ADP;
- grouped with one isotropic ADP per selected group;
- TLS;

Any combination of the above options (exception: an atom participating in TLS group cannot be a subject of individual anisotropic ADP refinement) can be applied to any selected part of a model. For example, if a model contains six chains A, B, C, D, E and F than it would require only one single refinement run to perform refinement of:

individual isotropic ADP for atoms in chain A, individual anisotropic ADP for atoms in chain B, grouped B with one B per all atoms in chain C, TLS refinement for chain D, TLS and individual isotropic refinement for chain E, TLS and grouped B refinement for chain F.

- individual isotropic ADP for atoms in chain A,
- individual anisotropic ADP for atoms in chain B,
- grouped B with one B per all atoms in chain C,
- TLS refinement for chain D,
- TLS and individual isotropic refinement for chain E,
- TLS and grouped B refinement for chain F.

Restraints are used for default ADP refinement of isotropic and anisotropic atoms. Completely unrestrained refinement is possible.

The total refinement target is defined as:

```
Etotal = wxu_scale * wxu * Exray + wu * Eadp
```

where: Exray is crystallographic refinement target (least-squares, maximum-likelihood, ...), Eadp is the ADP restraints term, wu is 1.0 by default and used to turn the restraints off, wxu and wxu_scale are defined similarly to coordinates refinement (see Refinement of Coordinates paragraph).

Atoms that participate in TLS refinement have ANISOU records in output PDB file. The anisotropic B-factor in ANISOU records is the total B-factor (Utls + Ulocal). The isotropic equivalent B-factor in ATOM records is the mean of the trace of the ANISOU matrix divided by 10000 and multiplied by 8*pi^2. It represents the isotropic equivalent of the total B-factor. To obtain the individual B-factors (Ulocal), one needs to compute the TLS component (Utls) using the TLS records in PDB file header and then subtract it from the total B-factors. This can be done using phenix.tls tool.

When performing TLS refinement along with individual isotropic refinement the restraints are applied to Ulocal only and not to the total ADP.

Group isotropic ADP refinement uses harmonic restraints on group B values to make overall B factors of adjacent residues less dissimilar.

TLS refinement alone does not use any restraints.

When ADP refinement is run without using selections then ADP of all atoms are refined. If selections are used, only ADP of selected atoms are refined.

phenix.refine will stop if an atom on special position is included in TLS group. The solution is to make a new TLS group selection containing no atoms on special positions.

- Refining group isotropic B-factors One B-factor per residue: % phenix.refine data.hkl model.pdb strategy=group_adp Two B-factors per residue: % phenix.refine data.hkl model.pdb strategy=group_adp \ group_adp_refinement_mode=two_adp_groups_per_residue This only applies to amino acid residues. One isotropic B per selected group of atoms: % phenix.refine data.hkl model.pdb strategy=group_adp \ group_adp_refinement_mode=group_selection \ adp.group="chain A" adp.group="chain B" This will refine one isotropic B for chain A and one B for chain B. Refinement of group isotropic B-factors in phenix.refine does not change the original distribution of B-factors within the group: the differences between B-factors for atoms within the group remain constant. Example: if 10,15,25 are B-factors of atoms subject to group B-factor refinement, then after refinement they may have B-factors such as 5,10,20 or 13,18,28. Atoms with anisotropic ADP are allowed to be within the group; anisotropy of such atoms will not be changes during group B-factor refinement.

Refining group isotropic B-factors

- One B-factor per residue: % phenix.refine data.hkl model.pdb strategy=group_adp Two B-factors per residue: % phenix.refine data.hkl model.pdb strategy=group_adp \ group_adp_refinement_mode=two_adp_groups_per_residue This only applies to amino acid residues.

One B-factor per residue:

```
% phenix.refine data.hkl model.pdb strategy=group_adp
```

Two B-factors per residue:

```
% phenix.refine data.hkl model.pdb strategy=group_adp \
 group_adp_refinement_mode=two_adp_groups_per_residue
```

This only applies to amino acid residues.

- One isotropic B per selected group of atoms: % phenix.refine data.hkl model.pdb strategy=group_adp \ group_adp_refinement_mode=group_selection \ adp.group="chain A" adp.group="chain B" This will refine one isotropic B for chain A and one B for chain B.

One isotropic B per selected group of atoms:

```
% phenix.refine data.hkl model.pdb strategy=group_adp \
 group_adp_refinement_mode=group_selection \
 adp.group="chain A" adp.group="chain B"
```

This will refine one isotropic B for chain A and one B for chain B.

Refinement of group isotropic B-factors in phenix.refine does not change the original distribution of B-factors within the group: the differences between B-factors for atoms within the group remain constant. Example: if 10,15,25 are B-factors of atoms subject to group B-factor refinement, then after refinement they may have B-factors such as 5,10,20 or 13,18,28.

Atoms with anisotropic ADP are allowed to be within the group; anisotropy of such atoms will not be changes during group B-factor refinement.

- Refinement of individual ADP (isotropic, anisotropic) By default atoms in a PDB file with ANISOU records are refined as anisotropic and atoms without ANISOU records are refined as isotropic. This behavior can be changed with appropriate keywords, and also a subject to automatic adjustments. If atoms have ANISOU records and are not a part of TLS group, and also the data resolution is lower than switch_to_isotropic_high_res_limit parameter, then ADPs of such atoms are reset to isotropic and will be refined as such. Default refinement of individual ADP: % phenix.refine data.hkl model.pdb strategy=individual_adp Note, atoms in input PDB file with ANISOU records will be refined as anisotropic and those without ANISOU - as isotropic (subject to automatic adjustments - see above). Refinement of individual isotropic ADP for a model previously refined as anisotropic or TLS: % phenix.refine data.hkl model.pdb strategy=individual_adp \ adp.individual.isotropic=all or equivalently: % phenix.refine data.hkl model.pdb strategy=individual_adp \ convert_to_isotropic=true All anisotropic atoms in input PDB file will be converted to isotropic before the refinement starts. Refinement of individual anisotropic ADP for a model previously refined as isotropic: % phenix.refine data.hkl model.pdb strategy=individual_adp \ adp.individual.anisotropic="not element H" This will refine all atoms as anisotropic except hydrogens. Refinement of mixed model (some atoms are isotropic, some are anisotropic): % phenix.refine data.hkl model.pdb strategy=individual_adp \ adp.individual.anisotropic="chain A and not element H" \ adp.individual.isotropic="chain B or element H" In this example atoms (except hydrogens if any) in chain A will be refined as anisotropic and atoms in chain B (and hydrogens if any) will be refined as isotropic. Often, the ADP of water and hydrogens are desired to be refined as isotropic while the other atoms - as anisotropic: % phenix.refine data.hkl model.pdb strategy=individual_adp \ adp.individual.anisotropic="not water and not element H" \ adp.individual.isotropic="water or element H" Exactly the same command using slightly shorter selection syntax: % phenix.refine data.hkl model.pdb strategy=individual_adp \ adp.individual.anisotropic="not (water or element H)" \ adp.individual.isotropic="water or element H" Shortcuts can be used: adp.individual.aniso or adp.individual.iso. To perform unrestrained individual ADP refinement (usually at ultra-high resolutions): % phenix.refine data.hkl model.pdb strategy=individual_adp wu=0 This assigns the contribution of the ADP restraints target to zero. However, it is still calculated for statistics output. When selections adp.individual.aniso or adp.individual.iso are used, B-factors of selected atoms are refined, while B-factors of other atoms are not refined.

Refinement of individual ADP (isotropic, anisotropic)

By default atoms in a PDB file with ANISOU records are refined as anisotropic and atoms without ANISOU records are refined as isotropic. This behavior can be changed with appropriate keywords, and also a subject to automatic adjustments.

If atoms have ANISOU records and are not a part of TLS group, and also the data resolution is lower than switch_to_isotropic_high_res_limit parameter, then ADPs of such atoms are reset to isotropic and will be refined as such.

- Default refinement of individual ADP: % phenix.refine data.hkl model.pdb strategy=individual_adp Note, atoms in input PDB file with ANISOU records will be refined as anisotropic and those without ANISOU - as isotropic (subject to automatic adjustments - see above).

Default refinement of individual ADP:

```
% phenix.refine data.hkl model.pdb strategy=individual_adp
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Note, atoms in input PDB file with ANISOU records will be refined as anisotropic and those without ANISOU - as isotropic (subject to automatic adjustments - see above).

- Refinement of individual isotropic ADP for a model previously refined as anisotropic or TLS: % phenix.refine data.hkl model.pdb strategy=individual_adp \ adp.individual.isotropic=all or equivalently: % phenix.refine data.hkl model.pdb strategy=individual_adp \ convert_to_isotropic=true All anisotropic atoms in input PDB file will be converted to isotropic before the refinement starts.

Refinement of individual isotropic ADP for a model previously refined as anisotropic or TLS:

```
% phenix.refine data.hkl model.pdb strategy=individual_adp \
 adp.individual.isotropic=all
```

or equivalently:

```
% phenix.refine data.hkl model.pdb strategy=individual_adp \
 convert_to_isotropic=true
```

All anisotropic atoms in input PDB file will be converted to isotropic before the refinement starts.

- Refinement of individual anisotropic ADP for a model previously refined as isotropic: % phenix.refine data.hkl model.pdb strategy=individual_adp \ adp.individual.anisotropic="not element H" This will refine all atoms as anisotropic except hydrogens.

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This will refine all atoms as anisotropic except hydrogens.

- Refinement of mixed model (some atoms are isotropic, some are anisotropic): % phenix.refine data.hkl model.pdb strategy=individual_adp \ adp.individual.anisotropic="chain A and not element H" \ adp.individual.isotropic="chain B or element H" In this example atoms (except hydrogens if any) in chain A will be refined as anisotropic and atoms in chain B (and hydrogens if any) will be refined as isotropic. Often, the ADP of water and hydrogens are desired to be refined as isotropic while the other atoms - as anisotropic: % phenix.refine data.hkl model.pdb strategy=individual_adp \ adp.individual.anisotropic="not water and not element H" \ adp.individual.isotropic="water or element H" Exactly the same command using slightly shorter selection syntax: % phenix.refine data.hkl model.pdb strategy=individual_adp \ adp.individual.anisotropic="not (water or element H)" \ adp.individual.isotropic="water or element H" Shortcuts can be used: adp.individual.aniso or adp.individual.iso.

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 adp.individual.anisotropic="not water and not element H" \
 adp.individual.isotropic="water or element H"
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Exactly the same command using slightly shorter selection syntax:

```
% phenix.refine data.hkl model.pdb strategy=individual_adp \
 adp.individual.anisotropic="not (water or element H)" \
 adp.individual.isotropic="water or element H"
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This assigns the contribution of the ADP restraints target to zero. However, it is still calculated for statistics output.

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When selections adp.individual.aniso or adp.individual.iso are used, B-factors of selected atoms are refined, while B-factors of other atoms are not refined.

- TLS refinement Refinement of TLS parameters only: % phenix.refine data.hkl model.pdb strategy=tls Refinement of TLS parameters only using multiple TLS groups: % phenix.refine data.hkl model.pdb strategy=tls tls_group_selections.params where, similar to the rigid body or group B-factor refinement, the selection for TLS groups has been made in a user-created parameter file (tls_group_selections.params) as following: refinement.refine.adp { tls = chain A tls = chain B } Alternatively, the selection for the TLS groups can be made from the command line (see rigid body refinement for an example). Note: TLS parameters will be refined only for selected fragments. This, for example, will allow to not include the solvent molecules into the TLS groups. Most useful is to perform combined TLS and individual or group isotropic ADP refinement: % phenix.refine data.hkl model.pdb strategy=tls+individual_adp or: % phenix.refine data.hkl model.pdb strategy=tls+group_adp This will allow to model global (TLS) and local (individual) components of the total ADP and also compensate for the model parts where TLS parametrization doesn't suite well. Partitioning model into TLS groups can be done automatically as part of refinement: % phenix.refine data.hkl model.pdb strategy=tls tls.find_automatically=True or using a dedicated tool: % phenix.find_tls_groups model.pdb Multiple CUPs (if available) can be used to speed-up the task: % phenix.find_tls_groups model.pdb nproc=48 Very important: automatic TLS group identication strongly relies on correctly refined B-factors and relatively good model geometry. Poor B-factors or model geometry may result in nonsensical TLS groups.

TLS refinement

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- Refinement of TLS parameters only using multiple TLS groups: % phenix.refine data.hkl model.pdb strategy=tls tls_group_selections.params where, similar to the rigid body or group B-factor refinement, the selection for TLS groups has been made in a user-created parameter file (tls_group_selections.params) as following: refinement.refine.adp { tls = chain A tls = chain B } Alternatively, the selection for the TLS groups can be made from the command line (see rigid body refinement for an example). Note: TLS parameters will be refined only for selected fragments. This, for example, will allow to not include the solvent molecules into the TLS groups.

Refinement of TLS parameters only using multiple TLS groups:

```
% phenix.refine data.hkl model.pdb strategy=tls tls_group_selections.params
```

where, similar to the rigid body or group B-factor refinement, the selection for TLS groups has been made in a user-created parameter file (tls_group_selections.params) as following:

```
refinement.refine.adp {
 tls = chain A
 tls = chain B
}
```

Alternatively, the selection for the TLS groups can be made from the command line (see rigid body refinement for an example).

Note: TLS parameters will be refined only for selected fragments. This, for example, will allow to not include the solvent molecules into the TLS groups.

- Most useful is to perform combined TLS and individual or group isotropic ADP refinement: % phenix.refine data.hkl model.pdb strategy=tls+individual_adp or: % phenix.refine data.hkl model.pdb strategy=tls+group_adp This will allow to model global (TLS) and local (individual) components of the total ADP and also compensate for the model parts where TLS parametrization doesn't suite well.

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- Partitioning model into TLS groups can be done automatically as part of refinement: % phenix.refine data.hkl model.pdb strategy=tls tls.find_automatically=True or using a dedicated tool: % phenix.find_tls_groups model.pdb Multiple CUPs (if available) can be used to speed-up the task: % phenix.find_tls_groups model.pdb nproc=48 Very important: automatic TLS group identication strongly relies on correctly refined B-factors and relatively good model geometry. Poor B-factors or model geometry may result in nonsensical TLS groups.

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Very important: automatic TLS group identication strongly relies on correctly refined B-factors and relatively good model geometry. Poor B-factors or model geometry may result in nonsensical TLS groups.

Occupancy refinement

List of facts about occupancy refinement in phenix.refine:

- phenix.refine can perform the following types of occupancy refinement: individual occupancy refinement - refinement of one occupancy per atom. In this case the refined occupancy value will be constrained between main.occupancy_min and main.occupancy_max, which is 0 and 1 by default. group (constrained) occupancy refinement. There are two typical uses of this option: refine one occupancy per selected set of atoms, such as a partially occupied ligand or ion. The refined occupancy will be constrained between 0 and 1. refine occupancies of atoms in alternative conformations. For example, if a residue has two alternative conformations, A and B, then there will be one refinable occupancy parameter. All occupancies within the conformer A will be equal to each other, and the same will be for conformer B. The sum of occupancies of A and B will add up to one. In general, for a constrained group containing N coupled conformers (for example, N=2 for alternative conformations A and B) there will be (N-1) refinable occupancies. There is no limit on the number of constrained groups (N can be any).

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occupancy parameter. All occupancies within the conformer A will be equal to each other, and the same will be for conformer B. The sum of occupancies of A and B will add up to one. In general, for a constrained group containing N coupled conformers (for example, N=2 for alternative conformations A and B) there will be (N-1) refinable occupancies. There is no limit on the number of constrained groups (N can be any).

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• Occupancy refinement is always enabled by default. This does not mean that occupancies of all atoms will be refined. Based on input PDB file, phenix.refine automatically finds which occupancies it will be refining. If no user defined selections is provided, phenix.refine will refine: individual occupancies for all atoms that have partial occupancy values in input PDB file (not equal to 0 or 1), for example: ATOM 1001 AU AU 500 14.333 3.856 26.301 0.23 7.97 occupancies of atoms in alternative conformations. Atoms in alternative conformations will be automatically determined based on altLoc identifiers (a one-letter code in front of three-letter residue name in ATOM record) in input PDB file and the group constrained occupancy refinement for these atoms will be performed. For example: ATOM 5085 N AALA 270 19.772 -6.267 40.250 0.75 5.17 ATOM 5086 CA AALA 270 19.927 -5.299 41.342 0.75 5.15 ATOM 5087 CB AALA 270 20.132 -6.108 42.617 0.75 6.92 ATOM 5088 C AALA 270 21.058 -4.290 41.124 0.75 5.06 ATOM 5089 O AALA 270 20.831 -3.090 41.384 0.75 5.54 ATOM 5090 N BALA 270 19.733 -6.282 40.242 0.25 5.04 ATOM 5091 CA BALA 270 19.592 -5.512 41.492 0.25 5.03 ATOM 5092 CB BALA 270 19.702 -6.389 42.726 0.25 6.62 ATOM 5093 C BALA 270 20.673 -4.426 41.454 0.25 4.77 ATOM 5094 O BALA 270 20.381 -3.268 41.761 0.25 5.70 Starting occupancy values can be any. Setting them to some reasonable values, like 0.75 and 0.25 above, may help refinement to converge faster. Refined occupancy values will be looking like in the above example: identical for all atoms within each conformer (in this example: 0.75 for conformer A and 0.25 for conformer B), and the sum of unique occupancies across all conformers will be exactly 1 (=0.25+0.75). If there are two or more consecutive residues in the same chain that have alternative conformations, then their occupancies will be automatically grouped and refined. For example: ATOM 0 N AGLY A 1 2.650 4.221 1.463 0.70 18.35 ATOM 1 CA AGLY A 1 2.206 4.688 2.763 0.70 17.27 ATOM 2 C AGLY A 1 3.296 4.604 3.813 0.70 4.90 ATOM 3 O AGLY A 1 4.143 3.711 3.772 0.70 10.35 ATOM 4 N BGLY A 1 3.650 6.221 4.463 0.30 18.35 ATOM 5 CA BGLY A 1 3.206 6.688 5.763 0.30 17.27 ATOM 6 C BGLY A 1 4.296 6.604 6.813 0.30 4.90 ATOM 7 O BGLY A 1 5.143 5.711 6.772 0.30 10.35 ATOM 8 N AALA A 2 3.276 5.538 4.758 0.70 8.03 ATOM 9 CA AALA A 2 2.260 6.584 4.782 0.70 4.94 ATOM 10 C AALA A 2 2.886 7.964 4.606 0.70 8.07 ATOM 11 O AALA A 2 3.307 8.594 5.576 0.70 2.01 ATOM 12 CB AALA A 2 1.465 6.522 6.077 0.70 4.52 ATOM 13 N BALA A 2 4.276 7.538 7.758 0.30 8.03 ATOM 14 CA BALA A 2 3.260 8.584 7.782 0.30 4.94 ATOM 15 C BALA A 2 3.886 9.964 7.606 0.30 8.07 ATOM 16 O BALA A 2 4.307 10.594 8.576 0.30 2.01 ATOM 17 CB BALA A 2 2.465 8.522 9.077 0.30 4.52 where the occupancies of conformer A (in residue numbers 1 and 2) are all equal to each other (0.7), the occupancies of conformer B are all equal to each other as well (0.3), and their sum is 1 (0.7+0.3). It is possible to have multiple different (having different residue name) alternative conformers within the same residue number of the same chain: ATOM 0 N APRO A 22 4.915 12.683 -3.102 0.25 11.83 ATOM 1 CA APRO A 22 6.042 13.429 -2.601 0.25 11.82 ATOM 2 C APRO A 22 6.387 13.122 -1.160 0.25 11.66 ATOM 3 O APRO A 22 5.480 13.006 -0.345 0.25 12.09 ATOM 4 CB APRO A

22 5.655 14.896 -2.744 0.25 12.86 ATOM 5 CG APRO A 22 4.661 14.854 -4.058 0.25 12.66 ATOM 6 CD APRO A 22 3.957 13.505 -3.910 0.25 12.27 ATOM 7 CA BSER A 22 6.034 13.399 -2.687 0.30 11.55 ATOM 8 C BSER A 22 6.367 13.062 -1.223 0.30 12.92 ATOM 9 O BSER A 22 5.412 13.050 -0.345 0.30 11.87 ATOM 10 CB BSER A 22 5.409 14.835 -2.876 0.30 12.60 ATOM 11 OG BSER A 22 4.760 15.243 -1.635 0.30 12.11 ATOM 12 CA CSER A 22 6.112 13.653 -2.656 0.45 12.31 ATOM 13 C CSER A 22 6.354 13.275 -1.187 0.45 11.92 ATOM 14 O CSER A 22 5.636 12.705 -0.270 0.45 11.77 ATOM 15 CB CSER A 22 5.605 15.097 -2.687 0.45 13.56 ATOM 16 OG CSER A 22 6.750 15.771 -2.280 0.45 16.38 If a structur...

Occupancy refinement is always enabled by default. This does not mean that occupancies of all atoms will be refined. Based on input PDB file, phenix.refine automatically finds which occupancies it will be refining. If no user defined selections is provided, phenix.refine will refine:

- individual occupancies for all atoms that have partial occupancy values in input PDB file (not equal to 0 or 1), for example: ATOM 1001 AU AU 500 14.333 3.856 26.301 0.23 7.97

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ATOM 1001 AU AU 500 14.333 3.856 26.301 0.23 7.97
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- occupancies of atoms in alternative conformations. Atoms in alternative conformations will be automatically determined based on altLoc identifiers (a one-letter code in front of three-letter residue name in ATOM record) in input PDB file and the group constrained occupancy refinement for these atoms will be performed. For example: ATOM 5085 N AALA 270 19.772 -6.267 40.250 0.75 5.17 ATOM 5086 CA AALA 270 19.927 -5.299 41.342 0.75 5.15 ATOM 5087 CB AALA 270 20.132 -6.108 42.617 0.75 6.92 ATOM 5088 C AALA 270 21.058 -4.290 41.124 0.75 5.06 ATOM 5089 O AALA 270 20.831 -3.090 41.384 0.75 5.54 ATOM 5090 N BALA 270 19.733 -6.282 40.242 0.25 5.04 ATOM 5091 CA BALA 270 19.592 -5.512 41.492 0.25 5.03 ATOM 5092 CB BALA 270 19.702 -6.389 42.726 0.25 6.62 ATOM 5093 C BALA 270 20.673 -4.426 41.454 0.25 4.77 ATOM 5094 O BALA 270 20.381 -3.268 41.761 0.25 5.70 Starting occupancy values can be any. Setting them to some reasonable values, like 0.75 and 0.25 above, may help refinement to converge faster. Refined occupancy values will be looking like in the above example: identical for all atoms within each conformer (in this example: 0.75 for conformer A and 0.25 for conformer B), and the sum of unique occupancies across all conformers will be exactly 1 (=0.25+0.75). If there are two or more consecutive residues in the same chain that have alternative conformations, then their occupancies will be automatically grouped and refined. For example: ATOM 0 N AGLY A 1 2.650 4.221 1.463 0.70 18.35 ATOM 1 CA AGLY A 1 2.206 4.688 2.763 0.70 17.27 ATOM 2 C AGLY A 1 3.296 4.604 3.813 0.70 4.90 ATOM 3 O AGLY A 1 4.143 3.711 3.772 0.70 10.35 ATOM 4 N BGLY A 1 3.650 6.221 4.463 0.30 18.35 ATOM 5 CA BGLY A 1 3.206 6.688 5.763 0.30 17.27 ATOM 6 C BGLY A 1 4.296 6.604 6.813 0.30 4.90 ATOM 7 O BGLY A 1 5.143 5.711 6.772 0.30 10.35 ATOM 8 N AALA A 2 3.276 5.538 4.758 0.70 8.03 ATOM 9 CA AALA A 2 2.260 6.584 4.782 0.70 4.94 ATOM 10 C AALA A 2 2.886 7.964 4.606 0.70 8.07 ATOM 11 O AALA A 2 3.307 8.594 5.576 0.70 2.01 ATOM 12 CB AALA A 2 1.465 6.522 6.077 0.70 4.52 ATOM 13 N BALA A 2 4.276 7.538 7.758 0.30 8.03 ATOM 14 CA BALA A 2 3.260 8.584 7.782 0.30 4.94 ATOM 15 C BALA A 2 3.886 9.964 7.606 0.30 8.07 ATOM 16 O BALA A 2 4.307 10.594 8.576 0.30 2.01 ATOM 17 CB BALA A 2 2.465 8.522 9.077 0.30 4.52 where the occupancies of conformer A (in residue numbers 1 and 2) are all equal to each other (0.7), the occupancies of conformer B are all equal to each other as well (0.3), and their sum is 1 (0.7+0.3). It is possible to have multiple different (having different residue name) alternative conformers within the same residue number of the same chain: ATOM 0 N APRO A 22 4.915 12.683 -3.102 0.25 11.83 ATOM 1 CA APRO A 22 6.042 13.429 -2.601 0.25 11.82 ATOM 2 C APRO A 22 6.387 13.122 -1.160 0.25 11.66 ATOM 3 O APRO A 22 5.480 13.006 -0.345 0.25 12.09 ATOM 4 CB APRO A 22 5.655 14.896 -2.744

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occupancies of atoms in alternative conformations. Atoms in alternative conformations will be automatically determined based on altLoc identifiers (a one-letter code in front of three-letter residue name in ATOM record) in input PDB file and the group constrained occupancy refinement for these atoms will be performed. For example:

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Starting occupancy values can be any. Setting them to some reasonable values, like 0.75 and 0.25 above, may help refinement to converge faster. Refined occupancy values will be looking like in the above example: identical for all atoms within each conformer (in this example: 0.75 for conformer A and 0.25 for conformer B), and the sum of unique occupancies across all conformers will be exactly 1 (=0.25+0.75).

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ATOM 9 CA AALA A 2 2.260 6.584 4.782 0.70 4.94
ATOM 10 C AALA A 2 2.886 7.964 4.606 0.70 8.07
ATOM 11 O AALA A 2 3.307 8.594 5.576 0.70 2.01
ATOM 12 CB AALA A 2 1.465 6.522 6.077 0.70 4.52
ATOM 13 N BALA A 2 4.276 7.538 7.758 0.30 8.03
ATOM 14 CA BALA A 2 3.260 8.584 7.782 0.30 4.94
ATOM 15 C BALA A 2 3.886 9.964 7.606 0.30 8.07
ATOM 16 O BALA A 2 4.307 10.594 8.576 0.30 2.01
ATOM 17 CB BALA A 2 2.465 8.522 9.077 0.30 4.52
```

where the occupancies of conformer A (in residue numbers 1 and 2) are all equal to each other (0.7), the occupancies of conformer B are all equal to each other as well (0.3), and their sum is 1 (0.7+0.3).

It is possible to have multiple different (having different residue name) alternative conformers within the same residue number of the same chain:

```
ATOM 0 N APRO A 22 4.915 12.683 -3.102 0.25 11.83
ATOM 1 CA APRO A 22 6.042 13.429 -2.601 0.25 11.82
ATOM 2 C APRO A 22 6.387 13.122 -1.160 0.25 11.66
ATOM 3 O APRO A 22 5.480 13.006 -0.345 0.25 12.09
ATOM 4 CB APRO A 22 5.655 14.896 -2.744 0.25 12.86
ATOM 5 CG APRO A 22 4.661 14.854 -4.058 0.25 12.66
ATOM 6 CD APRO A 22 3.957 13.505 -3.910 0.25 12.27
ATOM 7 CA BSER A 22 6.034 13.399 -2.687 0.30 11.55
ATOM 8 C BSER A 22 6.367 13.062 -1.223 0.30 12.92
ATOM 9 O BSER A 22 5.412 13.050 -0.345 0.30 11.87
ATOM 10 CB BSER A 22 5.409 14.835 -2.876 0.30 12.60
ATOM 11 OG BSER A 22 4.760 15.243 -1.635 0.30 12.11
ATOM 12 CA CSER A 22 6.112 13.653 -2.656 0.45 12.31
ATOM 13 C CSER A 22 6.354 13.275 -1.187 0.45 11.92
ATOM 14 O CSER A 22 5.636 12.705 -0.270 0.45 11.77
ATOM 15 CB CSER A 22 5.605 15.097 -2.687 0.45 13.56
ATOM 16 OG CSER A 22 6.750 15.771 -2.280 0.45 16.38
```

If a structure contains a residue or ligand with all equal non-zero occupancies, for example:

```
ATOM 6 S SO4 1 1.302 1.419 1.560 0.70 13.00
ATOM 7 O1 SO4 1 1.497 1.295 0.118 0.70 11.00
ATOM 8 O2 SO4 1 1.098 0.095 2.140 0.70 10.00
ATOM 9 O3 SO4 1 2.481 2.037 2.159 0.70 14.00
ATOM 10 O4 SO4 1 0.131 2.251 1.823 0.70 12.00
```

one occupancy per whole ligand will be refined automatically and it will be constrained between 0 and 1. If at least one occupancy is different from the rest:

```
ATOM 6 S SO4 1 1.302 1.419 1.560 0.70 13.00
ATOM 7 O1 SO4 1 1.497 1.295 0.118 0.21 11.00
ATOM 8 O2 SO4 1 1.098 0.095 2.140 0.70 10.00
ATOM 9 O3 SO4 1 2.481 2.037 2.159 0.70 14.00
ATOM 10 O4 SO4 1 0.131 2.251 1.823 0.70 12.00
```

then the occupancies of all atoms will refined individually. In this example:

```
ATOM 6 S SO4 1 1.302 1.419 1.560 0.70 13.00
ATOM 7 O1 SO4 1 1.497 1.295 0.118 0.70 11.00
ATOM 8 O2 SO4 1 1.098 0.095 2.140 0.00 10.00
ATOM 9 O3 SO4 1 2.481 2.037 2.159 0.70 14.00
ATOM 10 O4 SO4 1 0.131 2.251 1.823 0.70 12.00
```

all occupancies will be refined individually except for atom O2 where it will stay zero.

- A special case is refinement of a partially deuterated structure against neutron data. phenix.refine will treat exchangeable H/D sites as alternative conformations, for example: ATOM 54 N GLY L 2 2.908 -22.755 25.168 1.00 12.24 N ATOM 55 CA GLY L 2 2.957 -24.115 25.675 1.00 13.12 C ATOM 56 C GLY L 2 2.218 -24.358 26.958 1.00 15.55 C ATOM 57 O GLY L 2 2.343 -25.435 27.503 1.00 13.47 O ATOM 58 HA2 GLY L 2 2.590 -24.711 25.004 1.00 15.29 H ATOM 59 HA3 GLY L 2 3.885 -24.361 25.816 1.00 16.65 H ATOM 60 H AGLY L 2 2.349 -22.638 24.525 0.25 18.77 H ATOM 61 D BGLY L 2 2.349 -22.638 24.525 0.75 18.77 D This situation is detected automatically and the occupancies of H and D atoms are refined in such a way so their sum is one. In addition, B-factors and coordinates of such atoms are set to be identical.

A special case is refinement of a partially deuterated structure against neutron data. phenix.refine will treat exchangeable H/D sites as alternative conformations, for example:

```
ATOM 54 N GLY L 2 2.908 -22.755 25.168 1.00 12.24 N
ATOM 55 CA GLY L 2 2.957 -24.115 25.675 1.00 13.12 C
ATOM 56 C GLY L 2 2.218 -24.358 26.958 1.00 15.55 C
```

```
ATOM 57 O GLY L 2 2.343 -25.435 27.503 1.00 13.47 O
ATOM 58 HA2 GLY L 2 2.590 -24.711 25.004 1.00 15.29 H
ATOM 59 HA3 GLY L 2 3.885 -24.361 25.816 1.00 16.65 H
ATOM 60 H AGLY L 2 2.349 -22.638 24.525 0.25 18.77 H
ATOM 61 D BGLY L 2 2.349 -22.638 24.525 0.75 18.77 D
```

This situation is detected automatically and the occupancies of H and D atoms are refined in such a way so their sum is one. In addition, B-factors and coordinates of such atoms are set to be identical.

- Disabling occupancy refinement can be done by removing the star (*) from the corresponding keyword in strategy = ... *occupancies ... (in case parameter file is used).

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- If selections are provided by the user then the occupancy refinement for selected atoms will be performed as well as for those selected automatically.

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- User defined selections will override those defined by phenix.refine automatically.

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- User can withhold occupancy refinement for atoms that were automatically selected by phenix.refine for occupancy refinement.

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- The presence of user defined selections for occupancies to be refined is not enough to engage the occupancy refinement. It is important that the occupancy refinement is enabled by using the strategy = keyword.

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Examples:

- Running with all default parameters: % phenix.refine data.hkl model.pdb This will refine individual coordinates, individual B-factors (isotropic or anisotropic) and occupancies for atoms in alternative conformations or for atoms having partial occupancies. If there is no such atoms in input PDB file, then no occupancies will be refined.

Running with all default parameters:

```
% phenix.refine data.hkl model.pdb
```

This will refine individual coordinates, individual B-factors (isotropic or anisotropic) and occupancies for atoms in alternative conformations or for atoms having partial occupancies. If there is no such atoms in input PDB file, then no occupancies will be refined.

- Refinement of occupancies only: % phenix.refine data.hkl model.pdb strategy=occupancies This will only refine occupancies for atoms in alternative conformations or for atoms having partial occupancies. If there is no such atoms in input PDB file, then no occupancies will be refined. Other model parameters, such as B-factors or coordinates will not be refined.

Refinement of occupancies only:

```
% phenix.refine data.hkl model.pdb strategy=occupancies
```

This will only refine occupancies for atoms in alternative conformations or for atoms having partial occupancies. If there is no such atoms in input PDB file, then no occupancies will be refined. Other model parameters, such as B-factors or coordinates will not be refined.

- Refine individual occupancies of water molecules (in addition to atoms with partial occupancies and those in alternative conformations, if any): % phenix.refine data.hkl model.pdb refine.occupancies.individual="water" Similar refinement as above where all Zn atoms in chain X will be refined as well: % phenix.refine data.hkl model.pdb occupancies.individual="water" \ occupancies.individual="chain X and element Zn"

Refine individual occupancies of water molecules (in addition to atoms with partial occupancies and those in alternative conformations, if any):

```
% phenix.refine data.hkl model.pdb refine.occupancies.individual="water"
```

Similar refinement as above where all Zn atoms in chain X will be refined as well:

```
% phenix.refine data.hkl model.pdb occupancies.individual="water" \
 occupancies.individual="chain X and element Zn"
```

- Complex occupancy refinement strategy (combination of various available occupancy refinement types): % phenix.refine data.hkl model.pdb strategy=occupancies occ.params

Complex occupancy refinement strategy (combination of various available occupancy refinement types):

```
% phenix.refine data.hkl model.pdb strategy=occupancies occ.params
```

The amount of atom selections makes it inconvenient to type them all from the command line. This is why the parameter file occ.params is used and it contains following lines: refinement { refine { occupancies { individual = element BR or water individual = element Zn constrained_group { selection = chain A and resseq 1 } constrained_group { selection = chain A and resseq 2 selection = chain A and resseq 3 } constrained_group { selection = chain X and resname MAN selection = chain X and resseq 42 selection = chain X and resseq 121 } remove_selection = chain B and resseq 1 and name O remove_selection = chain B and resseq 3 and name O } } } which defines: individual occupancy refinement for all BR, Zn and water atoms; group occupancy refinement for residue number 1 in chain A (as selected with chain A and resseq 1). One occupancy for all atoms in this residue will be refined and it will be constrained between main.occupancy_min and main.occupancy_min, which by default is 0 and 1, correspondingly. another constrained occupancy group, where the occupancies of atoms in chain A and resseq 2 and chain A and resseq 3 will be coupled. That is all occupancies within chain A and resseq 2 will have the exact same value between 0 and 1, and same for chain A and resseq 3. The sum of occupancies of chain A and resseq 2 and chain A and resseq 3 will be 1.0, making it one constrained group. another constrained group contains three residues (number 42 and 121, and MAN) and their occupancies will be refined similarly as described above. occupancies of atoms O in residues 1 and 3 of chain B will not be refined as requested using remove_selection keyword (even though these atoms have partial occupancies in input PDB file and so they would normally be refined by default).

The amount of atom selections makes it inconvenient to type them all from the command line. This is why the parameter file occ.params is used and it contains following lines:

```
refinement {
 refine {
 occupancies {
 individual = element BR or water
 individual = element Zn
 constrained_group {
 selection = chain A and resseq 1
 }
 constrained_group {
 selection = chain A and resseq 2
```

```
  selection = chain A and resseq 3
  }
  constrained_group {
  selection = chain X and resname MAN
  selection = chain X and resseq 42
  selection = chain X and resseq 121
  }
  remove_selection = chain B and resseq 1 and name O
  remove_selection = chain B and resseq 3 and name O
  }
  }
}
```

which defines:

- individual occupancy refinement for all BR, Zn and water atoms;
- group occupancy refinement for residue number 1 in chain A (as selected with chain A and resseq 1). One occupancy for all atoms in this residue will be refined and it will be constrained between main.occupancy_min and main.occupancy_min, which by default is 0 and 1, correspondingly.
- another constrained occupancy group, where the occupancies of atoms in chain A and resseq 2 and chain A and resseq 3 will be coupled. That is all occupancies within chain A and resseq 2 will have the exact same value between 0 and 1, and same for chain A and resseq 3. The sum of occupancies of chain A and resseq 2 and chain A and resseq 3 will be 1.0, making it one constrained group.
- another constrained group contains three residues (number 42 and 121, and MAN) and their occupancies will be refined similarly as described above.
- occupancies of atoms O in residues 1 and 3 of chain B will not be refined as requested using remove_selection keyword (even though these atoms have partial occupancies in input PDB file and so they would normally be refined by default).

f' and f'' refinement

If the structure contains anomalous scatterers (e.g. Se in a SAD or MAD experiment), and if anomalous data are available, it is possible to refine the dispersive (f') and anomalous (f'') scattering contributions (see e.g. Ethan Merritt's tutorial for more information). In phenix.refine, each group of scatterers with common f' and f'' values is defined via an anomalous_scatterers scope, e.g.:

```
refinement.refine.anomalous_scatterers {
  group {
  selection = name BR
  f_prime = 0
  f_double_prime = 0
  refine = *f_prime *f_double_prime
  }
}
```

NOTE: The refinement of the f' and f'' values is carried out only if group_anomalous is included under refine.strategy! Otherwise the values are simply used as specified but not refined. So the refinement run with the parameters above included into group_anomalous_1.params:

```
% phenix.refine model.pdb data_anom.hkl group_anomalous_1.params \
  strategy=individual_sites+individual_adp+group_anomalous
```

If required, multiple scopes can be specified, one for each unique pair of f' and f'' values. These values are assigned to all selected atoms (see Phenix atom selection syntax for details). Often it is possible to start the refinement from zero. If the refinement is not stable, it may be necessary to start from better estimates, or even to fix some values. For example (file group_anomalous_2.params):

```
refinement.refine.anomalous_scatterers {
 group {
 selection = name BR
 f_prime = -5
 f_double_prime = 2
 refine = f_prime *f_double_prime
 }
}
```

```
% phenix.refine model.pdb data_anom.hkl group_anomalous_2.params \
 strategy=individual_sites+individual_adp+group_anomalous
```

Here f' is fixed at -5 (note the missing * in front of f_prime in the refine definition), and the refinement of f'' is initialized at 2.

The phenix.form_factor_query command is available for obtaining estimates of f' and f'' given an element type and a wavelength, e.g.:

```
% phenix.form_factor_query element=Br wavelength=0.8

Information from Sasaki table about Br (Z = 35) at 0.8 A
fp: -1.0333
fdp: 2.9928
```

Run without arguments for usage information:

```
% phenix.form_factor_query
```

Note that if you perform anomalous refinement, you may also want to include a log-likelihood gradient anomalous map (map_type=llg) in the output, as this will show any unmodeled anomalous scattering with greater sensitivity than the conventional anomalous difference map.

Using NCS in refinement

phenix.refine can use NCS as restraints or constraints. NCS restraints can be torsion- or Cartesian-based. NCS-related atoms can be identified automatically or be defined by the user.

The default NCS implementation in phenix.refine restrains NCS-related chains in torsion angle space.

Hydrogen atoms are excluded from NCS restraints.

Note that user-supplied NCS groups will be filtered at all times. More on this here.

Torsion NCS (default)

Torsion-based NCS restraints use a flexible target function that is smoothly shuts off as the difference between related torsions increases, allowing for local differences between NCS-related chains. The default behavior identifies related chains automatically, but users may also specify NCS groups.

Automatic rotamer outlier correction and rotamer consistency checks between NCS-related sidechains are carried out for refinements against data at 3.0 A and better.

No NCS restraints applied to B-factors.

- Refinement with automatic group determination: % phenix.refine data.hkl model.pdb ncs_search.enabled=True

Refinement with automatic group determination:

```
% phenix.refine data.hkl model.pdb ncs_search.enabled=True
```

- Refinement with user provided NCS selections: Create a torsion_ncs.params file with selections like: refinement.pdb_interpretation.ncs_group { reference = chain A selection = chain B selection = chain C }

- Specify torsion_ncs.params as an additional input when running phenix.refine: % phenix.refine data.hkl model.pdb ncs_search.enabled=True \ torsion_ncs.params

Refinement with user provided NCS selections:

Create a torsion_ncs.params file with selections like:

```
refinement.pdb_interpretation.ncs_group {
 reference = chain A
 selection = chain B
 selection = chain C
}
```

Specify torsion_ncs.params as an additional input when running phenix.refine:

```
% phenix.refine data.hkl model.pdb ncs_search.enabled=True \
 torsion_ncs.params
```

Cartesian NCS (optional)

Cartesian-based NCS restraints are also available in phenix.refine. Atoms in NCS-related chains are restrained to the average xyz position.

Gaps in selected sequences are allowed - a sequence alignment is performed to detect insertions or deletions. We recommend to check the automatically detected or adjusted NCS groups.

- Refinement with user provided NCS selections: Create a ncs_groups.params file with the NCS selections: refinement.pdb_interpretation.ncs_group { reference = chain A resid 1:4 selection = chain B and resid 1:3 selection = chain C } refinement.pdb_interpretation.ncs_group { reference = chain E selection = chain F }

Refinement with user provided NCS selections:

Create a ncs_groups.params file with the NCS selections:

```
refinement.pdb_interpretation.ncs_group {
 reference = chain A resid 1:4
 selection = chain B and resid 1:3
 selection = chain C
}
refinement.pdb_interpretation.ncs_group {
 reference = chain E
 selection = chain F
}
```

- Specify ncs_groups.params as an additional input when running phenix.refine: % phenix.refine data.hkl model.pdb ncs_groups.params \ ncs_search.enabled=True ncs.type=cartesian This will perform the default refinement round (individual coordinates and B-factors) using NCS restraints on coordinates and B-factors. Note: user specified NCS restraints in ncs_groups may be modified automatically if specified groups are not match each other. Automatic detection of NCS groups: % phenix.refine data.hkl model.pdb ncs_search.enabled=True \ ncs.type=cartesian This will perform the default refinement round (individual coordinates and B-factors) using NCS restraints automatically created based on input PDB file.

Specify ncs_groups.params as an additional input when running phenix.refine:

```
% phenix.refine data.hkl model.pdb ncs_groups.params \
 ncs_search.enabled=True ncs.type=cartesian
```

This will perform the default refinement round (individual coordinates and B-factors) using NCS restraints on coordinates and B-factors.

Note: user specified NCS restraints in ncs_groups may be modified automatically if specified groups are not match each other.

Automatic detection of NCS groups:

```
% phenix.refine data.hkl model.pdb ncs_search.enabled=True \
 ncs.type=cartesian
```

This will perform the default refinement round (individual coordinates and B-factors) using NCS restraints automatically created based on input PDB file.

NCS constraints (optional)

NCS constraints are used very similarly to NCS restraints (see above) except that ncs.type needs to be set to constraints:

```
% phenix.refine data.hkl model.pdb ncs_search.enabled=True \
 ncs.type=constraints
```

or in case NCS groups are provided:

```
% phenix.refine data.hkl model.pdb ncs_search.enabled=True \
 ncs.type=constraints ncs_groups.params
```

Using secondary structure restraints

Please read an overview about secondary structure restraints which provides overall description and ways to get correct secondary structure definitions. This part of documentation provides only specific to phenix.refine information.

At low resolutions it is often beneficial to restrain hydrogen bonding distances in helices, sheets, and nucleic acid base pairs. These can be used with or without explicit hydrogen atoms. Appropriate atom selections will be detected automatically if none are provided by the user, but in most cases careful manual annotation will probably yield better results, especially if the starting model is of low quality. To turn on the additional restraints a single extra parameter is sufficient:

```
% phenix.refine data.hkl model.pdb secondary_structure.enabled=True
```

You can also generate starting parameters for secondary structure restraints using a standalone utility

```
% phenix.secondary_structure_restraints model.pdb
```

This will print a set of parameters suitable for use in phenix.refine, which may be edited to correct errors or add undetected groups.

Like other restraints, the hydrogen bond distances have adjustable sigma and target values; these are defined in the secondary_structure.protein scope. The default potential mimics the covalent bond restraints. The sigma defaults to 0.05 Angstrom; the default target is 2.9A, but can be changed:

```
% phenix.refine data.hkl model.pdb secondary_structure.enabled=True \
 secondary_structure.protein.distance_ideal_n_o=2.0
```

Parameters such as sigma, slack and the usage of top_out potential are defined separately for each secondary structure element. A relatively strict outlier cutoff is applied by default, to prevent improperly restraining incorrectly annotated residues. Any bonds longer than the outlier cutoff (3.5A) will not be restrained. If you are certain of your annotations, you can increase the cutoff:

```
% phenix.refine data.hkl model.pdb secondary_structure.enabled=True \
 secondary_structure.protein.distance_cut_n_o=4.0
```

or disable outliers removal completely:

```
% phenix.refine data.hkl model.pdb secondary_structure_restraints=True \
 secondary_structure.protein.remove_outliers=False
```

Using a reference model in refinement

phenix.refine can be given a reference model that is used to steer refinement of the working model. This technique is advantageous in cases where the working data set is low resolution, but there is a known related structure solved at higher resolution. The higher resolution reference model is used to generate a set of dihedral restraints that are applied to each matching dihedral in the working model.

Reference chains are matched to working chains automatically, and sequences need not match exactly.

The default parameters are a good starting point:

```
% phenix.refine data.hkl model.pdb reference_model.enabled=True \
 reference_model.file=reference.pdb
```

The default sigma value for these reference dihedral restraints is 1.0 degrees. To increase the strength of these restraints, select a smaller sigma:

```
% phenix.refine data.hkl model.pdb reference_model.enabled=True \
 reference_model.file=reference.pdb reference_model.sigma=0.5
```

To decrease the strength of the restraints, select a larger sigma:

```
% phenix.refine data.hkl model.pdb reference_model.enabled=True \
 reference_model.file=reference.pdb reference_model.sigma=2.0
```

The reference restraints have a limit parameter which turns them off when the angle in the working model differs from the reference by an amount greater than limit. The default value is 15 degrees, but may be user-defined:

```
% phenix.refine data.hkl model.pdb reference_model.enabled=True \
 reference_model.file=reference.pdb reference_model.limit=10
```

For an optimal set of protein restraints, rotamer outliers in the working model that have rotameric counterparts in the reference model are automatically corrected to the rotamer from the reference model prior to refinement. In practice this step almost always improves the final model, but can be turned off if desired:

```
% phenix.refine data.hkl model.pdb reference_model.enabled=True \
 reference_model.file=reference.pdb reference_model.fix_outliers=False
```

Selections may also be used with reference_model restraints. Selections are useful in cases where multiple chains in the working model should be restrained to the same reference chain, the model or reference have insertions that change the register, only part of a chain is desirable to restrain, etc.

To specify selections, create a reference.params file with selections like:

```
refinement.reference_model.reference_group {
 reference = chain A and resseq 2:119
 selection = chain A and resseq 2:119
 }
refinement.reference_model.reference_group {
 reference = chain A and resseq 130:134
 selection = chain A and resseq 120:124
 }
refinement.reference_model.reference_group {
 reference = chain A
 selection = chain B
 }
```

Note that reference will be applied to reference model and selection will be applied to refined model.

Specify reference.params as an additional input when running phenix.refine:

```
% phenix.refine data.hkl model.pdb reference_model.enabled=True \
 reference_model.file=reference.pdb reference.params
```

Each selection (both reference and selection entries as above) may only specify one chainID and/or one resseq range.

Multiple reference models may also be specified in cases where a working complex has reference structures from different coordinate files. To specify multiple reference model input files, the command line or parameter file should contain:

```
refinement.reference_model.file = reference_A.pdb
refinement.reference_model.file = reference_B.pdb
```

Reference chain/model chain pairs are determined automatically, and details are written to the .log file and .eff file. If you want to specify your own matching, include a parameter file that contains:

```
refinement.reference_model.file = reference_A.pdb
refinement.reference_model.file = reference_B.pdb
refinement.reference_model.reference_group {
 reference = chain A
 selection = chain A
 file_name = reference_A.pdb
 }
refinement.reference_model.reference_group {
 reference = chain A
 selection = chain B
 file_name = reference_B.pdb
 }
```

Note that even though file_name parameter is provided in both reference groups, refinement.reference_model.file should still contain both file names.

The refinement.reference_model.reference_group.file_name parameter is only required when more than one reference file is used. This parameter allows the reference model restraint generation to disambiguate between reference files that contain chains with the same chainID.

Automatic Asn/Gln/His corrections

Asn, Gln, and His residues can often be fit favorably to the data in two orientations, related by a 180 degree rotation. In many cases, however, only one of these orientations is sterically and electrostatically favorable. phenix.refine uses Reduce to identify Asn, Gln, and His residues that should be flipped, and then flips them automatically. This feature is enabled by default.

To disable this feature:

```
% phenix.refine data.hkl model.pdb main.nqh_flips=True
```

Water picking

phenix.refine can add, remove and refine waters as part of a refinement run. This is the recommended procedure for adding water structure.

Normally, the default parameter settings are good for most cases:

```
% phenix.refine data.hkl model.pdb ordered_solvent=true
```

This will add new water, analyse existing waters (and delete bad ones if necessary) and refine individual coordinates and B-factors of both, macromolecule and water.

Water picking can be combined with all others protocols, like simulated annealing, TLS refinement, and more. Some useful commands are:

- Perform water update every macro-cycle. By default, water picking starts after a half of macro-cycles is done: % phenix.refine data.hkl model.pdb ordered_solvent=true \ ordered_solvent.mode=every_macro_cycle

Perform water update every macro-cycle.

By default, water picking starts after a half of macro-cycles is done:

```
% phenix.refine data.hkl model.pdb ordered_solvent=true \
 ordered_solvent.mode=every_macro_cycle
```

- Remove water only (based on specified criteria): % phenix.refine data.hkl model.pdb ordered_solvent=true \ ordered_solvent.mode=filter_only

Remove water only (based on specified criteria):

```
% phenix.refine data.hkl model.pdb ordered_solvent=true \
 ordered_solvent.mode=filter_only
```

- The following run illustrates the use of some important parameters: % phenix.refine data.hkl model.pdb ordered_solvent=true solvent.params where the parameter file solvent.params contains: refinement { ordered_solvent { low_resolution = 2.8 b_iso_min = 1.0 b_iso_max = 50.0 b_iso = 25.0 primary_map_type = mFobs-DFmodel primary_map_cutoff = 3.0 secondary_map_and_map_cc_filter { cc_map_2_type = 2mFobs-DFmodel } } peak_search { map_next_to_model { min_model_peak_dist = 1.8 max_model_peak_dist = 6.0 min_peak_peak_dist = 1.8 } } } This will skip water picking if the resolution of data is lower than 2.8A, it will remove waters with B < 1.0 or B > 50.0 A**2 or occupancy different from 1 or peak height at mFo-DFc map lower then 3 sigma. It will not select or will remove existing water if water-water or water-macromolecule distance is less than 1.8A or water-macromolecule distance is greater than 6.0 A. The initial occupancies and B-factors of newly placed waters will be 1.0 and 25.0 correspondingly. If b_iso = None, then b_iso will be the mean atomic B-factor.

The following run illustrates the use of some important parameters:

```
% phenix.refine data.hkl model.pdb ordered_solvent=true solvent.params
```

where the parameter file solvent.params contains:

```
refinement {
 ordered_solvent {
 low_resolution = 2.8
 b_iso_min = 1.0
 b_iso_max = 50.0
 b_iso = 25.0
 primary_map_type = mFobs-DFmodel
 primary_map_cutoff = 3.0
 secondary_map_and_map_cc_filter
 {
 cc_map_2_type = 2mFobs-DFmodel
 }
 }
 peak_search {
 map_next_to_model {
 min_model_peak_dist = 1.8
 max_model_peak_dist = 6.0
 min_peak_peak_dist = 1.8
 }
 }
}
```

This will skip water picking if the resolution of data is lower than 2.8A, it will remove waters with B < 1.0 or B > 50.0 A**2 or occupancy different from 1 or peak height at mFo-DFc map lower then 3 sigma. It will not select or will remove existing water if water-water or water-macromolecule distance is less than 1.8A or water-macromolecule distance is greater than 6.0 A. The initial occupancies and B-factors of newly placed waters will be 1.0 and 25.0 correspondingly. If b_iso = None, then b_iso will be the mean atomic B-factor.

Hydrogens in refinement

Depending on data type (X-ray or neutron), data quality (resolution, completeness) phenix.refine offers different options for parametrization of hydrogen atoms:

- riding model.
- complete refinement of H (H atoms will be refined as other atoms in the model).

Using the riding model does not add additional refinable parameters, since position of a hydrogen atom H in X-H bond is recalculated from the current position of atom X. Also, H atom inherits the occupancy of X atom and its B-factor. Sometime the B-factor of H atom is the product of B-factor of X atoms and a scale from 1 to 1.5. The riding model should be used to parametrize H atoms at almost all resolutions in X-ray refinement. An exception can be a subatomic resolution (~0.7A and higher), where the hydrogen's parameters can be refined individually.

Although the contribution of hydrogen atoms to X-ray scattering is weak, H atoms are present in real structures irrespective of the data quality. Including them as riding model at any resolution makes other model atoms aware of their positions resulting in better refined model parameters.

Scattering contribution of hydrogen atoms by default is always accounted for, however there is a parameter to disable this:

```
% phenix.refine model.pdb data.hkl hydrogens.contribute_to_f_calc=false
```

If neutron data is used then the parameters of H atoms should always be refined individually, except the cases where data resolution and/or completeness are poor. In that case riding model can be used. If partially deuterated structure is used in refinement then the constrained occupancies of exchangeable H/D sites are refined so they add up to 1.

phenix.refine does not add H atoms (except a few cases mentioned below). To use hydrogen atoms in refinement thet need to be added to the model first. This can be done by using phenix.ready_set program, which can add H, D or H/D atoms. Internally phenix.ready_set uses Reduce to add H to macromolecule (protein, DNA/RNA) and it uses its own resources to add hydrogens to ligands or water. Hydrogens are added to their ideal geometrical positions. Different dictionary X-H lengths can be used for X-ray and neutron data.

If a structure contains a ligand unknown to phenix.refine, ReadySet! will create a library CIF file which will include the definitions for all newly added hydrogens.

phenix.refine can build H or D atoms for water molecules only. To do so it uses residual density map, mFo-DFc. This option is normally used at relatively high resolution neutron data (~2.0...2.5A and higher) or at subatomic X-ray resolution:

```
% phenix.refine model.pdb data.hkl main.find_and_add_hydrogens=true
```

H atoms are automatically excluded from TLS groups and NCS restraints. However, if NCS selections are created manually and the structure contains H atoms, it might be a good idea to add and not (element H or element D) to all selection strings.

Below are some useful commands:

Add hydrogens: % phenix.ready_set model.pdb Add deuteriums: % phenix.ready_set model.pdb perdeuterate=true Add H and exchangeable H/D: % phenix.ready_set model.pdb neutron_exchange_hydrogens=true Add H to water: % phenix.ready_set model.pdb add_h_to_water=true Once hydrogens added to a model, by default they will be refined as riding model: % phenix.refine model.pdb data.hkl It is possible to refine individual parameters for H atoms (if neutron data is used or at ultra-high resolution): % phenix.refine model.pdb data.hkl hydrogens.refine=individual To refine individual coordinates and ADP of H atoms: % phenix.refine model.pdb data.hkl hydrogens.refine=individual To remove hydrogens from a model: % phenix.pdbtools model.pdb remove="element H or element D" or Reduce programs can be used for this: % phenix.reduce model_h.pdb -trim > model_noH.pdb We strongly recommend to not remove hydrogen atoms after refinement since it will make the refinement statistics (R-factors, etc...) unreproducible without repeating exactly the same refinement protocol. Yet another option to add hydrogens (rarely used in practice): % phenix.elbow --final-geometry=model.pdb --residue=MAN --output=model_h Output PDB file called model_h.pdb will contain the original ligand MAN with all hydrogen atoms added.

- Add hydrogens: % phenix.ready_set model.pdb

Add hydrogens:

```
% phenix.ready_set model.pdb
```

- Add deuteriums: % phenix.ready_set model.pdb perdeuterate=true

Add deuteriums:

```
% phenix.ready_set model.pdb perdeuterate=true
```

- Add H and exchangeable H/D: % phenix.ready_set model.pdb neutron_exchange_hydrogens=true

Add H and exchangeable H/D:

```
% phenix.ready_set model.pdb neutron_exchange_hydrogens=true
```

- Add H to water: % phenix.ready_set model.pdb add_h_to_water=true

Add H to water:

```
% phenix.ready_set model.pdb add_h_to_water=true
```

- Once hydrogens added to a model, by default they will be refined as riding model: % phenix.refine model.pdb data.hkl It is possible to refine individual parameters for H atoms (if neutron data is used or at ultra-high resolution): % phenix.refine model.pdb data.hkl hydrogens.refine=individual

Once hydrogens added to a model, by default they will be refined as riding model:

```
% phenix.refine model.pdb data.hkl
```

It is possible to refine individual parameters for H atoms (if neutron data is used or at ultra-high resolution):

```
% phenix.refine model.pdb data.hkl hydrogens.refine=individual
```

- To refine individual coordinates and ADP of H atoms: % phenix.refine model.pdb data.hkl hydrogens.refine=individual

To refine individual coordinates and ADP of H atoms:

```
% phenix.refine model.pdb data.hkl hydrogens.refine=individual
```

- To remove hydrogens from a model: % phenix.pdbtools model.pdb remove="element H or element D" or Reduce programs can be used for this: % phenix.reduce model_h.pdb -trim > model_noH.pdb We strongly recommend to not remove hydrogen atoms after refinement since it will make the refinement statistics (R-factors, etc...) unreproducible without repeating exactly the same refinement protocol.

To remove hydrogens from a model:

```
% phenix.pdbtools model.pdb remove="element H or element D"
```

or Reduce programs can be used for this:

```
% phenix.reduce model_h.pdb -trim &gt; model_noH.pdb
```

We strongly recommend to not remove hydrogen atoms after refinement since it will make the refinement statistics (R-factors, etc...) unreproducible without repeating exactly the same refinement protocol.

- Yet another option to add hydrogens (rarely used in practice): % phenix.elbow --final-geometry=model.pdb --residue=MAN --output=model_h Output PDB file called model_h.pdb will contain the original ligand MAN with all hydrogen atoms added.

Yet another option to add hydrogens (rarely used in practice):

```
% phenix.elbow --final-geometry=model.pdb --residue=MAN --output=model_h
```

Output PDB file called model_h.pdb will contain the original ligand MAN with all hydrogen atoms added.

Refinement using twinned data

phenix.refine can handle the refinement of hemihedrally twinned data (two twin domains). Least square twin refinement can be carried out using the following commands line instructions:

```
% phenix.refine data.hkl model.pdb xray_data.twin_law="-k,-h,-l"
```

The twin law (in this case -k,-h,-l) can be obtained from phenix.xtriage. If more than a single twin law is possible for the given unit cell and space group, using phenix.twin_map_utils might give clues which twin law is the most likely candidate to be used in refinement.

Correcting maps for anisotropy might be useful:

```
% phenix.refine data.hkl model.pdb xray_data.twin_law="-k,-h,-l" \
 detwin.map_types.aniso_correct=true
```

The detwinning mode is auto by default: it will perform algebraic detwinning for twin fraction below 40%, and detwinning using proportionality rules (SHELXL style) for fractions above 40%.

An important point to stress is that phenix.refine will only deal properly with twinning that involves two twin domains.

Neutron and joint X-ray and neutron refinement

The paradigm and implementation of the joint X-ray and neutron refinement were changed in the fall of 2023, as documented here: https://pubmed.ncbi.nlm.nih.gov/37942718/

Refinement using neutron data requires having H or/and D atoms added to the model. Use ReadySet! program to add all H, D or H/D atoms. See "Hydrogens in refinement" section for details.

- Running refinement with neutron data only: % phenix.refine data.hkl model.pdb main.scattering_table=neutron this will tell phenix.refine that the data in data.hkl file is coming from neutron scattering experiment and the appropriate scattering factors will be used in all calculations. All the examples and phenix.refine functionality presented in this document are valid and compatible with using neutron data.

Running refinement with neutron data only:

```
% phenix.refine data.hkl model.pdb main.scattering_table=neutron
```

this will tell phenix.refine that the data in data.hkl file is coming from neutron scattering experiment and the appropriate scattering factors will be used in all calculations. All the examples and phenix.refine functionality presented in this document are valid and compatible with using neutron data.

- Using X-ray and neutron data simultaneously (joint XN refinement). phenix.refine allows simultaneous use of both data sets, X-ray and neutron. The data sets are allowed to have different number of reflections, be collected at different resolutions and temperatures from crystalls having different space groups and unit cell parameters. Running a joint XN refinement requires a parameter file: % phenix.refine joint_xn.eff where the joint_xn.eff file contains the following lines: data_manager { miller_array { file = data.mtz labels { name = "F-obs,SIGF-obs" type = x_ray } labels { name = "R-free-flags" type = x_ray } labels { name = "F-obs-neutron,SIGF-obs-neutron" type = neutron } labels { name = "R-free-flags-neutron" type = neutron } } model { file = model_xray.pdb type = x_ray } model { file = model_neutron.pdb type = neutron } } xray { refinement { refine { strategy = *individual_sites *rigid_body } main { simulated_annealing = True ordered_solvent = True number_of_macro_cycles = 3 } } } neutron { refinement { refine { strategy = *individual_sites *individual_adp } main { simulated_annealing = False ordered_solvent = False number_of_macro_cycles = 5 } } } Parameters within the data_manager {...} scope define input data files, labels for X-ray and neutron data arrays, as well as labels for corresponding free-r flags. Note, two models are input: one is X-ray and another one is neutron model. Additionally, refinement strategies for X-ray and neutron models can be customized independently as illistrated in xray {...} and neutron {...} scopes of parameters.

Using X-ray and neutron data simultaneously (joint XN refinement).

phenix.refine allows simultaneous use of both data sets, X-ray and neutron. The data sets are allowed to have different number of reflections, be collected at different resolutions and temperatures from crystalls having different space groups and unit cell parameters.

Running a joint XN refinement requires a parameter file:

```
% phenix.refine joint_xn.eff
```

where the joint_xn.eff file contains the following lines:

```
data_manager {
 miller_array {
 file = data.mtz
 labels {
 name = "F-obs,SIGF-obs"
 type = x_ray
 }
 labels {
 name = "R-free-flags"
 type = x_ray
 }
 labels {
 name = "F-obs-neutron,SIGF-obs-neutron"
 type = neutron
 }
 labels {
 name = "R-free-flags-neutron"
 type = neutron
 }
 }
 model {
 file = model_xray.pdb
 type = x_ray
 }
 model {
 file = model_neutron.pdb
 type = neutron
```

```
  }
 }
 xray {
  refinement {
  refine {
  strategy = *individual_sites *rigid_body
  }
  main {
  simulated_annealing = True
  ordered_solvent = True
  number_of_macro_cycles = 3
  }
  }
 }
 neutron {
  refinement {
  refine {
  strategy = *individual_sites *individual_adp
  }
  main {
  simulated_annealing = False
  ordered_solvent = False
  number_of_macro_cycles = 5
  }
  }
 }
```

Parameters within the data_manager {...} scope define input data files, labels for X-ray and neutron data arrays, as well as labels for corresponding free-r flags. Note, two models are input: one is X-ray and another one is neutron model.

Additionally, refinement strategies for X-ray and neutron models can be customized independently as illistrated in xray {...} and neutron {...} scopes of parameters.

Optimizing target weights

phenix.refine uses automatic procedure to determine the weights between X-ray target and stereochemistry or ADP restraints. To optimize these weights (that is to find those resulting in lowest Rfree factors):

```
% phenix.refine data.hkl model.pdb optimize_xyz_weight=true optimize_adp_weight=true
```

where optimize_xyz_weight will turn on the optimization of X-ray/stereochemistry weight and optimize_adp_weight will turn on the optimization of X-ray/ADP weight. Note that this could be very slow since the procedure involves a grid search over an array of weights-candidates. It could be a good idea to run this overnight for a final model tune up.

Refinement at high resolution (higher than approx. 1.0 Angstrom)

Guidelines for structure refinement at high resolution:

make sure the model contains hydrogen atoms. If not, phenix.reduce can be used to add them: % phenix.reduce model.pdb > model_h.pdb By default, phenix.refine will refine positions of H atoms as riding model (H atom will exactly follow the atom it is attached to). Note that phenix.refine can also refine individual coordinates of H atoms (can be used for small molecules at ultra-high resolutions or for refinement against neutron data). This is governed by hydrogens.refine = individual *riding keyword and the default is to use riding model. hydrogens.refine defines how hydrogens' B-factors are refined (default is to refine one group B for all H atoms). At high resolution one should definitely try to use one_b_per_molecule or even individual choice (resolution permitting). Similar strategy should be used for refinement of H's occupancies, hydrogens.refine_occupancies keyword. most of the atoms should be refined with anisotropic ADP.

Exceptions could be model parts with high B-factors), atoms in alternative conformations, hydrogens and solvent molecules. However, at resolutions higher than 1.0A it's worth of trying to refine solvent with anisotropic ADP. it is a good idea to constantly monitor the existing solvent molecules and check for new ones by using ordered_solvent=true keyword. If it's decided to refine waters with anisotropic ADP then make sure that the newly added ones are also anisotropic; use ordered_solvent.new_solvent=anisotropic (default is isotropic). One can also ask phenix.refine to refine occupancies of water: ordered_solvent.refine_occupancies=true (default is False). at high resolution the alternative conformations can be visible for more than 20% of residues. phenix.refine automatically recognizes atoms in alternative conformations (based on PDB records) and by default does constrained refinement of occupancies for these atoms. Please note, that phenix.refine does not build or create the fragments in alternative conformations; the atoms in alternative conformations should be properly defined in input PDB file (using conformer identifiers) (if actually found in a structure). the default weights for stereochemical and ADP restraints are most likely too tight at this resolution, so most likely the corresponding values need to be relaxed. Use wxc_scale and wxu_scale for this; higher values, like 1, 1.5, 2, ... etc should be tried. phenix.refine allows automatically optimize these values (optimize_xyz_weight=True and optimize_adp_weight=True), however this is a very slow task so it may be considered for an over night run or even longer. At ultra-high resolutions (approx. 0.8A or higher) a complete unrestrained refinement should be definitely tried out for well ordered parts of the model (single conformations, low B-factors). at ultra-high resolution the residual maps show the electron density redistribution due to bonds formation as density peaks at interatomic bonds. phenix.refine has specific tools to model this density called IAS models (Afonine et al, Acta Cryst. (2007). D63, 1194-1197).

- make sure the model contains hydrogen atoms. If not, phenix.reduce can be used to add them: % phenix.reduce model.pdb > model_h.pdb By default, phenix.refine will refine positions of H atoms as riding model (H atom will exactly follow the atom it is attached to). Note that phenix.refine can also refine individual coordinates of H atoms (can be used for small molecules at ultra-high resolutions or for refinement against neutron data). This is governed by hydrogens.refine = individual *riding keyword and the default is to use riding model. hydrogens.refine defines how hydrogens' B-factors are refined (default is to refine one group B for all H atoms). At high resolution one should definitely try to use one_b_per_molecule or even individual choice (resolution permitting). Similar strategy should be used for refinement of H's occupancies, hydrogens.refine_occupancies keyword.

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```
% phenix.reduce model.pdb &gt; model_h.pdb
```

By default, phenix.refine will refine positions of H atoms as riding model (H atom will exactly follow the atom it is attached to). Note that phenix.refine can also refine individual coordinates of H atoms (can be used for small molecules at ultra-high resolutions or for refinement against neutron data). This is governed by hydrogens.refine = individual *riding keyword and the default is to use riding model. hydrogens.refine defines how hydrogens' B-factors are refined (default is to refine one group B for all H atoms). At high resolution one should definitely try to use one_b_per_molecule or even individual choice (resolution permitting). Similar strategy should be used for refinement of H's occupancies, hydrogens.refine_occupancies keyword.

- most of the atoms should be refined with anisotropic ADP. Exceptions could be model parts with high B-factors), atoms in alternative conformations, hydrogens and solvent molecules. However, at resolutions higher than 1.0A it's worth of trying to refine solvent with anisotropic ADP.

most of the atoms should be refined with anisotropic ADP. Exceptions could be model parts with high B-factors), atoms in alternative conformations, hydrogens and solvent molecules. However, at resolutions higher than 1.0A it's worth of trying to refine solvent with anisotropic ADP.

- it is a good idea to constantly monitor the existing solvent molecules and check for new ones by using ordered_solvent=true keyword. If it's decided to refine waters with anisotropic ADP then make sure that the newly added ones are also anisotropic; use ordered_solvent.new_solvent=anisotropic (default is isotropic). One can also ask phenix.refine to refine occupancies of water: ordered_solvent.refine_occupancies=true (default is False).

it is a good idea to constantly monitor the existing solvent molecules and check for new ones by using ordered_solvent=true keyword. If it's decided to refine waters with anisotropic ADP then make sure that the newly added ones are also anisotropic; use ordered_solvent.new_solvent=anisotropic (default is isotropic). One can also ask phenix.refine to refine occupancies of water: ordered_solvent.refine_occupancies=true (default is False).

- at high resolution the alternative conformations can be visible for more than 20% of residues. phenix.refine automatically recognizes atoms in alternative conformations (based on PDB records) and by default does constrained refinement of occupancies for these atoms. Please note, that phenix.refine does not build or create the fragments in alternative conformations; the atoms in alternative conformations should be properly defined in input PDB file (using conformer identifiers) (if actually found in a structure).

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- the default weights for stereochemical and ADP restraints are most likely too tight at this resolution, so most likely the corresponding values need to be relaxed. Use wxc_scale and wxu_scale for this; higher values, like 1, 1.5, 2, ... etc should be tried. phenix.refine allows automatically optimize these values (optimize_xyz_weight=True and optimize_adp_weight=True), however this is a very slow task so it may be considered for an over night run or even longer. At ultra-high resolutions (approx. 0.8A or higher) a complete unrestrained refinement should be definitely tried out for well ordered parts of the model (single conformations, low B-factors).

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- at ultra-high resolution the residual maps show the electron density redistribution due to bonds formation as density peaks at interatomic bonds. phenix.refine has specific tools to model this density called IAS models (Afonine et al, Acta Cryst. (2007). D63, 1194-1197).

at ultra-high resolution the residual maps show the electron density redistribution due to bonds formation as density peaks at interatomic bonds. phenix.refine has specific tools to model this density called IAS models (Afonine et al, Acta Cryst. (2007). D63, 1194-1197).

This example illustrates most of the above points:

```
% phenix.refine model_h.pdb data.hkl high_res.params
```

where the file high_res.params contains following lines (for more parameters under each scope look at complete list of parameters):

```
refinement.main {
 number_of_macro_cycles = 5
 ordered_solvent=true
}
refinement.refine {
 adp {
 individual {
 isotropic = element H
 anisotropic = not element H
 }
 }
}
refinement.target_weights {
 wxc_scale = 2
 wxu_scale = 2
}
refinement {
 ordered_solvent {
 mode = auto filter_only *every_macro_cycle
 new_solvent = isotropic *anisotropic
 refine_occupancies = True
 }
}
```

In the example above phenix.refine will perform 5 macro-cycles with ordered solvent update (add/remove) every macro-cycles, all atoms including newly added water will be refined with anisotropic B-factors (except hydrogens), riding model will be used for positional refinement of H atoms, one occupancy and isotropic B-factor will be refined per all hydrogens within a residue, occupancies of waters will be refined as well, the default stereochemistry and ADP restraints weights are scaled down by the factors of 0.25 and 0.3 respectively. If starting model is far enough from the "final" one, more macro-cycles may be required (than 5 used in this example).

Examples of frequently used refinement protocols, common problems

- Starting refinement from high R-factors: % phenix.refine data.hkl model.pdb ordered_solvent=true main.number_of_macro_cycles=10 \ simulated_annealing=true strategy=rigid_body+individual_sites+individual_adp \

Starting refinement from high R-factors:

```
% phenix.refine data.hkl model.pdb ordered_solvent=true main.number_of_macro_cycles=10 \
 simulated_annealing=true strategy=rigid_body+individual_sites+individual_adp \
```

Depending on data resolution, refinement of individual ADP may be replaced with grouped B refinement: % phenix.refine data.hkl model.pdb ordered_solvent=true simulated_annealing=true \ strategy=rigid_body+individual_sites+group_adp main.number_of_macro_cycles=10 Adding TLS refinement may be a good idea. Note, unlike other programs, phenix.refine does not require "good model" for doing TLS refinement; TLS refinement is always stable in phenix.refine (please report if noticed otherwise): % phenix.refine data.hkl model.pdb ordered_solvent=true simulated_annealing=true \ strategy=rigid_body+individual_sites+individual_adp+tls main.number_of_macro_cycles=10 If NCS is present - once can use it: % phenix.refine data.hkl model.pdb ordered_solvent=true simulated_annealing=true \ strategy=rigid_body+individual_sites+individual_adp+tls main.ncs=true \ main.number_of_macro_cycles=10 tls_group_selections.params \ rigid_body_selections.params where tls_groups_selections.txt, rigid_body_groups_selections.txt are the files TLS and rigid body groups selections, NCS will be determined automatically from input PDB file. See this document for details on how specify these selections. Note: in these four examples above we re-defined the default number of refinement macro-cycles from 3 to 10, since a start model with high R-factors most likely requires more cycles to become a good one. Also in these examples, the rigid body refinement will be run only once at first macro-cycle, the water picking will start after half of macro-cycles is done (after 5th), the SA will be done only twice - the first and before the last macro-cycles. Even though it is requested, the water picking may not be performed if the resolution is too low. All these default behaviors can be changed: see parameter's help for more details. The last command looks too long to type it in the command line. Look this document for an example of how to make it like this: % phenix.refine data.hkl model.pdb custom_par_1.params

Depending on data resolution, refinement of individual ADP may be replaced with grouped B refinement:

```
% phenix.refine data.hkl model.pdb ordered_solvent=true simulated_annealing=true \
 strategy=rigid_body+individual_sites+group_adp main.number_of_macro_cycles=10
```

Adding TLS refinement may be a good idea. Note, unlike other programs, phenix.refine does not require "good model" for doing TLS refinement; TLS refinement is always stable in phenix.refine (please report if noticed otherwise):

```
% phenix.refine data.hkl model.pdb ordered_solvent=true simulated_annealing=true \
 strategy=rigid_body+individual_sites+individual_adp+tls main.number_of_macro_cycles=10
```

If NCS is present - once can use it:

```
% phenix.refine data.hkl model.pdb ordered_solvent=true simulated_annealing=true \
 strategy=rigid_body+individual_sites+individual_adp+tls main.ncs=true \
 main.number_of_macro_cycles=10 tls_group_selections.params \
 rigid_body_selections.params
```

where tls_groups_selections.txt, rigid_body_groups_selections.txt are the files TLS and rigid body groups selections, NCS will be determined automatically from input PDB file. See this document for details on how specify these selections.

Note: in these four examples above we re-defined the default number of refinement macro-cycles from 3 to 10, since a start model with high R-factors most likely requires more cycles to become a good one. Also in these examples, the rigid body refinement will be run only once at first macro-cycle, the water picking will start after half of macro-cycles is done (after 5th), the SA will be done only twice - the first and before the last macro-cycles. Even though it is requested, the water picking may not be performed if the resolution is too low. All these default behaviors can be changed: see parameter's help for more details.

The last command looks too long to type it in the command line. Look this document for an example of how to make it like this:

```
% phenix.refine data.hkl model.pdb custom_par_1.params
```

- Refinement at "higher than medium" resolution - getting anisotropic.

Refining at higher resolution one may consider: At resolutions around 1.8 ... 1.7 A or higher it is a good idea to try refinement of anisotropic ADP for atoms at well ordered parts of the model. Well ordered parts can be identified by relatively small isotropic B-factors ~5-20A**2 of so. The riding model for H atoms should be used. Loosing stereochemistry and ADP restraints. Re-thing using the NCS (if present): it may turn out to be enough of data to not use NCS restrains. Try both, with and without NCS, and based on R-free vales decide the strategy. Supposing the H atoms were added to the model, below is an example of what may want to do at higher resolution: % phenix.refine data.hkl model.pdb adp.individual.anisotropic="resid 1:2 and not element H" \ adp.individual.isotropic="not (resid 1:2 and not element H)" wxc_scale=2 wxu_scale=2 In the command above phenix.refine will refine the ADP of atoms in residues from 1 to 2 as anisotropic, the rest (including all H atoms) will be isotropic, the X-ray target contribution is increased for both, coordinate and ADP refinement. IMPORTANT: Please make note of the selection used in the above command: selecting atoms in residues 1 and 2 to be refined as anisotropic, one need to exclude hydrogens, which should be refined as isotropic.

Refining at higher resolution one may consider:

At resolutions around 1.8 ... 1.7 A or higher it is a good idea to try refinement of anisotropic ADP for atoms at well ordered parts of the model. Well ordered parts can be identified by relatively small isotropic B-factors ~5-20A**2 of so. The riding model for H atoms should be used. Loosing stereochemistry and ADP restraints. Re-thing using the NCS (if present): it may turn out to be enough of data to not use NCS restrains. Try both, with and without NCS, and based on R-free vales decide the strategy.

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- Loosing stereochemistry and ADP restraints.
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Supposing the H atoms were added to the model, below is an example of what may want to do at higher resolution:

```
% phenix.refine data.hkl model.pdb adp.individual.anisotropic="resid 1:2 and not element H" \
 adp.individual.isotropic="not (resid 1:2 and not element H)" wxc_scale=2 wxu_scale=2
```

In the command above phenix.refine will refine the ADP of atoms in residues from 1 to 2 as anisotropic, the rest (including all H atoms) will be isotropic, the X-ray target contribution is increased for both, coordinate and ADP refinement. IMPORTANT: Please make note of the selection used in the above command: selecting atoms in residues 1 and 2 to be refined as anisotropic, one need to exclude hydrogens, which should be refined as isotropic.

- Stereochemistry looks too tightly / loosely restrained, or gap between R-free and R-work seems too big: playing with restraints contribution. Although the automatic calculation of weight between X-ray and stereochemistry or ADP restraint targets is good for most of cases, it may happen that rmsd deviations from ideal bonds length or angles are looking too tight or loose (depending on resolution). Or the difference between R-work and R-free is too big (significantly bigger than approx. 5%). In such cases one definitely need to try loose or tighten the restraints. Hers is how for coordinates refinement: % phenix.refine data.hkl model.pdb wxc_scale=5 The default value for wxc_scale is 0.5. Increasing wxc_scale will make the X-ray target contribution greater and restraints looser. Note: wxc_scale=0 will completely exclude the experimental data from the refinement resulting in idealization of the stereochemistry. For stereochemistry idealization use the separate command: % phenix.geometry_minimization model.pdb To see the options type: % phenix.geometry_minimization --help To play with ADP restraints contribution: % phenix.refine data.hkl model.pdb wxu_scale=3 The default value for wxu_scale is 1.0. Increasing wxu_scale will make the X-ray target contribution greater and therefore the B-factors restraints weaker. Also, one can completely ignore the automatically determined weights (for both, coordinates and ADP refinement) and use specific values instead: % phenix.refine data.hkl model.pdb fix_wxc=15.0 The refinement target will be: Etotal = 15.0 * Exray + Egeom Similarly for ADP refinement: % phenix.refine data.hkl model.pdb fix_wxu=25.0 The refinement target will be: Etotal = 25.0 * Exray + Eadp

Stereochemistry looks too tightly / loosely restrained, or gap between R-free and R-work seems too big: playing with restraints contribution.

Although the automatic calculation of weight between X-ray and stereochemistry or ADP restraint targets is good for most of cases, it may happen that rmsd deviations from ideal bonds length or angles are looking too tight or loose (depending on resolution). Or the difference between R-work and R-free is too big (significantly bigger than approx. 5%). In such cases one definitely need to try loose or tighten the restraints. Hers is how for coordinates refinement:

```
% phenix.refine data.hkl model.pdb wxc_scale=5
```

The default value for wxc_scale is 0.5. Increasing wxc_scale will make the X-ray target contribution greater and restraints looser. Note: wxc_scale=0 will completely exclude the experimental data from the refinement resulting in idealization of the stereochemistry. For stereochemistry idealization use the separate command:

```
% phenix.geometry_minimization model.pdb
```

To see the options type:

```
% phenix.geometry_minimization --help
```

To play with ADP restraints contribution:

```
% phenix.refine data.hkl model.pdb wxu_scale=3
```

The default value for wxu_scale is 1.0. Increasing wxu_scale will make the X-ray target contribution greater and therefore the B-factors restraints weaker.

Also, one can completely ignore the automatically determined weights (for both, coordinates and ADP refinement) and use specific values instead:

```
% phenix.refine data.hkl model.pdb fix_wxc=15.0
```

The refinement target will be: Etotal = 15.0 * Exray + Egeom

Similarly for ADP refinement:

```
% phenix.refine data.hkl model.pdb fix_wxu=25.0
```

The refinement target will be: Etotal = 25.0 * Exray + Eadp

- Having unknown to phenix.refine item in PDB file (novel ligand, etc...). phenix.refine uses the CCP4 Monomer Library as the source of stereochemical information for building geometry restraints and reporting statistics. If phenix.refine is unable to match an item in input PDB file against the Monomer Library it will stop with "Sorry" message explaining what to do and listing the problem atoms. If this happened, it is necessary to obtain a cif file (parameter file, describing unknown molecule) by either making it manually or having eLBOW program to generate it: phenix.elbow model.pdb --do-all --output=all_ligands this will ask eLBOW to inspect the model_new.pdb file, find all unknown items in it and create one cif file for them all_ligands.cif. Alternatively, one can specify a three-letters name for the unknown residue: phenix.elbow model.pdb --residue=MAN --output=man Once the cif file is created, the new run of phenix.refine will be: phenix.refine model.pdb data.pdb man.cif Consult eLBOW documentation for more details.

Having unknown to phenix.refine item in PDB file (novel ligand, etc...).

phenix.refine uses the CCP4 Monomer Library as the source of stereochemical information for building geometry restraints and reporting statistics.

If phenix.refine is unable to match an item in input PDB file against the Monomer Library it will stop with "Sorry" message explaining what to do and listing the problem atoms. If this happened, it is necessary to obtain a cif file (parameter file, describing unknown molecule) by either making it manually or having eLBOW program to generate it:

```
phenix.elbow model.pdb --do-all --output=all_ligands
```

this will ask eLBOW to inspect the model_new.pdb file, find all unknown items in it and create one cif file for them all_ligands.cif. Alternatively, one can specify a three-letters name for the unknown residue:

```
phenix.elbow model.pdb --residue=MAN --output=man
```

Once the cif file is created, the new run of phenix.refine will be:

```
phenix.refine model.pdb data.pdb man.cif
```

Consult eLBOW documentation for more details.

Useful options

Changing the number of refinement cycles and minimizer iterations

```
% phenix.refine data.hkl model.pdb main.number_of_macro_cycles=5 \
 main.max_number_of_iterations=20
```

Parallelizing for multi-core systems

```
% phenix.refine data.hkl model.pdb optimize_xyz_weight=True nproc=4
```

The nproc parameter instructures phenix.refine to use multiple processors for several highly parallel routines. Currently this applies to the following optional procedures:

Automatic TLS identification (tls.find_automatically=True) Bulk solvent mask optimization (optimize_mask=True) XYZ restraints weight optimization (optimize_xyz_weight=True) ADP restraints weight optimization (optimize_adp_weight=True)

- Automatic TLS identification (tls.find_automatically=True)
- Bulk solvent mask optimization (optimize_mask=True)
- XYZ restraints weight optimization (optimize_xyz_weight=True)
- ADP restraints weight optimization (optimize_adp_weight=True)

When used with the default settings, nproc will have a minimal effect on overall runtime, but when the optimization grid searches are enabled, a speedup of 4-5x is possible. Values of nproc above 18 are unlikely to yield further speed improvement.

Note: this parallelization method is not compatible with OpenMP, and is limited to Mac and Linux systems. (It is, however, available in the Phenix GUI.)

Creating R-free flags (if not present in the input reflection files)

```
% phenix.refine data.hkl model.pdb xray_data.r_free_flags.generate=True
```

It is important to understand that reflections selected for test set must be never used in any refinement of any parameters. If the newly selected test reflections were used in refinement before then the corresponding R-free statistics will be wrong. In such case "refinement memory" removal procedure must be applied to recover proper statistics.

To change the default maximal number of test flags to be generated and the fraction:

```
% phenix.refine data.hkl model.pdb xray_data.r_free_flags.generate=True \
 xray_data.r_free_flags.fraction=0.05 xray_data.r_free_flags.max_free=500
```

Specify the name for output files

```
% phenix.refine data.hkl model.pdb output.prefix=lysozyme
```

Choosing data arrays from input file

If input file(s) contains multiple sets of reflection data and/or free-r flags so that the program cannot unambiguously choose them for refinement, it will stop and ask to selected the desired arays. Example below shows how to specify labels for the data and flags:

```
% phenix.refine data.hkl model.pdb miller_array.labels.name="IOBS" \
 miller_array.labels.name="R-free-flags"
```

Reflection output

At the end of refinement a file with Fobs, Fmodel, Fcalc, Fmask, FOM, R-free_flags can be written out (in MTZ format):

```
% phenix.refine data.hkl model.pdb export_final_f_model=true
```

Note: Fmodel is the total model structure factor including all scales:

Fmodel = scale_k1 * exp(-h*U_overall*ht) * (Fcalc + k_sol * exp(-B_sol*s^2) * Fmask)

Setting the resolution range for the refinement

```
% phenix.refine data.hkl model.pdb xray_data.low_resolution=15.0 xray_data.high_resolution=2.0
```

Bulk solvent correction and anisotropic scaling

By default phenix.refine always starts with bulk solvent modeling and anisotropic scaling. Here is the list of command that may be of use in some cases:

- Perform bulk-solvent modeling and anisotropic scaling only: % phenix.refine data.hkl model.pdb strategy=none

Perform bulk-solvent modeling and anisotropic scaling only:

```
% phenix.refine data.hkl model.pdb strategy=none
```

- Bulk-solvent modeling only (no anisotropic scaling): % phenix.refine data.hkl model.pdb strategy=none bulk_solvent_and_scale.anisotropic_scaling=false

Bulk-solvent modeling only (no anisotropic scaling):

```
% phenix.refine data.hkl model.pdb strategy=none bulk_solvent_and_scale.anisotropic_scaling=false
```

- Anisotropic scaling only (no bulk-solvent modeling): % phenix.refine data.hkl model.pdb strategy=none bulk_solvent_and_scale.bulk_solvent=false

Anisotropic scaling only (no bulk-solvent modeling):

```
% phenix.refine data.hkl model.pdb strategy=none bulk_solvent_and_scale.bulk_solvent=false
```

- Turn off bulk-solvent modeling and anisotropic scaling: % phenix.refine data.hkl model.pdb main.bulk_solvent_and_scale=false

Turn off bulk-solvent modeling and anisotropic scaling:

```
% phenix.refine data.hkl model.pdb main.bulk_solvent_and_scale=false
```

- Fixing bulk-solvent and anisotropic scale parameters to user defined values: % phenix.refine data.hkl model.pdb bulk_solvent_and_scale.params where bulk_solvent_and_scale.params is the file containing these lines: refinement { bulk_solvent_and_scale { k_sol_b_sol_grid_search = False minimization_k_sol_b_sol = False minimization_b_cart = False fix_k_sol = 0.45 fix_b_sol = 56.0 fix_b_cart { b11 = 1.2 b22 = 2.3 b33 = 3.6 b12 = 0.0 b13 = 0.0 b23 = 0.0 } } }

Fixing bulk-solvent and anisotropic scale parameters to user defined values:

```
% phenix.refine data.hkl model.pdb bulk_solvent_and_scale.params
```

where bulk_solvent_and_scale.params is the file containing these lines:

```
refinement {
 bulk_solvent_and_scale {
 k_sol_b_sol_grid_search = False
 minimization_k_sol_b_sol = False
 minimization_b_cart = False
 fix_k_sol = 0.45
 fix_b_sol = 56.0
 fix_b_cart {
 b11 = 1.2
 b22 = 2.3
 b33 = 3.6
 b12 = 0.0
 b13 = 0.0
 b23 = 0.0
 }
 }
}
```

- Mask parameters: Bulk solvent modeling involves the mask calculation. There are three principal parameters controlling it: solvent_radius, shrink_truncation_radius and grid_step_factor. Normally, these parameters are not supposed to be changed but can be changed: % phenix.refine data.hkl model.pdb refinement.mask.solvent_radius=1.0 \ refinement.mask.shrink_truncation_radius=1.0 refinement.mask.grid_step_factor=3 If one wants to gain some more drop in R-factors (somewhere between 0.0 and 1.0%) it is possible to run fairly time consuming (depending on structure size and resolution) procedure of mask parameters optimization: % phenix.refine data.hkl model.pdb optimize_mask=true This will perform the grid search for solvent_radius and shrink_truncation_radius and select the values giving the best R-factor.

Mask parameters:

Bulk solvent modeling involves the mask calculation. There are three principal parameters controlling it: solvent_radius, shrink_truncation_radius and grid_step_factor. Normally, these parameters are not supposed to be changed but can be changed:

```
% phenix.refine data.hkl model.pdb refinement.mask.solvent_radius=1.0 \
 refinement.mask.shrink_truncation_radius=1.0 refinement.mask.grid_step_factor=3
```

If one wants to gain some more drop in R-factors (somewhere between 0.0 and 1.0%) it is possible to run fairly time consuming (depending on structure size and resolution) procedure of mask parameters optimization:

```
% phenix.refine data.hkl model.pdb optimize_mask=true
```

This will perform the grid search for solvent_radius and shrink_truncation_radius and select the values giving the best R-factor.

By default phenix.refine adds isotropic component of overall anisotropic scale matrix to atomic B-factors, leaving the trace of overall anisotropic scale matrix equals to zero. This is the reason why one can observe the ADP changed even though the only anisotropic scaling was done and no ADP refinement performed.

Default refinement with user specified X-ray target function

- Refinement with least-squares target: % phenix.refine data.hkl model.pdb main.target=ls

Refinement with least-squares target:

```
% phenix.refine data.hkl model.pdb main.target=ls
```

- Refinement with maximum-likelihood target (default): % phenix.refine data.hkl model.pdb main.target=ml

Refinement with maximum-likelihood target (default):

```
% phenix.refine data.hkl model.pdb main.target=ml
```

- Refinement with phased maximum-likelihood target: % phenix.refine data.hkl model.pdb main.target=mlhl If phenix.refine finds Hendrickson-Lattman coefficients in input reflection file, it will automatically switch to mlhl target. To disable this: % phenix.refine data.hkl model.pdb main.use_experimental_phases=false

Refinement with phased maximum-likelihood target:

```
% phenix.refine data.hkl model.pdb main.target=mlhl
```

If phenix.refine finds Hendrickson-Lattman coefficients in input reflection file, it will automatically switch to mlhl target. To disable this:

```
% phenix.refine data.hkl model.pdb main.use_experimental_phases=false
```

Modifying the initial model before refinement starts

phenix.refine offers several options to modify input model before refinement starts:

- shaking of coordinates (adding a random shift to coordinates): % phenix.refine data.hkl model.pdb sites.shake=0.3

shaking of coordinates (adding a random shift to coordinates):

```
% phenix.refine data.hkl model.pdb sites.shake=0.3
```

- rotation-translation shift of coordinates: % phenix.refine data.hkl model.pdb sites.rotate="1 2 3" sites.translate="4 5 6"

rotation-translation shift of coordinates:

```
% phenix.refine data.hkl model.pdb sites.rotate="1 2 3" sites.translate="4 5 6"
```

- shaking of occupancies: % phenix.refine data.hkl model.pdb occupancies.randomize=true

shaking of occupancies:

```
% phenix.refine data.hkl model.pdb occupancies.randomize=true
```

- shaking of ADP: % phenix.refine data.hkl model.pdb adp.randomize=true

shaking of ADP:

```
% phenix.refine data.hkl model.pdb adp.randomize=true
```

- shifting of ADP (adding a constant value): % phenix.refine data.hkl model.pdb adp.shift_b_iso=10.0

shifting of ADP (adding a constant value):

```
% phenix.refine data.hkl model.pdb adp.shift_b_iso=10.0
```

• scaling of ADP (multiplying by a constant value): % phenix.refine data.hkl model.pdb adp.scale_adp=0.5

scaling of ADP (multiplying by a constant value):

```
% phenix.refine data.hkl model.pdb adp.scale_adp=0.5
```

• setting a value to ADP: % phenix.refine data.hkl model.pdb adp.set_b_iso=25

setting a value to ADP:

```
% phenix.refine data.hkl model.pdb adp.set_b_iso=25
```

• converting to isotropic: % phenix.refine data.hkl model.pdb adp.convert_to_isotropic=true

converting to isotropic:

```
% phenix.refine data.hkl model.pdb adp.convert_to_isotropic=true
```

• converting to anisotropic: % phenix.refine data.hkl model.pdb adp.convert_to_anisotropic=true \ modify_start_model.selection="not element H" When converting atoms into anisotropic, it is important to make sure that hydrogens (if present in the model) are not converted into anisotropic.

converting to anisotropic:

```
% phenix.refine data.hkl model.pdb adp.convert_to_anisotropic=true \
 modify_start_model.selection="not element H"
```

When converting atoms into anisotropic, it is important to make sure that hydrogens (if present in the model) are not converted into anisotropic.

By default, the specified manipulations will be applied to all atoms. However, it is possible to apply them to only selected atoms:

```
% phenix.refine data.hkl model.pdb adp.set_b_iso=25 modify_start_model.selection="chain A"
```

To write out the modified model (without any refinement), add: main.number_of_macro_cycles=0, e.g.:

```
% phenix.refine data.hkl model.pdb adp.set_b_iso=25 \
 main.number_of_macro_cycles=0
```

All the commands listed above plus some more are available from phenix.pdbtools utility which in fact is used internally in phenix.refine to perform these manipulations. For more information on phenix.pdbtools type:

```
% phenix.pdbtools --help
```

Documentation on phenix.pdbtools is also available.

Refinement using FFT or direct structure factor calculation algorithm

```
% phenix.refine data.hkl model.pdb \
 structure_factors_and_gradients_accuracy.algorithm=fft
```

or:

```
% phenix.refine data.hkl model.pdb \
 structure_factors_and_gradients_accuracy.algorithm=direct
```

Ignoring test (free) flags in refinement

Sometimes one needs to use all reflections ("work" and "test") in the refinement; for example, at very low resolution where each single reflection counts, or at subatomic resolution where the risk of overfitting is very low. In the example below all the reflections are used in the refinement:

```
% phenix.refine data.hkl model.pdb xray_data.r_free_flags.ignore_r_free_flags=true
```

Note: 1) the corresponding statistics (R-factors, ...) will be identical for "work" and "test" sets; 2) it is still necessary to have test flags presented in input reflection file (or automatically generated by phenix.refine).

Using phenix.refine to calculate structure factors

The total structure factor used in phenix.refine nearly in all calculations is defined as:

```
Fmodel = scale_k1 * exp(-h*U_overall*ht) * (Fcalc + k_sol * exp(-B_sol*s^2) * Fmask)
```

- Calculate Fcalc from atomic model and output in MTZ file (no solvent modeling or scaling): % phenix.refine data.hkl model.pdb main.number_of_macro_cycles=0 \ main.bulk_solvent_and_scale=false export_final_f_model=true

Calculate Fcalc from atomic model and output in MTZ file (no solvent modeling or scaling):

```
% phenix.refine data.hkl model.pdb main.number_of_macro_cycles=0 \
 main.bulk_solvent_and_scale=false export_final_f_model=true
```

- Calculate Fcalc from atomic model including bulk solvent and all scales: % phenix.refine data.hkl model.pdb main.number_of_macro_cycles=1 \ strategy=none export_final_f_model=true

Calculate Fcalc from atomic model including bulk solvent and all scales:

```
% phenix.refine data.hkl model.pdb main.number_of_macro_cycles=1 \
 strategy=none export_final_f_model=true
```

- Resolution limits can be applied: % phenix.refine data.hkl model.pdb main.number_of_macro_cycles=1 \ strategy=none xray_data.low_resolution=15.0 xray_data.high_resolution=2.0

Resolution limits can be applied:

```
% phenix.refine data.hkl model.pdb main.number_of_macro_cycles=1 \
 strategy=none xray_data.low_resolution=15.0 xray_data.high_resolution=2.0
```

Note:

- The number of calculated structure factors will the same as the number of observed data (Fobs) provided in the input reflection files or less since resolution and sigma cutoffs may be applied to Fobs or some Fobs may be automatically removed by outliers detection procedure.
- The set of calculated structure factors has the same completeness as the set of provided Fobs.

Scattering factors

There are four choices for the scattering table to be used in phenix.refine:

- wk1995: Waasmaier & Kirfel table;
- it1992: International Crystallographic Tables (1992)
- n_gaussian: dynamic n-gaussian approximation
- neutron: table for neutron scattering

The default is n_gaussian. To switch to different table:

```
% phenix.refine data.hkl model.pdb main.scattering_table=neutron
```

Suppressing the output of certain files

The following command will tell phenix,refine to not write .eff, .geo, .def, maps and map coefficients files:

```
% phenix.refine data.hkl model.pdb write_eff_file=false write_geo_file=false \
 write_def_file=false write_maps=false write_map_coefficients=false
```

The only output will be: .log and .pdb files.

Random seed

To change random seed:

```
% phenix.refine data.hkl model.pdb main.random_seed=7112384
```

The results of certain refinement protocols, such as restrained refinement of coordinates (with SA or LBFGS minimization), are sensitive to the random seed. This is because: 1) for SA the refinement starts with random assignment of velocities to atoms; 2) the X-ray/geometry target weight calculation involves model shaking with some Cartesian dynamics. As result, running such refinement jobs with exactly the same parameters but different random seeds will produce different refinement statistics. The author's experience includes the case where the difference in R-factors was about 2.0% between two SA runs.

Also, this opens a possibility to perform multi-start SA refinement to create an ensemble of slightly different models in average but sometimes containing significant variations in certain parts.

Electron density maps

By default phenix.refine outputs two likelihood-weighted maps: 2mFo-DFc and mFo-DFc. These are the map coefficients generated for use in Coot. The user can also choose between likelihood-weighted or regular maps with any specified coefficients, for example: 2mFo-DFc, 2.7mFo-1.3DFc, Fo-Fc, 3Fo-2Fc. Any number of maps can be created. Optionally, the result can be output as binary CCP4 format. The example below illustrates the main options:

```
% phenix.refine data.hkl model.pdb map.params write_maps=true
```

where map.params contains:

```
refinement {
 electron_density_maps {
 map_coefficients {
 mtz_label_amplitudes = 2FOFCWT
 mtz_label_phases = PH2FOFCWT
 map_type = 2mFo-DFc
 }
 map_coefficients {
 mtz_label_amplitudes = FOFCWT
 mtz_label_phases = PHFOFCWT
 map_type = mFo-DFc
 }
 map_coefficients {
 mtz_label_amplitudes = 3FO2FCWT
 mtz_label_phases = PH3FO2FCWT
 map_type = 3Fo-2Fc
 }
 map {
 map_type = 2mFo-DFc
 grid_resolution_factor = 1/4.
 region = *selection cell
 atom_selection = chain A and resseq 1
 }
 }
}
```

This will output one file with map coefficients for 2mFo-DFc, mFo-DFc and 3Fo-2Fc maps, and one X-plor formatted file containing 2mFo-DFc map computed around residue 1 in chain A. The map finesse will be (data resolution)*grid_resolution_factor. If atom_selection is set to None or all then map will be computed for all atoms.

Refining with anomalous data (or what phenix.refine does with Fobs+ and Fobs-).

The way phenix.refine uses Fobs+ and Fobs- is controlled by xray_data.force_anomalous_flag_to_be_equal_to parameter.

Here are 3 possibilities:

- Default behavior: phenix.refine will use all Fobs: Fobs+ and Fobs- as independent reflections: % phenix.refine model.pdb data_anom.hkl

Default behavior: phenix.refine will use all Fobs: Fobs+ and Fobs- as independent reflections:

```
% phenix.refine model.pdb data_anom.hkl
```

- phenix.refine will generate missing Bijvoet mates and use all Fobs+ and Fobs- as independent reflections if: % phenix.refine model.pdb data_anom.hkl xray_data.force_anomalous_flag_to_be_equal_to=true

phenix.refine will generate missing Bijvoet mates and use all Fobs+ and Fobs- as independent reflections if:

```
% phenix.refine model.pdb data_anom.hkl xray_data.force_anomalous_flag_to_be_equal_to=true
```

- phenix.refine will merge Fobs+ and Fobs-, that is instead of two separate Fobs+ and Fobs- it will use one value F_mean = (Fobs+ + Fobs-)/2 if: % phenix.refine model.pdb data_anom.hkl xray_data.force_anomalous_flag_to_be_equal_to=false

phenix.refine will merge Fobs+ and Fobs-, that is instead of two separate Fobs+ and Fobs- it will use one value F_mean = (Fobs+ + Fobs-)/2 if:

```
% phenix.refine model.pdb data_anom.hkl xray_data.force_anomalous_flag_to_be_equal_to=false
```

Look this documentation to see how to use and refine f' and f''.

Rejecting reflections by sigma

Reflections can be rejected by sigma cutoff criterion applied to amplitudes Fobs <= sigma_fobs_rejection_criterion * sigma(Fobs):

```
% phenix.refine model.pdb data_anom.hkl xray_data.sigma_fobs_rejection_criterion=2
```

or/and intensities Iobs <= sigma_iobs_rejection_criterion * sigma(Iobs):

```
% phenix.refine model.pdb data_anom.hkl xray_data.sigma_iobs_rejection_criterion=2
```

Internally, phenix.refine uses amplitudes. If both sigma_fobs_rejection_criterion and sigma_iobs_rejection_criterion are given as non-zero values, then both criteria will be applied: first to Iobs, then to Fobs (after truncated Iobs got converted to Fobs):

```
% phenix.refine model.pdb data_anom.hkl xray_data.sigma_fobs_rejection_criterion=2 \
 xray_data.sigma_iobs_rejection_criterion=2
```

By default, both sigma_fobs_rejection_criterion and sigma_iobs_rejection_criterion are set to zero (no reflections rejected) and, unless strongly motivated, we encourage to not change these values. If amplitudes provided at input then sigma_fobs_rejection_criterion is ignored.

Developer's tools

phenix.refine offers a broad functionality for experimenting that may not be useful in everyday practice but handy for testing ideas.

Substitute input Fobs with calculated Fcalc, shake model and refine it

Instead of using Fobs from input data file one can ask phenix.refine to use the calculated structure factors Fcalc using the input model. Obviously, the R-factors will be zero throughout the refinement. One can also shake various model parameters (see this document for details), then refinement will start with some bad statistics (big R-factors at least) and hopefully will converge to unmodified start model (if not shaken too well). Also it's possible to simulate Flat bulk solvent model contribution and anisotropic scaling: % phenix.refine model.pdb data.hkl experiment.params where experiment.params contains the following: refinement { main { fake_f_obs = True } modify_start_model { selection = "chain A" sites { shake = 0.5 } } fake_f_obs { fmodel { k_sol = 0.35 b_sol = 45.0 b_cart = 1.25 3.78 1.25 0.0 0.0 0.0 scale = 358.0 } } } In this example, the input Fobs will be substituted with the same amount of Fcalc (absolute values of Fcalc), then the coordinates of the structure will be shaken to achieve rmsd=0.5 and finally the default run of refinement will be done. The bulk solvent and anisotropic scale and overall scalar scales are also added to thus obtained Fcalc in accordance with Fmodel definition (see this document for definition of total structure factor, Fmodel). Expected refinement behavior: R-factors will drop from something big to zero.

Instead of using Fobs from input data file one can ask phenix.refine to use the calculated structure factors Fcalc using the input model. Obviously, the R-factors will be zero throughout the refinement. One can also shake various model parameters (see this document for details), then refinement will start with some bad statistics (big R-factors at least) and hopefully will converge to unmodified start model (if not shaken too well).

Also it's possible to simulate Flat bulk solvent model contribution and anisotropic scaling:

```
% phenix.refine model.pdb data.hkl experiment.params
```

where experiment.params contains the following:

```
refinement {
 main {
 fake_f_obs = True
 }
 modify_start_model {
 selection = "chain A"
 sites {
 shake = 0.5
 }
 }
 fake_f_obs {
 fmodel {
 k_sol = 0.35
 b_sol = 45.0
 b_cart = 1.25 3.78 1.25 0.0 0.0 0.0
 scale = 358.0
 }
 }
}
```

In this example, the input Fobs will be substituted with the same amount of Fcalc (absolute values of Fcalc), then the coordinates of the structure will be shaken to achieve rmsd=0.5 and finally the default run of refinement will be done. The bulk solvent and anisotropic scale and overall scalar scales are also added to thus obtained Fcalc in accordance with Fmodel definition (see this document for definition of total structure factor, Fmodel). Expected refinement behavior: R-factors will drop from something big to zero.

Automatic linking

phenix.refine has an automatic option for generating links within the same chain. It will look for carbohydrate links, both within the sugar polymer and linking to the protein. Covalent bonded ligands can also be linked with this option.

automatic_linking.link_all = True

There are a number of parameters that allow the tailoring of the various bond class cutoffs.

```
automatic_linking {
 metal_coordination_cutoff = 3.5
 amino_acid_bond_cutoff = 1.9
 rna_dna_bond_cutoff = 3.5
 inter_residue_bond_cutoff = 2.5
 carbohydrate_bond_cutoff = 1.99
}
```

There are also options within the automatic_linking scope for various bond length cutoffs.

CIF modifications and links

phenix.refine uses the CCP4 monomer library to build geometry restraints (bond, angle, dihedral, chirality and planarity restraints). The CCP4 monomer library comes with a set of "modifications" and "links" which are defined in the file mon_lib_list.cif. Some of these are used automatically when phenix.refine builds the geometry restraints (e.g. the peptide and RNA/DNA chain links). Other links and modifications have to be applied manually, e.g. (cif_modification.params file):

```
refinement.pdb_interpretation.apply_cif_modification {
 data_mod = 5pho
 residue_selection = resname GUA and name O5T
}
```

Here a custom 5pho modification is applied to all GUA residues with an O5T atom. I.e. the modification can be applied to multiple residues with a single apply_cif_modification block. The CIF modification is supplied as a separate file on the phenix.refine command line, e.g. (data_mod_5pho.cif file):

```
data_mod_5pho
#
loop_
_chem_mod_atom.mod_id
_chem_mod_atom.function
_chem_mod_atom.atom_id
_chem_mod_atom.new_atom_id
_chem_mod_atom.new_type_symbol
_chem_mod_atom.new_type_energy
_chem_mod_atom.new_partial_charge
 5pho add . O5T O OH .
loop_
_chem_mod_bond.mod_id
_chem_mod_bond.function
_chem_mod_bond.atom_id_1
_chem_mod_bond.atom_id_2
_chem_mod_bond.new_type
_chem_mod_bond.new_value_dist
_chem_mod_bond.new_value_dist_esd
 5pho add O5T P coval 1.520 0.020
```

The whole command will be:

```
% phenix.refine model_o5t.pdb data.hkl data_mod_5pho.cif cif_modification.params
```

Similarly, a link can be applied like this (cif_link.params file):

```
refinement.pdb_interpretation.apply_cif_link {
 data_link = MAN-THR
 residue_selection_1 = chain X and resname MAN and resid 900
 residue_selection_2 = chain X and resname THR and resid 42
}

% phenix.refine model.pdb data.hkl cif_link.params
```

The residue selections for links must select exactly one residue each. The MAN-THR link is pre-defined in mon_lib_list.cif. Custom links can be supplied as additional files on the phenix.refine command line. See mon_lib_list.cif for examples. The full path to this file can be obtained with the command:

```
% phenix.where_mon_lib_list_cif
```

All apply_cif_modification and apply_cif_link definitions will be included into the .def files. I.e. it is not necessary to specify the definitions again if further refinement runs are started with .def files.

Note that all LINK, SSBOND, HYDBND, SLTBRG and CISPEP records in the input PDB files are ignored.

Definition of custom bonds and angles

Most geometry restraints (bonds, angles, etc.) are generated automatically based on the CCP4 monomer library. Additional custom bond and angle restraints, e.g. between protein and a ligand or ion, can be specified in this way:

```
refinement.geometry_restraints.edits {
 zn_selection = chain X and resname ZN and resid 200 and name ZN
 his117_selection = chain X and resname HIS and resid 117 and name NE2
 asp130_selection = chain X and resname ASP and resid 130 and name OD1
 bond {
 action = *add
 atom_selection_1 = $zn_selection
 atom_selection_2 = $his117_selection
 symmetry_operation = None
 distance_ideal = 2.1
 sigma = 0.02
 slack = None
 }
 bond {
 action = *add
 atom_selection_1 = $zn_selection
 atom_selection_2 = $asp130_selection
 symmetry_operation = None
 distance_ideal = 2.1
 sigma = 0.02
 slack = None
 }
 angle {
 action = *add
 atom_selection_1 = $his117_selection
 atom_selection_2 = $zn_selection
 atom_selection_3 = $asp130_selection
 angle_ideal = 109.47
 sigma = 5
 }
}
```

The atom selections must uniquely select a single atom. Save the geometry_restraints.edits to a file and specify the file name as an additional argument when running phenix.refine for the first time. For example:

```
% phenix.refine model.pdb data.hkl restraints_edits.params
```

The edits will be included into the .def files. I.e. it is not necessary to manually specify them again if further refinement runs are started with .def files.

For bonds to symmetry copies, specify the symmetry operation in xyz notation, for example:

```
symmetry_operation = -x-1/2,y-1/2,-z+1/2
```

To obtain the symmetry_operation, either use Coot (turn on drawing on symmetry copies, then click on the copy and look for the symmetry operation in the status bar), or run this command:

```
iotbx.show_distances your.pdb &gt; all_distances
```

This will produce a potentially long all_distances file, but if you search for sym= there will probably only be a few matches from which it is easy to pick the one you are interested in, based on the pdb atom labels.

The bond.slack parameter above can be used to disable a bond restraint within the slack tolerance around distance_ideal. This is useful for hydrogen bond restraints, or when refining with very high-resolution data (e.g. better than 1 A). The bond restraint is activated only if the discrepancy between the model bond distance and distance_ideal is greater than the slack value. The slack is subtracted from the discrepancy. The resulting potential is called a "square-well potential" by some authors. The formula for the contribution to the refinement target function is:

```
weight * delta_slack**2
```

with:

```
delta_slack = sign(delta) * max(0, (abs(delta) - slack))
delta = distance_ideal - distance_model
weight = 1 / sigma**2
```

The slack value must be greater than or equal to zero (it can also be None, which is equivalent to zero in this case).

Making sure atom charge is used in refinement

To use atom charge in refinement it must be present in input PDB file as in this example:

```
HETATM 3241 SN SN C 3 5.000 5.000 5.000 0.25 41.55 SN4+
```

To verify it was actually recognized and used by phenix.refine find lines like these in the log file:

```
(...)
Number of scattering types: 4
 Type Number sf(0) Gaussians
 Sn4+ 1 45.79 1
(...)
```

What is a reasonable gap between R and Rfree?

Normally the value of the Free R value (R-free, R value for the reflections in the test set) will be in the range of 0.01 to 0.03 higher than the working R value (R-work, R value for the reflections used in refinement). If the Rfree value is much higher than Rwork (more than about 0.05), this may indicate overfitting of the data.

Interpreting the R values at the end of the phenix refinement log file

At the end of your phenix refinement log file you will find a table of R values like this:

```
start: r(all,work,free)=0.4314 0.4232 0.4972 n_refl.: 7217
 re-set all scales: r(all,work,free)=0.4314 0.4232 0.4972 n_refl.: 7217
 remove outliers: r(all,work,free)=0.4314 0.4232 0.4972 n_refl.: 7217
 overall B=0.00 to atoms: r(all,work,free)=0.4314 0.4232 0.4972 n_refl.: 7217
bulk-solvent and scaling: r(all,work,free)=0.3404 0.3341 0.3962 n_refl.: 7217
 remove outliers: r(all,work,free)=0.3404 0.3341 0.3962 n_refl.: 7217
```

The way to interpret these R values is that each line contains R values for all reflections, for work reflections only, and for free (test set) reflections only, at a certain stage of processing.

The first line, labeled start, corresponds to values before applying any scaling. The next lines correspond to resetting scale values, removing outliers, applying B values, applying bulk solvent and scaling, and finall, removing outliers.

Your final R values are the ones in the last line, labeled remove outliers.

What to do next after running refinement with PhenixRefine

The next step after refinement with PhenixRefine is usually validation with Comprehensive Validation. A common step after successful refinement is looking for ligands with LigandFit. A common step after refinement that yields some non-optimal validation metrics is rebuilding manually with Coot or automatically with AutoBuild.

After refinement you will want to examine your log file to make sure that there are no unusual circumstances and to check that you have obtained reasonable values of R and Rfree and reasonable values of all validation criteria.

Depositing refined structure with PDB

phenix.refine reports a comprehensive statistics in PDB file header of refined model. This statistics consists of two parts: the first (upper, formatted with REMARK record) part is relevant to the current refinement run and contains the information about input data and model files, time stamp, start and final R-factors, refinement statistics from macro-cycle to macro-cycle, etc. The second (lower, formatted with REMARK 3 record) part is abstracted from a particular refinement run (no intermediate statistics, time, no file names, etc.). This part is supposed to go in PDB and the first part should be removed manually.

Feedback, more information

- For help or bugs write to: help@phenix-online.org

For help or bugs write to: help@phenix-online.org

- Questions: phenixbb@phenix-online.org

Questions: phenixbb@phenix-online.org

- More information: www.phenix-online.org or type: phenix.about

More information: www.phenix-online.org or type:

```
phenix.about
```


List of all available keywords

refinementScope of parameters for structure refinement with phenix.refinedry_run = False Read and process inputs and quit (no refinement)custom_scattering_factors = None Use custom scattering factors and replaces default values entirelycrystal_symmetryScope of space group and unit cell parametersunit_cell = None space_group = None inputScope of input file names, labels, processing directionssymmetry_safety_check = *error warning Check for consistency of crystal symmetry from model and data filesoutputScope for output filesjob_title = None Job title in PHENIX GUI, not used on command linewrite_eff_file = True write_geo_file = True write_final_geo_file = False write_def_file = True write_model_cif_file = True write_final_pdb_file = True write_mtz_file = True write_files = True write_reflection_cif_file = False export_final_f_model = False Write Fobs, Fmodel, various scales and more to MTZ filewrite_maps = False write_map_coefficients = True write_map_coefficients_only = False pickle_fmodel = False Dump final fmodel object into a pickle file.n_resolution_bins = None Sets the number of bins used for resolution-dependent statistics in output files and the Phenix GUI. If None, the binning will be determined automatically.electron_density_mapsElectron density maps calculation parametersapply_default_maps = None map_coefficientsmap_type = None format = *mtz phs mtz_label_amplitudes = None mtz_label_phases = None fill_missing_f_obs = False acentrics_scale

= 2.0 Scale terms corresponding to acentric reflections (residual maps only: k==n)centrics_pre_scale = 1.0 Centric reflections, k!=n and k*n != 0: max(k-centrics_pre_scale,0)*Fo-max(n-centrics_pre_scale,0)*Fcsharpening = False Apply B-factor sharpeningsharpening_b_factor = None Optional sharpening B-factor valueexclude_free_r_reflections = False Exclude free-R selected reflections from output map coefficientsisotropize = True devcomplete_set_up_to_d_min = False aply_same_incompleteness_to_complete_set_at = randomly low high mapmap_type = None format = xplor *ccp4 file_name = None fill_missing_f_obs = False grid_resolution_factor = 1/4. scale = *sigma volume region = *selection cell atom_selection = None atom_selection_buffer = 3 acentrics_scale = 2.0 Scale terms corresponding to acentric reflections (residual maps only: k==n)centrics_pre_scale = 1.0 Centric reflections, k!=n and k*n != 0: max(k-centrics_pre_scale,0)*Fo-max(n-centrics_pre_scale,0)*Fcsharpening = False Apply B-factor sharpeningsharpening_b_factor = None Optional sharpening B-factor valueexclude_free_r_reflections = False Exclude free-R selected reflections from map calculationisotropize = True refineScope of refinement flags (=flags defining what to refine) and atom selections (=atoms to be refined)strategy = *individual_sites *individual_sites_real_space rigid_body *individual_adp group_adp tls *occupancies group_anomalous den Atomic parameters to be refinedsitesScope of atom selections for coordinates refinementindividual = None Atom selections for individual atomstorsion_angles = None Atom selections for Torsion Angle Refinement and Dynamicsrigid_body = None Atom selections for rigid groupsadpScope of atom selections for ADP (Atomic Displacement Parameters) refinementgroup_adp_refinement_mode = *one_adp_group_per_residue two_adp_groups_per_residue group_selection Select one of three available modes for group B-factors refinement. For two groups per residue, the groups will be main-chain and side-chain atoms. Provide selections for groups if group_selection is chosen.group = None One isotropic ADP for group of selected here atoms will be refinedtls = None Selection(s) for TLS group(s)individualScope of atom selections for refinement of individual ADPisotropic = None Selections for atoms to be refinement with isotropic ADPanisotropic = None Selections for atoms to be refinement with anisotropic ADPoccupanciesScope of atom selections for occupancy refinementindividual = None Selection(s) for individual atoms. None is de...

- refinementScope of parameters for structure refinement with phenix.refinedry_run = False Read and process inputs and quit (no refinement)custom_scattering_factors = None Use custom scattering factors and replaces default values entirelycrystal_symmetryScope of space group and unit cell parametersunit_cell = None space_group = None inputScope of input file names, labels, processing directionssymmetry_safety_check = *error warning Check for consistency of crystal symmetry from model and data filesoutputScope for output filesjob_title = None Job title in PHENIX GUI, not used on command linewrite_eff_file = True write_geo_file = True write_final_geo_file = False write_def_file = True write_model_cif_file = True write_final_pdb_file = True write_mtz_file = True write_files = True write_reflection_cif_file = False export_final_f_model = False Write Fobs, Fmodel, various scales and more to MTZ filewrite_maps = False write_map_coefficients = True write_map_coefficients_only = False pickle_fmodel = False Dump final fmodel object into a pickle file.n_resolution_bins = None Sets the number of bins used for resolution-dependent statistics in output files and the Phenix GUI. If None, the binning will be determined automatically.electron_density_mapsElectron density maps calculation parametersapply_default_maps = None map_coefficientsmap_type = None format = *mtz phs mtz_label_amplitudes = None mtz_label_phases = None fill_missing_f_obs = False acentrics_scale = 2.0 Scale terms corresponding to acentric reflections (residual maps only: k==n)centrics_pre_scale = 1.0 Centric reflections, k!=n and k*n != 0: max(k-centrics_pre_scale,0)*Fo-max(n-centrics_pre_scale,0)*Fcsharpening = False Apply B-factor sharpeningsharpening_b_factor = None Optional sharpening B-factor valueexclude_free_r_reflections = False Exclude free-R selected reflections from output map coefficientsisotropize = True devcomplete_set_up_to_d_min = False aply_same_incompleteness_to_complete_set_at = randomly low high mapmap_type = None format = xplor *ccp4 file_name = None fill_missing_f_obs = False grid_resolution_factor = 1/4. scale = *sigma volume region = *selection cell atom_selection = None atom_selection_buffer = 3 acentrics_scale = 2.0 Scale terms corresponding to acentric reflections (residual maps only: k==n)centrics_pre_scale = 1.0 Centric reflections, k!=n and k*n != 0: max(k-centrics_pre_scale,0)*Fo-max(n-centrics_pre_scale,0)*Fcsharpening = False Apply B-factor sharpeningsharpening_b_factor = None Optional sharpening B-factor valueexclude_free_r_reflections = False Exclude free-R selected reflections from map calculationisotropize = True refineScope of refinement flags (=flags defining what to refine) and atom selections (=atoms to be refined)strategy = *individual_sites *individual_sites_real_space rigid_body *individual_adp group_adp tls *occupancies group_anomalous den Atomic parameters to be refinedsitesScope of atom selections for coordinates refinementindividual = None Atom selections for individual atomstorsion_angles = None Atom selections for Torsion Angle Refinement and Dynamicsrigid_body = None Atom selections for rigid groupsadpScope of atom selections for ADP (Atomic Displacement Parameters) refinementgroup_adp_refinement_mode = *one_adp_group_per_residue two_adp_groups_per_residue group_selection Select one of three available modes for group B-factors refinement. For two groups per residue, the groups will be main-chain and side-chain atoms. Provide selections for groups if group_selection is chosen.group = None One isotropic ADP for group of selected here atoms will be refinedtls = None Selection(s) for TLS group(s)individualScope of atom selections for refinement of individual ADPisotropic = None Selections for atoms to be refinement with isotropic ADPanisotropic = None Selections for atoms to be refinement with anisotropic ADPoccupanciesScope of atom selections for occupancy refinementindividual = None Selection(s) for individual atoms. None is de...

- dry_run = False Read and process inputs and quit (no refinement)
- custom_scattering_factors = None Use custom scattering factors and replaces default values entirely
- crystal_symmetryScope of space group and unit cell parametersunit_cell = None space_group = None
- unit_cell = None
- space_group = None
- inputScope of input file names, labels, processing directionssymmetry_safety_check = *error warning Check for consistency of crystal symmetry from model and data files
- symmetry_safety_check = *error warning Check for consistency of crystal symmetry from model and data files
- outputScope for output filesjob_title = None Job title in PHENIX GUI, not used on command linewrite_eff_file = True write_geo_file = True write_final_geo_file = False write_def_file = True write_model_cif_file = True write_final_pdb_file = True write_mtz_file = True write_files = True write_reflection_cif_file = False export_final_f_model = False Write Fobs, Fmodel, various scales and more to MTZ filewrite_maps = False write_map_coefficients = True write_map_coefficients_only = False pickle_fmodel = False Dump final fmodel object into a pickle file.n_resolution_bins = None Sets the number of bins used for resolution-dependent statistics in output files and the Phenix GUI. If None, the binning will be determined automatically.
- job_title = None Job title in PHENIX GUI, not used on command line
- write_eff_file = True
- write_geo_file = True
- write_final_geo_file = False
- write_def_file = True
- write_model_cif_file = True
- write_final_pdb_file = True

- write_mtz_file = True
- write_files = True
- write_reflection_cif_file = False
- export_final_f_model = False Write Fobs, Fmodel, various scales and more to MTZ file
- write_maps = False
- write_map_coefficients = True
- write_map_coefficients_only = False
- pickle_fmodel = False Dump final fmodel object into a pickle file.
- n_resolution_bins = None Sets the number of bins used for resolution-dependent statistics in output files and the Phenix GUI. If None, the binning will be determined automatically.
- electron_density_mapsElectron density maps calculation parametersapply_default_maps = None map_coefficientsmap_type = None format = *mtz phs mtz_label_amplitudes = None mtz_label_phases = None fill_missing_f_obs = False acentrics_scale = 2.0 Scale terms corresponding to acentric reflections (residual maps only: k==n)centrics_pre_scale = 1.0 Centric reflections, k!=n and k*n != 0: max(k-centrics_pre_scale,0)*Fo-max(n-centrics_pre_scale,0)*Fcsharpening = False Apply B-factor sharpeningsharpening_b_factor = None Optional sharpening B-factor valueexclude_free_r_reflections = False Exclude free-R selected reflections from output map coefficientsisotropize = True devcomplete_set_up_to_d_min = False aply_same_incompleteness_to_complete_set_at = randomly low high mapmap_type = None format = xplor *ccp4 file_name = None fill_missing_f_obs = False grid_resolution_factor = 1/4. scale = *sigma volume region = *selection cell atom_selection = None atom_selection_buffer = 3 acentrics_scale = 2.0 Scale terms corresponding to acentric reflections (residual maps only: k==n)centrics_pre_scale = 1.0 Centric reflections, k!=n and k*n != 0: max(k-centrics_pre_scale,0)*Fo-max(n-centrics_pre_scale,0)*Fcsharpening = False Apply B-factor sharpeningsharpening_b_factor = None Optional sharpening B-factor valueexclude_free_r_reflections = False Exclude free-R selected reflections from map calculationisotropize = True
- apply_default_maps = None
- map_coefficientsmap_type = None format = *mtz phs mtz_label_amplitudes = None mtz_label_phases = None fill_missing_f_obs = False acentrics_scale = 2.0 Scale terms corresponding to acentric reflections (residual maps only: k==n)centrics_pre_scale = 1.0 Centric reflections, k!=n and k*n != 0: max(k-centrics_pre_scale,0)*Fo-max(n-centrics_pre_scale,0)*Fcsharpening = False Apply B-factor sharpeningsharpening_b_factor = None Optional sharpening B-factor valueexclude_free_r_reflections = False Exclude free-R selected reflections from output map coefficientsisotropize = True devcomplete_set_up_to_d_min = False aply_same_incompleteness_to_complete_set_at = randomly low high
- map_type = None
- format = *mtz phs
- mtz_label_amplitudes = None
- mtz_label_phases = None
- fill_missing_f_obs = False
- acentrics_scale = 2.0 Scale terms corresponding to acentric reflections (residual maps only: k==n)
- centrics_pre_scale = 1.0 Centric reflections, k!=n and k*n != 0: max(k-centrics_pre_scale,0)*Fo-max(n-centrics_pre_scale,0)*Fc
- sharpening = False Apply B-factor sharpening

- sharpening_b_factor = None Optional sharpening B-factor value
- exclude_free_r_reflections = False Exclude free-R selected reflections from output map coefficients
- isotropize = True
- devcomplete_set_up_to_d_min = False aply_same_incompleteness_to_complete_set_at = randomly low high
- complete_set_up_to_d_min = False
- aply_same_incompleteness_to_complete_set_at = randomly low high
- mapmap_type = None format = xplor *ccp4 file_name = None fill_missing_f_obs = False grid_resolution_factor = 1/4. scale = *sigma volume region = *selection cell atom_selection = None atom_selection_buffer = 3 acentrics_scale = 2.0 Scale terms corresponding to acentric reflections (residual maps only: k==n)centrics_pre_scale = 1.0 Centric reflections, k!=n and k*n != 0: max(k-centrics_pre_scale,0)*Fo-max(n-centrics_pre_scale,0)*Fcsharpening = False Apply B-factor sharpeningsharpening_b_factor = None Optional sharpening B-factor valueexclude_free_r_reflections = False Exclude free-R selected reflections from map calculationisotropize = True
- map_type = None
- format = xplor *ccp4
- file_name = None
- fill_missing_f_obs = False
- grid_resolution_factor = 1/4.
- scale = *sigma volume
- region = *selection cell
- atom_selection = None
- atom_selection_buffer = 3
- acentrics_scale = 2.0 Scale terms corresponding to acentric reflections (residual maps only: k==n)
- centrics_pre_scale = 1.0 Centric reflections, k!=n and k*n != 0: max(k-centrics_pre_scale,0)*Fo-max(n-centrics_pre_scale,0)*Fc
- sharpening = False Apply B-factor sharpening
- sharpening_b_factor = None Optional sharpening B-factor value
- exclude_free_r_reflections = False Exclude free-R selected reflections from map calculation
- isotropize = True
- refineScope of refinement flags (=flags defining what to refine) and atom selections (=atoms to be refined)strategy = *individual_sites *individual_sites_real_space rigid_body *individual_adp group_adp tls *occupancies group_anomalous den Atomic parameters to be refinedsitesScope of atom selections for coordinates refinementindividual = None Atom selections for individual atomstorsion_angles = None Atom selections for Torsion Angle Refinement and Dynamicsrigid_body = None Atom selections for rigid groupsadpScope of atom selections for ADP (Atomic Displacement Parameters) refinementgroup_adp_refinement_mode = *one_adp_group_per_residue two_adp_groups_per_residue group_selection Select one of three available modes for group B-factors refinement. For two groups per residue, the groups will be main-chain and side-chain atoms. Provide selections for groups if group_selection is chosen.group = None One isotropic ADP for group of selected here atoms will be refinedtls = None Selection(s) for TLS group(s)individualScope of atom selections for refinement of individual ADPisotropic = None Selections for atoms to be refinement with isotropic ADPanisotropic =

None Selections for atoms to be refinement with anisotropic ADPoccupanciesScope of atom selections for occupancy refinementindividual = None Selection(s) for individual atoms. None is default which is to refine the individual occupancies for atoms in alternative conformations or for atoms with partial occupancies only.remove_selection = None Occupancies of selected atoms will not be refined (even though they might satisfy the default criteria for occupancy refinement).constrained_groupSelections to define constrained occupancies. If only one selection is provided then one occupancy factor per selected atoms will be refined and it will be constrained between predefined max and min values.selection = None Atom selection string.anomalous_scatterersgroupselection = None f_prime = 0 f_double_prime = 0 refine = *f_prime *f_double_prime

- strategy = *individual_sites *individual_sites_real_space rigid_body *individual_adp group_adp tls *occupancies group_anomalous den Atomic parameters to be refined
- sitesScope of atom selections for coordinates refinementindividual = None Atom selections for individual atomstorsion_angles = None Atom selections for Torsion Angle Refinement and Dynamicsrigid_body = None Atom selections for rigid groups
- individual = None Atom selections for individual atoms
- torsion_angles = None Atom selections for Torsion Angle Refinement and Dynamics
- rigid_body = None Atom selections for rigid groups
- adpScope of atom selections for ADP (Atomic Displacement Parameters) refinementgroup_adp_refinement_mode = *one_adp_group_per_residue two_adp_groups_per_residue group_selection Select one of three available modes for group B-factors refinement. For two groups per residue, the groups will be main-chain and side-chain atoms. Provide selections for groups if group_selection is chosen.group = None One isotropic ADP for group of selected here atoms will be refinedtls = None Selection(s) for TLS group(s)individualScope of atom selections for refinement of individual ADPisotropic = None Selections for atoms to be refinement with isotropic ADPanisotropic = None Selections for atoms to be refinement with anisotropic ADP
- group_adp_refinement_mode = *one_adp_group_per_residue two_adp_groups_per_residue group_selection Select one of three available modes for group B-factors refinement. For two groups per residue, the groups will be main-chain and side-chain atoms. Provide selections for groups if group_selection is chosen.
- group = None One isotropic ADP for group of selected here atoms will be refined
- tls = None Selection(s) for TLS group(s)
- individualScope of atom selections for refinement of individual ADPisotropic = None Selections for atoms to be refinement with isotropic ADPanisotropic = None Selections for atoms to be refinement with anisotropic ADP
- isotropic = None Selections for atoms to be refinement with isotropic ADP
- anisotropic = None Selections for atoms to be refinement with anisotropic ADP
- occupanciesScope of atom selections for occupancy refinementindividual = None Selection(s) for individual atoms. None is default which is to refine the individual occupancies for atoms in alternative conformations or for atoms with partial occupancies only.remove_selection = None Occupancies of selected atoms will not be refined (even though they might satisfy the default criteria for occupancy refinement).constrained_groupSelections to define constrained occupancies. If only one selection is provided then one occupancy factor per selected atoms will be refined and it will be constrained between predefined max and min values.selection = None Atom selection string.

• individual = None Selection(s) for individual atoms. None is default which is to refine the individual occupancies for atoms in alternative conformations or for atoms with partial occupancies only.

• remove_selection = None Occupancies of selected atoms will not be refined (even though they might satisfy the default criteria for occupancy refinement).

• constrained_groupSelections to define constrained occupancies. If only one selection is provided then one occupancy factor per selected atoms will be refined and it will be constrained between predefined max and min values.selection = None Atom selection string.

• selection = None Atom selection string.

• anomalous_scatterersgroupselection = None f_prime = 0 f_double_prime = 0 refine = *f_prime *f_double_prime

• groupselection = None f_prime = 0 f_double_prime = 0 refine = *f_prime *f_double_prime

• selection = None

• f_prime = 0

• f_double_prime = 0

• refine = *f_prime *f_double_prime

• mainScope for most common and frequently used parametersfit_altlocs_method = masking sampling show_process_info = False Track memory consumption. Works on Linux onlystop_for_unknowns = True Stop if undefined entity is ecountered (such as novel ligand)bulk_solvent_and_scale = True Do bulk solvent correction and anisotropic scalingapply_overall_isotropic_scale_to_adp = True flip_peptides = False nqh_flips = True simulated_annealing = False Do simulated annealingsimulated_annealing_torsion = False Do simulated annealing in torsion angle spaceordered_solvent = False Add (or/and remove) and refine ordered solvent molecules (water)real_space_refinement = *local global rotamer_restraints = False place_ions = None If enabled, Phenix will attempt to replace specific solvent atoms with elemental ions. A value of Auto will search for Mg, Ca, Zn, and Cl using relatively restrictive rules, but an explicit list of elements is recommended. See associated ion_placement parameters for more options.backbone_sample = False Sample CB (aka backrub sampling) as part of rotamer fittingfix_cablam = *off once first_half Try cablam idealization using parameters defined in cablam_idealization scope. once - before first macro-cycle, first_half - before first half of macro-cycles, rounding down (e.g. if 3 macro-cycles - do it before first only).update_ss = *off first_half every_macro_cycle Recalculate secondary structure restraints according to pdb_interpretation.secondary_structure scope first_half - before the first half of macro-cycles, rounding down (e.g. if 3 macro-cycles - do it before first only). every_macro_cycle - before every macro cycle.ias = False Build and use IAS (interatomic scatterers) model (at resolutions higher than approx. 0.9 A)number_of_macro_cycles = 3 Number of macro-cycles to be performedmax_number_of_iterations = 25 use_form_factor_weights = False tan_u_iso = False Use tan() reparameterization in ADP refinement (currently disabeled)use_geometry_restraints = True use_convergence_test = False Determine if refinement converged and stop thentarget = *auto ml mlhl ml_sad ls mli Choices for refinement targetmin_number_of_test_set_reflections_for_max_likelihood_target = 50 minimum number of test reflections required for use of ML targetmax_number_of_resolution_bins = 30 use_experimental_phases = None Use experimental phases if available. If true, the target function must be set to mlhl , and a file containing Hendrickson-Lattman coefficients must be supplied.random_seed = 2679941 Ransom seedscattering_table = wk1995 it1992 *n_gaussian electron neutron Choices of scattering table for structure factors calculationswavelength = None X-ray wavelength, currently used only when ion placement is enabled (and anomalous data are supplied).use_normalized_geometry_target = True target_weights_only = False Calculate target weights only and exit refinementuse_f_model_scaled =

False Use Fmodel structure factors multiplied by overall scale factor scale_k1max_d_min = 0.25 Highest allowable resolution limit for refinementfake_f_obs = False Substitute real experimental Fobs with those calculated from input model (scales and solvent can be added)optimize_mask = False Refine mask parameters (solvent_radius and shrink_truncation_radius)occupancy_max = 1.0 Maximum allowable occupancy of an atomoccupancy_min = 0.0 Minimum allowable occupancy of an atomstir = None Stepwise increase of resolution: start refinement at lower resolution and gradually proceed with higher resolutionshow_residual_map_peaks_and_holes = False Show highest peaks and deepest holes in residual_map.fft_vs_direct = False Check accuracy of approximations used in Fcalc calculationsswitch_to_isotropic_high_res_limit = 1.5 If the resolution is lower than this limit, all atoms selected for individual ADP refinement and not participating in TLS groups will be automatically converted to isotropic, whether or not ANISOU records are present in the input PDB file.find_and_add_hydrogens = False Find ...

- fit_altlocs_method = masking sampling
- show_process_info = False Track memory consumption. Works on Linux only
- stop_for_unknowns = True Stop if undefined entity is ecountered (such as novel ligand)
- bulk_solvent_and_scale = True Do bulk solvent correction and anisotropic scaling
- apply_overall_isotropic_scale_to_adp = True
- flip_peptides = False
- nqh_flips = True
- simulated_annealing = False Do simulated annealing
- simulated_annealing_torsion = False Do simulated annealing in torsion angle space
- ordered_solvent = False Add (or/and remove) and refine ordered solvent molecules (water)
- real_space_refinement = *local global
- rotamer_restraints = False
- place_ions = None If enabled, Phenix will attempt to replace specific solvent atoms with elemental ions. A value of Auto will search for Mg, Ca, Zn, and Cl using relatively restrictive rules, but an explicit list of elements is recommended. See associated ion_placement parameters for more options.
- backbone_sample = False Sample CB (aka backrub sampling) as part of rotamer fitting
- fix_cablam = *off once first_half Try cablam idealization using parameters defined in cablam_idealization scope. once - before first macro-cycle, first_half - before first half of macro-cycles, rounding down (e.g. if 3 macro-cycles - do it before first only).
- update_ss = *off first_half every_macro_cycle Recalculate secondary structure restraints according to pdb_interpretation.secondary_structure scope first_half - before the first half of macro-cycles, rounding down (e.g. if 3 macro-cycles - do it before first only). every_macro_cycle - before every macro cycle.
- ias = False Build and use IAS (interatomic scatterers) model (at resolutions higher than approx. 0.9 A)
- number_of_macro_cycles = 3 Number of macro-cycles to be performed
- max_number_of_iterations = 25
- use_form_factor_weights = False
- tan_u_iso = False Use tan() reparameterization in ADP refinement (currently disabeled)
- use_geometry_restraints = True
- use_convergence_test = False Determine if refinement converged and stop then

- target = *auto ml mlhl ml_sad ls mli Choices for refinement target
- min_number_of_test_set_reflections_for_max_likelihood_target = 50 minimum number of test reflections required for use of ML target
- max_number_of_resolution_bins = 30
- use_experimental_phases = None Use experimental phases if available. If true, the target function must be set to mlhl , and a file containing Hendrickson-Lattman coefficients must be supplied.
- random_seed = 2679941 Ransom seed
- scattering_table = wk1995 it1992 *n_gaussian electron neutron Choices of scattering table for structure factors calculations
- wavelength = None X-ray wavelength, currently used only when ion placement is enabled (and anomalous data are supplied).
- use_normalized_geometry_target = True
- target_weights_only = False Calculate target weights only and exit refinement
- use_f_model_scaled = False Use Fmodel structure factors multiplied by overall scale factor scale_k1
- max_d_min = 0.25 Highest allowable resolution limit for refinement
- fake_f_obs = False Substitute real experimental Fobs with those calculated from input model (scales and solvent can be added)
- optimize_mask = False Refine mask parameters (solvent_radius and shrink_truncation_radius)
- occupancy_max = 1.0 Maximum allowable occupancy of an atom
- occupancy_min = 0.0 Minimum allowable occupancy of an atom
- stir = None Stepwise increase of resolution: start refinement at lower resolution and gradually proceed with higher resolution
- show_residual_map_peaks_and_holes = False Show highest peaks and deepest holes in residual_map.
- fft_vs_direct = False Check accuracy of approximations used in Fcalc calculations
- switch_to_isotropic_high_res_limit = 1.5 If the resolution is lower than this limit, all atoms selected for individual ADP refinement and not participating in TLS groups will be automatically converted to isotropic, whether or not ANISOU records are present in the input PDB file.
- find_and_add_hydrogens = False Find H or D atoms using difference map and add them to the model. This option should be used if ultra-high resolution data is available or when refining againts neutron data.
- correct_special_position_tolerance = 1.0
- nproc = 1 Determines number of processor cores to use in parallel routines. Currently, this only applies to automatic TLS group identification.
- update_f_part1 = True
- tncs_correction = False Account for translational NCS (tNCS) in ML refinement
- ddrenable = False weighting_factor = 0.5 cc_minimum = 0. map_value_minimum = 0.5 bond_weighting = True angle_weighting = True trans_peptide = True
- enable = False
- weighting_factor = 0.5
- cc_minimum = 0.

- map_value_minimum = 0.5
- bond_weighting = True
- angle_weighting = True
- trans_peptide = True
- statistical_model_for_missing_atomssolvent_content = 0.5 map_type = *2mFo-DFc resolution_factor = 0.25 probability_mask = True diff_map_cutoff = 1.5 output_all_masks = False use_dm_map = False
- solvent_content = 0.5
- map_type = *2mFo-DFc
- resolution_factor = 0.25
- probability_mask = True
- diff_map_cutoff = 1.5
- output_all_masks = False
- use_dm_map = False
- modify_start_modelScope of parameters to modify initial model before refinementmodifyremove = None Selection for the atoms to be removedkeep = None Select atoms to keepput_into_box_with_buffer = None Move molecule into center of boxselection = None Selection for atoms to be modifiedflip_symmetric_amino_acids = False Flip symmetric amino acid side chainschange_of_basis = None Apply change-of-basis operator (e.g. reindexing operator) to the coordinates and symmetry. Example: a,c,b .renumber_residues = False Re-number residuesincrement_resseq = None Increment residue numbertruncate_to_polyala = False Truncate a model to poly-Ala.truncate_to_polygly = False Truncate a model to poly-Gly.remove_alt_confs = False Deletes atoms whose altloc identifier is not blank or A , and resets the occupancies of the remaining atoms to 1.0.always_keep_one_conformer = False Modifies behavior of remove_alt_confs so that residues with no conformer labeled blank or A are not deleted. Silent if remove_alt_confs is False.keep_occupancy = False Do not reset occupancy to 1 after removing altlocsaltloc_to_keep = None Modifies behavior of remove_alt_confs so that the altloc identifier to keep in addition to blank is this one (instead of A). Silent if remove_alt_confs is False.average_alt_confs = False Averages the coordinates of altloc atoms with 1.0 angstrom.set_chemical_element_simple_if_necessary = None Make a simple guess about what the chemical element is (based on atom name and the way how it is formatted) and write it into output file.set_seg_id_to_chain_id = False Sets the segID field to the chain ID (padded with spaces).clear_seg_id = False Erases the segID field.convert_semet_to_met = False convert_met_to_semet = False neutralize_scatterers = False remove_fraction = None random_seed = None Random seedmove_waters_last = False Transfer waters to the end of the model. Addresses some limitations of water picking in phenix.refine.renumber_and_move_waters = False Transfer waters to the end of respective chains. Renumber them so that numbering starts at the nearest 100 after the last non-water in the chain.adpScope of options to modify ADP of selected atomsatom_selection = None Selection for atoms to be modified. Overrides parent-level selection.randomize = False Randomize ADP within a certain rangeset_b_iso = None Set ADP of atoms to set_b_isoconvert_to_isotropic = False Convert atoms to isotropicconvert_to_anisotropic = False Convert atoms to anisotropicshift_b_iso = None Add shift_b_iso value to ADPscale_adp = None Multiply ADP by scale_adpsitesScope of options to modify coordinates of selected atomsatom_selection = None Selection for atoms to be modified. Overrides parent-level selection.shake = None Randomize coordinates with mean error value equal to shakeswitch_rotamers = max_distant min_distant exact_match fix_outliers translate = 0 0 0 Translational shift (x,y,z)rotate = 0 0 0 Rotational shift (x,y,z)euler_angle_convention = *xyz zyz Euler angles convention to be used for rotationoccupanciesScope of options to modify occupancies of selected

atomsatom_selection = None Selection for atoms to be modified. Overrides parent-level selection.randomize = False Randomize occupancies within a certain rangeset = None Set all or selected occupancies to given valuerotate_about_axisaxis = None angle = None atom_selection = None rename_chain_idRename chainsold_id = None new_id = None set_chargecharge_selection = None charge = None outputWrite out file with modified model (file name is defined in write_modified)file_name = None Default is the original file name with the file extension replaced by _modified.pdb .format = *pdb mmcif Choose the output format of coordinate file (PDB or mmCIF)

• modifyremove = None Selection for the atoms to be removedkeep = None Select atoms to keepput_into_box_with_buffer = None Move molecule into center of boxselection = None Selection for atoms to be modifiedflip_symmetric_amino_acids = False Flip symmetric amino acid side chainschange_of_basis = None Apply change-of-basis operator (e.g. reindexing operator) to the coordinates and symmetry. Example: a,c,b .renumber_residues = False Re-number residuesincrement_resseq = None Increment residue numbertruncate_to_polyala = False Truncate a model to poly-Ala.truncate_to_polygly = False Truncate a model to poly-Gly.remove_alt_confs = False Deletes atoms whose altloc identifier is not blank or A , and resets the occupancies of the remaining atoms to 1.0.always_keep_one_conformer = False Modifies behavior of remove_alt_confs so that residues with no conformer labeled blank or A are not deleted. Silent if remove_alt_confs is False.keep_occupancy = False Do not reset occupancy to 1 after removing altlocsaltloc_to_keep = None Modifies behavior of remove_alt_confs so that the altloc identifier to keep in addition to blank is this one (instead of A). Silent if remove_alt_confs is False.average_alt_confs = False Averages the coordinates of altloc atoms with 1.0 angstrom.set_chemical_element_simple_if_necessary = None Make a simple guess about what the chemical element is (based on atom name and the way how it is formatted) and write it into output file.set_seg_id_to_chain_id = False Sets the segID field to the chain ID (padded with spaces).clear_seg_id = False Erases the segID field.convert_semet_to_met = False convert_met_to_semet = False neutralize_scatterers = False remove_fraction = None random_seed = None Random seedmove_waters_last = False Transfer waters to the end of the model. Addresses some limitations of water picking in phenix.refine.renumber_and_move_waters = False Transfer waters to the end of respective chains. Renumber them so that numbering starts at the nearest 100 after the last non-water in the chain.adpScope of options to modify ADP of selected atomsatom_selection = None Selection for atoms to be modified. Overrides parent-level selection.randomize = False Randomize ADP within a certain rangeset_b_iso = None Set ADP of atoms to set_b_isoconvert_to_isotropic = False Convert atoms to isotropicconvert_to_anisotropic = False Convert atoms to anisotropicshift_b_iso = None Add shift_b_iso value to ADPscale_adp = None Multiply ADP by scale_adpsitesScope of options to modify coordinates of selected atomsatom_selection = None Selection for atoms to be modified. Overrides parent-level selection.shake = None Randomize coordinates with mean error value equal to shakeswitch_rotamers = max_distant min_distant exact_match fix_outliers translate = 0 0 0 Translational shift (x,y,z)rotate = 0 0 0 Rotational shift (x,y,z)euler_angle_convention = *xyz zyz Euler angles convention to be used for rotationoccupanciesScope of options to modify occupancies of selected atomsatom_selection = None Selection for atoms to be modified. Overrides parent-level selection.randomize = False Randomize occupancies within a certain rangeset = None Set all or selected occupancies to given valuerotate_about_axisaxis = None angle = None atom_selection = None rename_chain_idRename chainsold_id = None new_id = None set_chargecharge_selection = None charge = None

• remove = None Selection for the atoms to be removed

• keep = None Select atoms to keep

• put_into_box_with_buffer = None Move molecule into center of box

• selection = None Selection for atoms to be modified

- flip_symmetric_amino_acids = False Flip symmetric amino acid side chains
- change_of_basis = None Apply change-of-basis operator (e.g. reindexing operator) to the coordinates and symmetry. Example: a,c,b .
- renumber_residues = False Re-number residues
- increment_resseq = None Increment residue number
- truncate_to_polyala = False Truncate a model to poly-Ala.
- truncate_to_polygly = False Truncate a model to poly-Gly.
- remove_alt_confs = False Deletes atoms whose altloc identifier is not blank or A , and resets the occupancies of the remaining atoms to 1.0.
- always_keep_one_conformer = False Modifies behavior of remove_alt_confs so that residues with no conformer labeled blank or A are not deleted. Silent if remove_alt_confs is False.
- keep_occupancy = False Do not reset occupancy to 1 after removing altlocs
- altloc_to_keep = None Modifies behavior of remove_alt_confs so that the altloc identifier to keep in addition to blank is this one (instead of A). Silent if remove_alt_confs is False.
- average_alt_confs = False Averages the coordinates of altloc atoms with 1.0 angstrom.
- set_chemical_element_simple_if_necessary = None Make a simple guess about what the chemical element is (based on atom name and the way how it is formatted) and write it into output file.
- set_seg_id_to_chain_id = False Sets the segID field to the chain ID (padded with spaces).
- clear_seg_id = False Erases the segID field.
- convert_semet_to_met = False
- convert_met_to_semet = False
- neutralize_scatterers = False
- remove_fraction = None
- random_seed = None Random seed
- move_waters_last = False Transfer waters to the end of the model. Addresses some limitations of water picking in phenix.refine.
- renumber_and_move_waters = False Transfer waters to the end of respective chains. Renumber them so that numbering starts at the nearest 100 after the last non-water in the chain.
- adpScope of options to modify ADP of selected atomsatom_selection = None Selection for atoms to be modified. Overrides parent-level selection.randomize = False Randomize ADP within a certain rangeset_b_iso = None Set ADP of atoms to set_b_isoconvert_to_isotropic = False Convert atoms to isotropicconvert_to_anisotropic = False Convert atoms to anisotropicshift_b_iso = None Add shift_b_iso value to ADPscale_adp = None Multiply ADP by scale_adp
- atom_selection = None Selection for atoms to be modified. Overrides parent-level selection.
- randomize = False Randomize ADP within a certain range
- set_b_iso = None Set ADP of atoms to set_b_iso
- convert_to_isotropic = False Convert atoms to isotropic
- convert_to_anisotropic = False Convert atoms to anisotropic
- shift_b_iso = None Add shift_b_iso value to ADP
- scale_adp = None Multiply ADP by scale_adp

- sitesScope of options to modify coordinates of selected atomsatom_selection = None Selection for atoms to be modified. Overrides parent-level selection.shake = None Randomize coordinates with mean error value equal to shakeswitch_rotamers = max_distant min_distant exact_match fix_outliers translate = 0 0 0 Translational shift (x,y,z)rotate = 0 0 0 Rotational shift (x,y,z)euler_angle_convention = *xyz zyz Euler angles convention to be used for rotation
- atom_selection = None Selection for atoms to be modified. Overrides parent-level selection.
- shake = None Randomize coordinates with mean error value equal to shake
- switch_rotamers = max_distant min_distant exact_match fix_outliers
- translate = 0 0 0 Translational shift (x,y,z)
- rotate = 0 0 0 Rotational shift (x,y,z)
- euler_angle_convention = *xyz zyz Euler angles convention to be used for rotation
- occupanciesScope of options to modify occupancies of selected atomsatom_selection = None Selection for atoms to be modified. Overrides parent-level selection.randomize = False Randomize occupancies within a certain rangeset = None Set all or selected occupancies to given value
- atom_selection = None Selection for atoms to be modified. Overrides parent-level selection.
- randomize = False Randomize occupancies within a certain range
- set = None Set all or selected occupancies to given value
- rotate_about_axisaxis = None angle = None atom_selection = None
- axis = None
- angle = None
- atom_selection = None
- rename_chain_idRename chainsold_id = None new_id = None
- old_id = None
- new_id = None
- set_chargecharge_selection = None charge = None
- charge_selection = None
- charge = None
- outputWrite out file with modified model (file name is defined in write_modified)file_name = None Default is the original file name with the file extension replaced by _modified.pdb .format = *pdb mmcif Choose the output format of coordinate file (PDB or mmCIF)
- file_name = None Default is the original file name with the file extension replaced by _modified.pdb .
- format = *pdb mmcif Choose the output format of coordinate file (PDB or mmCIF)
- fake_f_obsScope of parameters to simulate Fobsr_free_flags_fraction = None scattering_table = wk1995 it1992 *n_gaussian neutron Choices of scattering table for structure factors calculationsfmodelk_sol = 0.0 Bulk solvent k_sol valuesb_sol = 0.0 Bulk solvent b_sol valuesb_cart = 0 0 0 0 0 0 Anisotropic scale matrixscale = 1.0 Overall scale factorstructure_factors_accuracyalgorithm = *fft direct taam cos_sin_table = False grid_resolution_factor = 1/3. quality_factor = None u_base = None b_base = None wing_cutoff = None exp_table_one_over_step_size = None extraExtra parameters for additional algorithms, e.g. TAAM using DiSCaMBmaskuse_asu_masks = True solvent_radius = 1.1 shrink_truncation_radius = 0.9 use_resolution_based_gridding = False If enabled, mask grid step = d_min / grid_step_factorstep = 0.6 Grid step (in A) for solvent mask calculationn_real = None Gridding for

mask calculationgrid_step_factor = 4.0 Grid step is resolution divided by grid_step_factorverbose = 1 mean_shift_for_mask_update = 0.1 Value of overall model shift in refinement to updates the mask.ignore_zero_occupancy_atoms = True Include atoms with zero occupancy into mask calculationignore_hydrogens = True Ignore H or D atoms in mask calculationn_radial_shells = 1 Number of shells in a radial shell bulk solvent modelradial_shell_width = 1.3 Radial shell widthFmask_res_high = 3.0 Assume Fmask=0 for reflections outside the range inf>d>=Fmask_res_high

• r_free_flags_fraction = None

• scattering_table = wk1995 it1992 *n_gaussian neutron Choices of scattering table for structure factors calculations

• fmodelk_sol = 0.0 Bulk solvent k_sol valuesb_sol = 0.0 Bulk solvent b_sol valuesb_cart = 0 0 0 0 0 0 Anisotropic scale matrixscale = 1.0 Overall scale factor

• k_sol = 0.0 Bulk solvent k_sol values

• b_sol = 0.0 Bulk solvent b_sol values

• b_cart = 0 0 0 0 0 0 Anisotropic scale matrix

• scale = 1.0 Overall scale factor

• structure_factors_accuracyalgorithm = *fft direct taam cos_sin_table = False grid_resolution_factor = 1/3. quality_factor = None u_base = None b_base = None wing_cutoff = None exp_table_one_over_step_size = None extraExtra parameters for additional algorithms, e.g. TAAM using DiSCaMB

• algorithm = *fft direct taam

• cos_sin_table = False

• grid_resolution_factor = 1/3.

• quality_factor = None

• u_base = None

• b_base = None

• wing_cutoff = None

• exp_table_one_over_step_size = None

• extraExtra parameters for additional algorithms, e.g. TAAM using DiSCaMB

• maskuse_asu_masks = True solvent_radius = 1.1 shrink_truncation_radius = 0.9 use_resolution_based_gridding = False If enabled, mask grid step = d_min / grid_step_factorstep = 0.6 Grid step (in A) for solvent mask calculationn_real = None Gridding for mask calculationgrid_step_factor = 4.0 Grid step is resolution divided by grid_step_factorverbose = 1 mean_shift_for_mask_update = 0.1 Value of overall model shift in refinement to updates the mask.ignore_zero_occupancy_atoms = True Include atoms with zero occupancy into mask calculationignore_hydrogens = True Ignore H or D atoms in mask calculationn_radial_shells = 1 Number of shells in a radial shell bulk solvent modelradial_shell_width = 1.3 Radial shell widthFmask_res_high = 3.0 Assume Fmask=0 for reflections outside the range inf>d>=Fmask_res_high

• use_asu_masks = True

• solvent_radius = 1.1

• shrink_truncation_radius = 0.9

• use_resolution_based_gridding = False If enabled, mask grid step = d_min / grid_step_factor

- step = 0.6 Grid step (in A) for solvent mask calculation
- n_real = None Gridding for mask calculation
- grid_step_factor = 4.0 Grid step is resolution divided by grid_step_factor
- verbose = 1
- mean_shift_for_mask_update = 0.1 Value of overall model shift in refinement to updates the mask.
- ignore_zero_occupancy_atoms = True Include atoms with zero occupancy into mask calculation
- ignore_hydrogens = True Ignore H or D atoms in mask calculation
- n_radial_shells = 1 Number of shells in a radial shell bulk solvent model
- radial_shell_width = 1.3 Radial shell width
- Fmask_res_high = 3.0 Assume Fmask=0 for reflections outside the range inf>d>=Fmask_res_high
- hydrogensScope of parameters for H atoms refinementrefine = individual riding *Auto Choice for refinement: riding model or full (H is refined as other atoms, useful at very high resolutions only)force_riding_adp = None optimize_scattering_contribution = True contribute_to_f_calc = True Add H contribution to Xray (Fcalc) calculationshigh_resolution_limit_to_include_scattering_from_h = 1.6 real_space_optimize_x_h_orientation = True xh_bond_distance_deviation_limit = 0.0 Idealize XH bond distances if deviation from ideal is greater than xh_bond_distance_deviation_limitlocal_real_space_fit_angular_step = 0.5 buildmap_type = mFobs-DFmodel Map type to be used to find hydrogensmap_cutoff = 2.0 Map cutoffsecondary_map_type = 2mFobs-DFmodel Map type to be used to validate peaks in primary mapsecondary_map_cutoff = 1.4 Secondary map cutoffangular_step = 3.0 Step in degrees for 6D rigid body search for best fitdod_and_od = False Build DOD/OD/O types of waters for neutron modelsfilter_dod = False Filter DOD/OD/O by correlationmin_od_dist = 0.60 Minimum O-D distance when building water from peaksmax_od_dist = 1.35 Maximum O-D distance when building water from peaksmin_dod_angle = 85.0 Minimum D-O-D angle when building water from peaksmax_dod_angle = 170.0 Maximum D-O-D angle when building water from peaksh_bond_min_mac = 1.8 h_bond_max = 3.9 use_sigma_scaled_maps = True Default is sigma scaled map, map in absolute scale is used otherwise.grid_step = 0.6 Grid step for map samplingmax_number_of_peaks = None map_next_to_modelmin_model_peak_dist = 0.7 max_model_peak_dist = 1.05 min_peak_peak_dist = 0.7 use_hydrogens = False peak_searchpeak_search_level = 1 max_peaks = 0 interpolate = True min_distance_sym_equiv = None general_positions_only = True min_cross_distance = 1.0 min_cubicle_edge = 5
- refine = individual riding *Auto Choice for refinement: riding model or full (H is refined as other atoms, useful at very high resolutions only)
- force_riding_adp = None
- optimize_scattering_contribution = True
- contribute_to_f_calc = True Add H contribution to Xray (Fcalc) calculations
- high_resolution_limit_to_include_scattering_from_h = 1.6
- real_space_optimize_x_h_orientation = True
- xh_bond_distance_deviation_limit = 0.0 Idealize XH bond distances if deviation from ideal is greater than xh_bond_distance_deviation_limit
- local_real_space_fit_angular_step = 0.5
- buildmap_type = mFobs-DFmodel Map type to be used to find hydrogensmap_cutoff = 2.0 Map cutoffsecondary_map_type = 2mFobs-DFmodel Map type to be used to validate peaks in primary

mapsecondary_map_cutoff = 1.4 Secondary map cutoffangular_step = 3.0 Step in degrees for 6D rigid body search for best fitdod_and_od = False Build DOD/OD/O types of waters for neutron modelsfilter_dod = False Filter DOD/OD/O by correlationmin_od_dist = 0.60 Minimum O-D distance when building water from peaksmax_od_dist = 1.35 Maximum O-D distance when building water from peaksmin_dod_angle = 85.0 Minimum D-O-D angle when building water from peaksmax_dod_angle = 170.0 Maximum D-O-D angle when building water from peaksh_bond_min_mac = 1.8 h_bond_max = 3.9 use_sigma_scaled_maps = True Default is sigma scaled map, map in absolute scale is used otherwise.grid_step = 0.6 Grid step for map samplingmax_number_of_peaks = None map_next_to_modelmin_model_peak_dist = 0.7 max_model_peak_dist = 1.05 min_peak_peak_dist = 0.7 use_hydrogens = False peak_searchpeak_search_level = 1 max_peaks = 0 interpolate = True min_distance_sym_equiv = None general_positions_only = True min_cross_distance = 1.0 min_cubicle_edge = 5

- map_type = mFobs-DFmodel Map type to be used to find hydrogens
- map_cutoff = 2.0 Map cutoff
- secondary_map_type = 2mFobs-DFmodel Map type to be used to validate peaks in primary map
- secondary_map_cutoff = 1.4 Secondary map cutoff
- angular_step = 3.0 Step in degrees for 6D rigid body search for best fit
- dod_and_od = False Build DOD/OD/O types of waters for neutron models
- filter_dod = False Filter DOD/OD/O by correlation
- min_od_dist = 0.60 Minimum O-D distance when building water from peaks
- max_od_dist = 1.35 Maximum O-D distance when building water from peaks
- min_dod_angle = 85.0 Minimum D-O-D angle when building water from peaks
- max_dod_angle = 170.0 Maximum D-O-D angle when building water from peaks
- h_bond_min_mac = 1.8
- h_bond_max = 3.9
- use_sigma_scaled_maps = True Default is sigma scaled map, map in absolute scale is used otherwise.
- grid_step = 0.6 Grid step for map sampling
- max_number_of_peaks = None
- map_next_to_modelmin_model_peak_dist = 0.7 max_model_peak_dist = 1.05 min_peak_peak_dist = 0.7 use_hydrogens = False
- min_model_peak_dist = 0.7
- max_model_peak_dist = 1.05
- min_peak_peak_dist = 0.7
- use_hydrogens = False
- peak_searchpeak_search_level = 1 max_peaks = 0 interpolate = True min_distance_sym_equiv = None general_positions_only = True min_cross_distance = 1.0 min_cubicle_edge = 5
- peak_search_level = 1
- max_peaks = 0
- interpolate = True
- min_distance_sym_equiv = None

- general_positions_only = True
- min_cross_distance = 1.0
- min_cubicle_edge = 5
- group_b_isonumber_of_macro_cycles = 3 max_number_of_iterations = 25 convergence_test = False run_finite_differences_test = False use_restraints = True restraints_weight = None
- number_of_macro_cycles = 3
- max_number_of_iterations = 25
- convergence_test = False
- run_finite_differences_test = False
- use_restraints = True
- restraints_weight = None
- adpisomax_number_of_iterations = 25 scalingscale_max = 3.0 scale_min = 10.0
- isomax_number_of_iterations = 25 scalingscale_max = 3.0 scale_min = 10.0
- max_number_of_iterations = 25
- scalingscale_max = 3.0 scale_min = 10.0
- scale_max = 3.0
- scale_min = 10.0
- tlsfind_automatically = None one_residue_one_group = None refine_T = True refine_L = True refine_S = True number_of_macro_cycles = 2 max_number_of_iterations = 25 start_tls_value = None run_finite_differences_test = False Test with finite differences instead of gradients. FOR DEVELOPMENT PURPOSES ONLY.eps = 1.e-6 Finite difference setting.min_tls_group_size = 5 min number of atoms allowed per TLS groupverbose = True
- find_automatically = None
- one_residue_one_group = None
- refine_T = True
- refine_L = True
- refine_S = True
- number_of_macro_cycles = 2
- max_number_of_iterations = 25
- start_tls_value = None
- run_finite_differences_test = False Test with finite differences instead of gradients. FOR DEVELOPMENT PURPOSES ONLY.
- eps = 1.e-6 Finite difference setting.
- min_tls_group_size = 5 min number of atoms allowed per TLS group
- verbose = True
- adp_restraintsisouse_u_local_only = False sphere_radius = 5.0 distance_power = 1.69 average_power = 1.03 wilson_b_weight_auto = False wilson_b_weight = None plain_pairs_radius = 5.0 refine_ap_and_dp = False

- isouse_u_local_only = False sphere_radius = 5.0 distance_power = 1.69 average_power = 1.03 wilson_b_weight_auto = False wilson_b_weight = None plain_pairs_radius = 5.0 refine_ap_and_dp = False
- use_u_local_only = False
- sphere_radius = 5.0
- distance_power = 1.69
- average_power = 1.03
- wilson_b_weight_auto = False
- wilson_b_weight = None
- plain_pairs_radius = 5.0
- refine_ap_and_dp = False
- group_occupancynumber_of_macro_cycles = 3 max_number_of_iterations = 25 convergence_test = False run_finite_differences_test = False constrain_correlated_3d_groups = False DEVELOPER OPTION: constrain occupancies for correlated alternate conformations.
- number_of_macro_cycles = 3
- max_number_of_iterations = 25
- convergence_test = False
- run_finite_differences_test = False
- constrain_correlated_3d_groups = False DEVELOPER OPTION: constrain occupancies for correlated alternate conformations.
- group_anomalousnumber_of_minimizer_cycles = 3 lbfgs_max_iterations = 20 number_of_finite_difference_tests = 0 find_automatically = False
- number_of_minimizer_cycles = 3
- lbfgs_max_iterations = 20
- number_of_finite_difference_tests = 0
- find_automatically = False
- rigid_bodyScope of parameters for rigid body refinementmode = *first_macro_cycle_only every_macro_cycle Defines how many times the rigid body refinement is performed during refinement run. first_macro_cycle_only to run only once at first macrocycle, every_macro_cycle to do rigid body refinement main.number_of_macro_cycles timestarget = ls_wunit_k1 ml *auto Rigid body refinement target function: least-squares or maximum-likelihoodtarget_auto_switch_resolution = 6.0 Used if target=auto, use optimal target for given working resolution.disable_final_r_factor_check = False If True, the R-factor check after refinement will not revert to the previous model, even if the R-factors have increased.refine_rotation = True Only rotation is refined (translation is fixed).refine_translation = True Only translation is refined (rotation is fixed).max_iterations = 25 Number of LBFGS minimization iterationsbulk_solvent_and_scale = True Bulk-solvent and scaling within rigid body refinement (needed since large rigid body shifts invalidate the mask).euler_angle_convention = *xyz zyz Euler angles conventionlbfgs_line_search_max_function_evaluations = 10 min_number_of_reflections = 200 Number of reflections that defines the first lowest resolution zone for the multiple_zones protocol. If very large displacements are expected, decreasing this parameter to 100 may lead to a larger convergence radius.multi_body_factor = 1 zone_exponent = 3.0 high_resolution = 3.0 High resolution cutoff (used for rigid body refinement only)max_low_high_res_limit = None Maximum value for high resolution cutoff for

the first lowest resolution zonenumber_of_zones = 5 Number of resolution zones for MZ protocol

- mode = *first_macro_cycle_only every_macro_cycle Defines how many times the rigid body refinement is performed during refinement run. first_macro_cycle_only to run only once at first macrocycle, every_macro_cycle to do rigid body refinement main.number_of_macro_cycles times
- target = ls_wunit_k1 ml *auto Rigid body refinement target function: least-squares or maximum-likelihood
- target_auto_switch_resolution = 6.0 Used if target=auto, use optimal target for given working resolution.
- disable_final_r_factor_check = False If True, the R-factor check after refinement will not revert to the previous model, even if the R-factors have increased.
- refine_rotation = True Only rotation is refined (translation is fixed).
- refine_translation = True Only translation is refined (rotation is fixed).
- max_iterations = 25 Number of LBFGS minimization iterations
- bulk_solvent_and_scale = True Bulk-solvent and scaling within rigid body refinement (needed since large rigid body shifts invalidate the mask).
- euler_angle_convention = *xyz zyz Euler angles convention
- lbfgs_line_search_max_function_evaluations = 10
- min_number_of_reflections = 200 Number of reflections that defines the first lowest resolution zone for the multiple_zones protocol. If very large displacements are expected, decreasing this parameter to 100 may lead to a larger convergence radius.
- multi_body_factor = 1
- zone_exponent = 3.0
- high_resolution = 3.0 High resolution cutoff (used for rigid body refinement only)
- max_low_high_res_limit = None Maximum value for high resolution cutoff for the first lowest resolution zone
- number_of_zones = 5 Number of resolution zones for MZ protocol
- ncstype = *torsion cartesian constraints coordinate_sigma = 0.05 restrain_b_factors = False If enabled, b-factors will be restrained for NCS-related atoms. Otherwise, atomic b-factors will be refined independently, and b_factor_weight will be set to 0.0b_factor_weight = 10 excessive_distance_limit = 1.5 special_position_warnings_only = False constraintsrefine_operators = True apply_to_coordinates = True apply_to_adp = True torsionsigma = 2.5 limit = 15.0 fix_outliers = False check_rotamer_consistency = Auto Check for rotamer differences between NCS matched sidechains and search for best fit amongst candidate rotamerstarget_damping = False damping_limit = 10.0 filter_phi_psi_outliers = True restrain_to_master_chain = False silence_warnings = False
- type = *torsion cartesian constraints
- coordinate_sigma = 0.05
- restrain_b_factors = False If enabled, b-factors will be restrained for NCS-related atoms. Otherwise, atomic b-factors will be refined independently, and b_factor_weight will be set to 0.0
- b_factor_weight = 10
- excessive_distance_limit = 1.5
- special_position_warnings_only = False
- constraintsrefine_operators = True apply_to_coordinates = True apply_to_adp = True

- refine_operators = True
- apply_to_coordinates = True
- apply_to_adp = True
- torsionsigma = 2.5 limit = 15.0 fix_outliers = False check_rotamer_consistency = Auto Check for rotamer differences between NCS matched sidechains and search for best fit amongst candidate rotamerstarget_damping = False damping_limit = 10.0 filter_phi_psi_outliers = True restrain_to_master_chain = False silence_warnings = False
- sigma = 2.5
- limit = 15.0
- fix_outliers = False
- check_rotamer_consistency = Auto Check for rotamer differences between NCS matched sidechains and search for best fit amongst candidate rotamers
- target_damping = False
- damping_limit = 10.0
- filter_phi_psi_outliers = True
- restrain_to_master_chain = False
- silence_warnings = False
- modify_f_obsremove = random strong weak strong_and_weak low other remove_fraction = 0.1 fill_mode = fobs_mean_mixed_with_dfmodel random fobs_mean *dfmodel
- remove = random strong weak strong_and_weak low other
- remove_fraction = 0.1
- fill_mode = fobs_mean_mixed_with_dfmodel random fobs_mean *dfmodel
- pdb_interpretationsort_atoms = True use_ncs_to_build_restraints = True show_restraints_histograms = True flip_symmetric_amino_acids = True superpose_ideal_ligand = *None all SF4 F3S DVT disable_uc_volume_vs_n_atoms_check = False allow_polymer_cross_special_position = False correct_hydrogens = True c_beta_restraints = True use_neutron_distances = False Use neutron X-H distances (which are longer than X-ray ones)disulfide_bond_exclusions_selection_string = None exclusion_distance_cutoff = 3 If SG of CYS forming SS bond is closer than this distance to an atom that it may coordinate then this SG is excluded from SS bond.link_distance_cutoff = 3 Length of link between the linked residuesdisulfide_distance_cutoff = 3 add_angle_and_dihedral_restraints_for_disulfides = True dihedral_function_type = *determined_by_sign_of_periodicity all_sinusoidal all_harmonic chir_volume_esd = 0.2 max_reasonable_bond_distance = 50.0 nonbonded_distance_cutoff = None default_vdw_distance = 1 min_vdw_distance = 1 nonbonded_buffer = 1 **EXPERIMENTAL, developers only**nonbonded_weight = None Weighting of nonbonded restraints term. By default, this will be set to 16 if explicit hydrogens are used (this was the default in earlier versions of Phenix), or 100 if hydrogens are missing.const_shrink_donor_acceptor = 0 **EXPERIMENTAL, developers only**vdw_1_4_factor = 0.8 min_distance_sym_equiv = 0.5 custom_nonbonded_symmetry_exclusions = None translate_cns_dna_rna_residue_names = None proceed_with_excessive_length_bonds = False restraints_librarycdl = True Use Conformation Dependent Library (CDL) for geometry restraintsinclude_modified_amino_acid_in_cdl = False mcl = True Use Metal Coordination Library (MCL) for tetrahedral Zn++ and iron-sulfur clusters SF4, FES, F3S, ...cis_pro_eh99 = True omega_cdl = False Use Omega Conformation Dependent Library (omega-CDL) for geometry restraintscdl_svl = False rdl = False rdl_selection = *all TRP hpdl = False cdl_nucleotidesenable = False Use RestraintsLib for DNA and

RNA for geometry restraintsesd = *phenix csd factor = 2.0 user_suppliedpath = None Root directory of user supplied restraintsaction = *pre post Choice to look here before GeoStd, the default, or aftersecondary_structuress_by_chain = True Only applies if search_method = from_ca. Find secondary structure only within individual chains. Alternative is to allow H-bonds between chains. Can be much slower with ss_by_chain=False. If your model is complete use ss_by_chain=True. If your model is many fragments, use ss_by_chain=False.from_ca_conservative = False various parameters changed to make from_ca method more conservative, hopefully to closer resemble ksdssp.max_rmsd = 1 Only applies if search_method = from_ca. Maximum rmsd to consider two chains with identical sequences as the same for ss identificationuse_representative_chains = True Only applies if search_method = from_ca. Use a representative of all chains with the same sequence. Alternative is to examine each chain individually. Can be much slower with use_representative_of_chain=False if there are many symmetry copies. Ignored unless ss_by_chain is True.max_representative_chains = 100 Only applies if search_method = from_ca. Maximum number of representative chainsenabled = False Turn on secondary structure restraints (main switch)proteinenabled = True Turn on secondary structure restraints for proteinsearch_method = *ksdssp from_ca cablam Particular method to search protein secondary structure.distance_ideal_n_o = 2.9 Target length for N-O hydrogen bonddistance_cut_n_o = 3.5 Hydrogen bond with length exceeding this value will not be establishedremove_outliers = True If true, h-bonds exceeding distance_cut_n_o length will not be establishedrestrain_hbond_angles = True helixserial_number = None helix_identifier = None enabled = True Restrain this particular helixselection = None helix_type = *alpha pi 3_10 unknown Type of helix, defaults to alpha. Only alpha, pi, and 3_10 helices are used for hydrogen-bond res...

- sort_atoms = True
- use_ncs_to_build_restraints = True
- show_restraints_histograms = True
- flip_symmetric_amino_acids = True
- superpose_ideal_ligand = *None all SF4 F3S DVT
- disable_uc_volume_vs_n_atoms_check = False
- allow_polymer_cross_special_position = False
- correct_hydrogens = True
- c_beta_restraints = True
- use_neutron_distances = False Use neutron X-H distances (which are longer than X-ray ones)
- disulfide_bond_exclusions_selection_string = None
- exclusion_distance_cutoff = 3 If SG of CYS forming SS bond is closer than this distance to an atom that it may coordinate then this SG is excluded from SS bond.
- link_distance_cutoff = 3 Length of link between the linked residues
- disulfide_distance_cutoff = 3
- add_angle_and_dihedral_restraints_for_disulfides = True
- dihedral_function_type = *determined_by_sign_of_periodicity all_sinusoidal all_harmonic
- chir_volume_esd = 0.2
- max_reasonable_bond_distance = 50.0
- nonbonded_distance_cutoff = None
- default_vdw_distance = 1

- min_vdw_distance = 1
- nonbonded_buffer = 1 **EXPERIMENTAL, developers only**
- nonbonded_weight = None Weighting of nonbonded restraints term. By default, this will be set to 16 if explicit hydrogens are used (this was the default in earlier versions of Phenix), or 100 if hydrogens are missing.
- const_shrink_donor_acceptor = 0 **EXPERIMENTAL, developers only**
- vdw_1_4_factor = 0.8
- min_distance_sym_equiv = 0.5
- custom_nonbonded_symmetry_exclusions = None
- translate_cns_dna_rna_residue_names = None
- proceed_with_excessive_length_bonds = False
- restraints_librarycdl = True Use Conformation Dependent Library (CDL) for geometry restraintsinclude_modified_amino_acid_in_cdl = False mcl = True Use Metal Coordination Library (MCL) for tetrahedral Zn++ and iron-sulfur clusters SF4, FES, F3S, ...cis_pro_eh99 = True omega_cdl = False Use Omega Conformation Dependent Library (omega-CDL) for geometry restraintscdl_svl = False rdl = False rdl_selection = *all TRP hpdl = False cdl_nucleotidesenable = False Use RestraintsLib for DNA and RNA for geometry restraintsesd = *phenix csd factor = 2.0 user_suppliedpath = None Root directory of user supplied restraintsaction = *pre post Choice to look here before GeoStd, the default, or after
- cdl = True Use Conformation Dependent Library (CDL) for geometry restraints
- include_modified_amino_acid_in_cdl = False
- mcl = True Use Metal Coordination Library (MCL) for tetrahedral Zn++ and iron-sulfur clusters SF4, FES, F3S, ...
- cis_pro_eh99 = True
- omega_cdl = False Use Omega Conformation Dependent Library (omega-CDL) for geometry restraints
- cdl_svl = False
- rdl = False
- rdl_selection = *all TRP
- hpdl = False
- cdl_nucleotidesenable = False Use RestraintsLib for DNA and RNA for geometry restraintsesd = *phenix csd factor = 2.0
- enable = False Use RestraintsLib for DNA and RNA for geometry restraints
- esd = *phenix csd
- factor = 2.0
- user_suppliedpath = None Root directory of user supplied restraintsaction = *pre post Choice to look here before GeoStd, the default, or after
- path = None Root directory of user supplied restraints
- action = *pre post Choice to look here before GeoStd, the default, or after
- secondary_structuress_by_chain = True Only applies if search_method = from_ca. Find secondary structure only within individual chains. Alternative is to allow H-bonds between chains. Can be much slower with ss_by_chain=False. If your model is complete use ss_by_chain=True. If your model is many fragments, use ss_by_chain=False.from_ca_conservative = False various parameters changed to make

from_ca method more conservative, hopefully to closer resemble ksdssp.max_rmsd = 1 Only applies if search_method = from_ca. Maximum rmsd to consider two chains with identical sequences as the same for ss identificationuse_representative_chains = True Only applies if search_method = from_ca. Use a representative of all chains with the same sequence. Alternative is to examine each chain individually. Can be much slower with use_representative_of_chain=False if there are many symmetry copies. Ignored unless ss_by_chain is True.max_representative_chains = 100 Only applies if search_method = from_ca. Maximum number of representative chainsenabled = False Turn on secondary structure restraints (main switch)proteinenabled = True Turn on secondary structure restraints for proteinsearch_method = *ksdssp from_ca cablam Particular method to search protein secondary structure.distance_ideal_n_o = 2.9 Target length for N-O hydrogen bonddistance_cut_n_o = 3.5 Hydrogen bond with length exceeding this value will not be establishedremove_outliers = True If true, h-bonds exceeding distance_cut_n_o length will not be establishedrestrain_hbond_angles = True helixserial_number = None helix_identifier = None enabled = True Restrain this particular helixselection = None helix_type = *alpha pi 3_10 unknown Type of helix, defaults to alpha. Only alpha, pi, and 3_10 helices are used for hydrogen-bond restraints.sigma = 0.05 slack = 0 top_out = False angle_sigma_scale = 1 Multiply sigmas for h-bond angles by this value. Original sigmas range from 5 to 10.angle_sigma_set = None Use this parameter to set sigmas for h-bond angles to a particular valuehbonddonor = None acceptor = None sheetenabled = True Restrain this particular sheetfirst_strand = None sheet_id = None sigma = 0.05 slack = 0 top_out = False angle_sigma_scale = 1 Multiply sigmas for h-bond angles by this value. Original sigmas range from 5 to 10.angle_sigma_set = None Use this parameter to set sigmas for h-bond angles to a particular valuestrandselection = None sense = parallel antiparallel *unknown bond_start_current = None bond_start_previous = None hbonddonor = None acceptor = None nucleic_acidenabled = True Turn on secondary structure restraints for nucleic acidshbond_distance_cutoff = 3.4 Hydrogen bonds with length exceeding this limit will not be establishedscale_bonds_sigma = 1. All sigmas for h-bond length will be multiplied by this number. The smaller number is tighter restraints.base_pairenabled = True Restraint this particular base-pairbase1 = None Selection string selecting at least one atom in the desired residuebase2 = None Selection string selecting at least one atom in the desired residuesaenger_class = 0 Type of base-pairingrestrain_planarity = False Apply planarity restraint to this base-pairplanarity_sigma = 0.176 restrain_hbonds = True Restrain hydrogen bonds length for this base-pairrestrain_hb_angles = True Restrain angles around hydrogen bonds for this base-pairrestrain_parallelity = True Apply parallelity restraint to this base-pairparallelity_target = 0 parallelity_sigma = 0.0335 stacking_pairenabled = True Restraint this particular base-pairbase1 = None Selection string selecting at least one atom in the desired residuebase2 = None Selection string selecting at least one atom in the desired residueangle = 0 sigma = 0.027

- ss_by_chain = True Only applies if search_method = from_ca. Find secondary structure only within individual chains. Alternative is to allow H-bonds between chains. Can be much slower with ss_by_chain=False. If your model is complete use ss_by_chain=True. If your model is many fragments, use ss_by_chain=False.
- from_ca_conservative = False various parameters changed to make from_ca method more conservative, hopefully to closer resemble ksdssp.
- max_rmsd = 1 Only applies if search_method = from_ca. Maximum rmsd to consider two chains with identical sequences as the same for ss identification
- use_representative_chains = True Only applies if search_method = from_ca. Use a representative of all chains with the same sequence. Alternative is to examine each chain individually. Can be much slower with use_representative_of_chain=False if there are many symmetry copies. Ignored unless ss_by_chain is True.

• max_representative_chains = 100 Only applies if search_method = from_ca. Maximum number of representative chains

• enabled = False Turn on secondary structure restraints (main switch)

• proteinenabled = True Turn on secondary structure restraints for proteinsearch_method = *ksdssp from_ca cablam Particular method to search protein secondary structure.distance_ideal_n_o = 2.9 Target length for N-O hydrogen bonddistance_cut_n_o = 3.5 Hydrogen bond with length exceeding this value will not be establishedremove_outliers = True If true, h-bonds exceeding distance_cut_n_o length will not be establishedrestrain_hbond_angles = True helixserial_number = None helix_identifier = None enabled = True Restrain this particular helixselection = None helix_type = *alpha pi 3_10 unknown Type of helix, defaults to alpha. Only alpha, pi, and 3_10 helices are used for hydrogen-bond restraints.sigma = 0.05 slack = 0 top_out = False angle_sigma_scale = 1 Multiply sigmas for h-bond angles by this value. Original sigmas range from 5 to 10.angle_sigma_set = None Use this parameter to set sigmas for h-bond angles to a particular valuehbonddonor = None acceptor = None sheetenabled = True Restrain this particular sheetfirst_strand = None sheet_id = None sigma = 0.05 slack = 0 top_out = False angle_sigma_scale = 1 Multiply sigmas for h-bond angles by this value. Original sigmas range from 5 to 10.angle_sigma_set = None Use this parameter to set sigmas for h-bond angles to a particular valuestrandselection = None sense = parallel antiparallel *unknown bond_start_current = None bond_start_previous = None hbonddonor = None acceptor = None

• enabled = True Turn on secondary structure restraints for protein

• search_method = *ksdssp from_ca cablam Particular method to search protein secondary structure.

• distance_ideal_n_o = 2.9 Target length for N-O hydrogen bond

• distance_cut_n_o = 3.5 Hydrogen bond with length exceeding this value will not be established

• remove_outliers = True If true, h-bonds exceeding distance_cut_n_o length will not be established

• restrain_hbond_angles = True

• helixserial_number = None helix_identifier = None enabled = True Restrain this particular helixselection = None helix_type = *alpha pi 3_10 unknown Type of helix, defaults to alpha. Only alpha, pi, and 3_10 helices are used for hydrogen-bond restraints.sigma = 0.05 slack = 0 top_out = False angle_sigma_scale = 1 Multiply sigmas for h-bond angles by this value. Original sigmas range from 5 to 10.angle_sigma_set = None Use this parameter to set sigmas for h-bond angles to a particular valuehbonddonor = None acceptor = None

• serial_number = None

• helix_identifier = None

• enabled = True Restrain this particular helix

• selection = None

• helix_type = *alpha pi 3_10 unknown Type of helix, defaults to alpha. Only alpha, pi, and 3_10 helices are used for hydrogen-bond restraints.

• sigma = 0.05

• slack = 0

• top_out = False

• angle_sigma_scale = 1 Multiply sigmas for h-bond angles by this value. Original sigmas range from 5 to 10.

• angle_sigma_set = None Use this parameter to set sigmas for h-bond angles to a particular value

- hbonddonor = None acceptor = None
- donor = None
- acceptor = None
- sheetenabled = True Restrain this particular sheetfirst_strand = None sheet_id = None sigma = 0.05 slack = 0 top_out = False angle_sigma_scale = 1 Multiply sigmas for h-bond angles by this value. Original sigmas range from 5 to 10.angle_sigma_set = None Use this parameter to set sigmas for h-bond angles to a particular valuestrandselection = None sense = parallel antiparallel *unknown bond_start_current = None bond_start_previous = None hbonddonor = None acceptor = None
- enabled = True Restrain this particular sheet
- first_strand = None
- sheet_id = None
- sigma = 0.05
- slack = 0
- top_out = False
- angle_sigma_scale = 1 Multiply sigmas for h-bond angles by this value. Original sigmas range from 5 to 10.
- angle_sigma_set = None Use this parameter to set sigmas for h-bond angles to a particular value
- strandselection = None sense = parallel antiparallel *unknown bond_start_current = None bond_start_previous = None
- selection = None
- sense = parallel antiparallel *unknown
- bond_start_current = None
- bond_start_previous = None
- hbonddonor = None acceptor = None
- donor = None
- acceptor = None
- nucleic_acidenabled = True Turn on secondary structure restraints for nucleic acidshbond_distance_cutoff = 3.4 Hydrogen bonds with length exceeding this limit will not be establishedscale_bonds_sigma = 1. All sigmas for h-bond length will be multiplied by this number. The smaller number is tighter restraints.base_pairenabled = True Restraint this particular base-pairbase1 = None Selection string selecting at least one atom in the desired residuebase2 = None Selection string selecting at least one atom in the desired residuesaenger_class = 0 Type of base-pairingrestrain_planarity = False Apply planarity restraint to this base-pairplanarity_sigma = 0.176 restrain_hbonds = True Restrain hydrogen bonds length for this base-pairrestrain_hb_angles = True Restrain angles around hydrogen bonds for this base-pairrestrain_parallelity = True Apply parallelity restraint to this base-pairparallelity_target = 0 parallelity_sigma = 0.0335 stacking_pairenabled = True Restraint this particular base-pairbase1 = None Selection string selecting at least one atom in the desired residuebase2 = None Selection string selecting at least one atom in the desired residueangle = 0 sigma = 0.027
- enabled = True Turn on secondary structure restraints for nucleic acids
- hbond_distance_cutoff = 3.4 Hydrogen bonds with length exceeding this limit will not be established

- scale_bonds_sigma = 1. All sigmas for h-bond length will be multiplied by this number. The smaller number is tighter restraints.
- base_pairenabled = True Restraint this particular base-pairbase1 = None Selection string selecting at least one atom in the desired residuebase2 = None Selection string selecting at least one atom in the desired residuesaenger_class = 0 Type of base-pairingrestrain_planarity = False Apply planarity restraint to this base-pairplanarity_sigma = 0.176 restrain_hbonds = True Restrain hydrogen bonds length for this base-pairrestrain_hb_angles = True Restrain angles around hydrogen bonds for this base-pairrestrain_parallelity = True Apply parallelity restraint to this base-pairparallelity_target = 0 parallelity_sigma = 0.0335
- enabled = True Restraint this particular base-pair
- base1 = None Selection string selecting at least one atom in the desired residue
- base2 = None Selection string selecting at least one atom in the desired residue
- saenger_class = 0 Type of base-pairing
- restrain_planarity = False Apply planarity restraint to this base-pair
- planarity_sigma = 0.176
- restrain_hbonds = True Restrain hydrogen bonds length for this base-pair
- restrain_hb_angles = True Restrain angles around hydrogen bonds for this base-pair
- restrain_parallelity = True Apply parallelity restraint to this base-pair
- parallelity_target = 0
- parallelity_sigma = 0.0335
- stacking_pairenabled = True Restraint this particular base-pairbase1 = None Selection string selecting at least one atom in the desired residuebase2 = None Selection string selecting at least one atom in the desired residueangle = 0 sigma = 0.027
- enabled = True Restraint this particular base-pair
- base1 = None Selection string selecting at least one atom in the desired residue
- base2 = None Selection string selecting at least one atom in the desired residue
- angle = 0
- sigma = 0.027
- reference_coordinate_restraintsRestrains coordinates in Cartesian space to stay near their starting positions. This is intended for use in generating simulated annealing omit maps, to prevent refined atoms from collapsing in on the region missing atoms. For conserving geometry quality at low resolution, the more flexible reference model restraints should be used instead.enabled = False exclude_outliers = True selection = all sigma = 0.2 limit = 1.0 top_out = False reference_is_average_alt_confs = False Sets the reference coordinate to the average of two alt. confs. Ignores selection option.
- enabled = False
- exclude_outliers = True
- selection = all
- sigma = 0.2
- limit = 1.0
- top_out = False

- reference_is_average_alt_confs = False Sets the reference coordinate to the average of two alt. confs. Ignores selection option.
- automatic_restraintsside_chain_parallelity = False
- side_chain_parallelity = False
- automatic_linkinglink_all = False If True, bond restraints will be generated for any appropriate ligand-protein or ligand-nucleic acid covalent bonds. This includes sugars, amino acid modifications, and other prosthetic groups.link_none = False link_metals = Auto link_residues = True link_amino_acid_rna_dna = False link_carbohydrates = True link_ligands = True link_small_molecules = False metal_coordination_cutoff = 3.0 amino_acid_bond_cutoff = 1.9 inter_residue_bond_cutoff = 2.2 buffer_for_second_row_elements = 0.5 carbohydrate_bond_cutoff = 1.99 ligand_bond_cutoff = 1.99 small_molecule_bond_cutoff = 1.98 exclude_hydrogens_from_bonding_decisions = False
- link_all = False If True, bond restraints will be generated for any appropriate ligand-protein or ligand-nucleic acid covalent bonds. This includes sugars, amino acid modifications, and other prosthetic groups.
- link_none = False
- link_metals = Auto
- link_residues = True
- link_amino_acid_rna_dna = False
- link_carbohydrates = True
- link_ligands = True
- link_small_molecules = False
- metal_coordination_cutoff = 3.0
- amino_acid_bond_cutoff = 1.9
- inter_residue_bond_cutoff = 2.2
- buffer_for_second_row_elements = 0.5
- carbohydrate_bond_cutoff = 1.99
- ligand_bond_cutoff = 1.99
- small_molecule_bond_cutoff = 1.98
- exclude_hydrogens_from_bonding_decisions = False
- include_in_automatic_linkingselection_1 = None selection_2 = None bond_cutoff = 4.5
- selection_1 = None
- selection_2 = None
- bond_cutoff = 4.5
- exclude_from_automatic_linkingselection_1 = None selection_2 = None
- selection_1 = None
- selection_2 = None
- apply_cis_trans_specificationcis_trans_mod = cis *trans residue_selection = None Residues containing C-alpha atom of omega dihedral
- cis_trans_mod = cis *trans
- residue_selection = None Residues containing C-alpha atom of omega dihedral

- apply_cif_restraintsrestraints_file_name = None residue_selection = None
- restraints_file_name = None
- residue_selection = None
- apply_cif_modificationdata_mod = None residue_selection = None
- data_mod = None
- residue_selection = None
- apply_cif_linkdata_link = None residue_selection_1 = None residue_selection_2 = None
- data_link = None
- residue_selection_1 = None
- residue_selection_2 = None
- peptide_linkramachandran_restraints = False !!! OBSOLETED. Kept for backward compatibility only !!! Restrains peptide backbone to fall within allowed regions of Ramachandran plot. Although it does not eliminate outliers, it can significantly improve the percent favored and percent outliers at low resolution. Probably not useful (and maybe even harmful) at resolutions much higher than 3.5A.cis_threshold = 45 apply_all_trans = False discard_omega = False discard_psi_phi = True apply_peptide_plane = False omega_esd_override_value = None rama_weight = 1.0 scale_allowed = 1.0 rama_potential = *oldfield emsley rama_selection = None Selection of part of the model for which Ramachandran restraints will be set up.restrain_rama_outliers = True Apply restraints to Ramachandran outliersrestrain_rama_allowed = True Apply restraints to residues in allowed region on Ramachandran plotrestrain_allowed_outliers_with_emsley = False In case of restrain_rama_outliers=True and/or restrain_rama_allowed=True still restrain these residues with emsley. Make sense only in case of using oldfield potential.oldfieldesd = 10.0 weight_scale = 1.0 dist_weight_max = 10.0 weight = None plot_cutoff = 0.027
- ramachandran_restraints = False !!! OBSOLETED. Kept for backward compatibility only !!! Restrains peptide backbone to fall within allowed regions of Ramachandran plot. Although it does not eliminate outliers, it can significantly improve the percent favored and percent outliers at low resolution. Probably not useful (and maybe even harmful) at resolutions much higher than 3.5A.
- cis_threshold = 45
- apply_all_trans = False
- discard_omega = False
- discard_psi_phi = True
- apply_peptide_plane = False
- omega_esd_override_value = None
- rama_weight = 1.0
- scale_allowed = 1.0
- rama_potential = *oldfield emsley
- rama_selection = None Selection of part of the model for which Ramachandran restraints will be set up.
- restrain_rama_outliers = True Apply restraints to Ramachandran outliers
- restrain_rama_allowed = True Apply restraints to residues in allowed region on Ramachandran plot
- restrain_allowed_outliers_with_emsley = False In case of restrain_rama_outliers=True and/or restrain_rama_allowed=True still restrain these residues with emsley. Make sense only in case of using

oldfield potential.

- oldfieldesd = 10.0 weight_scale = 1.0 dist_weight_max = 10.0 weight = None plot_cutoff = 0.027
- esd = 10.0
- weight_scale = 1.0
- dist_weight_max = 10.0
- weight = None
- plot_cutoff = 0.027
- ramachandran_plot_restraintsenabled = False favored = *oldfield emsley emsley8k phi_psi_2 allowed = *oldfield emsley emsley8k phi_psi_2 outlier = *oldfield emsley emsley8k phi_psi_2 selection = None Selection of part of the model for which Ramachandran restraints will be set up.inject_emsley8k_into_oldfield_favored = True Backdoor to disable temporary dirty hack to use botholdfieldweight = 0. Direct weight value. If 0 the weight will be calculated as following: (w, op.esd, op.dist_weight_max, 2.0, op.weight_scale) 1 / esd^2 * max(2.0, min(current_distance_to_allowed, dist_weight_max)) * weight_scale max(2.0, current_distance_to_allowed) 1 / esd^2 * weight_scale * max(distance_to_allowed_cutoff, current_distance_to_allowed) weight_scale(=0.01) * max(distance_weight_min(=2.), min(distance_weight_max(=10.), current_distance_to_allowed))weight_scale = 0.01 distance_weight_min = 2.0 minimum coefficient when scaling depending on how far the residue is from allowed region.distance_weight_max = 10.0 maximum coefficient when scaling depending on how far the residue is from allowed region.plot_cutoff = 0.027 emsleyweight = 1.0 scale_allowed = 1.0 emsley8kweight_favored = 5.0 weight_allowed = 10.0 weight_outlier = 10.0 phi_psi_2favored_strategy = *closest highest_probability random weighted_random allowed_strategy = *closest highest_probability random weighted_random outlier_strategy = *closest highest_probability random weighted_random
- enabled = False
- favored = *oldfield emsley emsley8k phi_psi_2
- allowed = *oldfield emsley emsley8k phi_psi_2
- outlier = *oldfield emsley emsley8k phi_psi_2
- selection = None Selection of part of the model for which Ramachandran restraints will be set up.
- inject_emsley8k_into_oldfield_favored = True Backdoor to disable temporary dirty hack to use both
- oldfieldweight = 0. Direct weight value. If 0 the weight will be calculated as following: (w, op.esd, op.dist_weight_max, 2.0, op.weight_scale) 1 / esd^2 * max(2.0, min(current_distance_to_allowed, dist_weight_max)) * weight_scale max(2.0, current_distance_to_allowed) 1 / esd^2 * weight_scale * max(distance_to_allowed_cutoff, current_distance_to_allowed) weight_scale(=0.01) * max(distance_weight_min(=2.), min(distance_weight_max(=10.), current_distance_to_allowed))weight_scale = 0.01 distance_weight_min = 2.0 minimum coefficient when scaling depending on how far the residue is from allowed region.distance_weight_max = 10.0 maximum coefficient when scaling depending on how far the residue is from allowed region.plot_cutoff = 0.027
- weight = 0. Direct weight value. If 0 the weight will be calculated as following: (w, op.esd, op.dist_weight_max, 2.0, op.weight_scale) 1 / esd^2 * max(2.0, min(current_distance_to_allowed, dist_weight_max)) * weight_scale max(2.0, current_distance_to_allowed) 1 / esd^2 * weight_scale * max(distance_to_allowed_cutoff, current_distance_to_allowed) weight_scale(=0.01) * max(distance_weight_min(=2.), min(distance_weight_max(=10.), current_distance_to_allowed))
- weight_scale = 0.01

- distance_weight_min = 2.0 minimum coefficient when scaling depending on how far the residue is from allowed region.
- distance_weight_max = 10.0 maximum coefficient when scaling depending on how far the residue is from allowed region.
- plot_cutoff = 0.027
- emsleyweight = 1.0 scale_allowed = 1.0
- weight = 1.0
- scale_allowed = 1.0
- emsley8kweight_favored = 5.0 weight_allowed = 10.0 weight_outlier = 10.0
- weight_favored = 5.0
- weight_allowed = 10.0
- weight_outlier = 10.0
- phi_psi_2favored_strategy = *closest highest_probability random weighted_random allowed_strategy = *closest highest_probability random weighted_random outlier_strategy = *closest highest_probability random weighted_random
- favored_strategy = *closest highest_probability random weighted_random
- allowed_strategy = *closest highest_probability random weighted_random
- outlier_strategy = *closest highest_probability random weighted_random
- rna_sugar_pucker_analysisbond_min_distance = 1.2 bond_max_distance = 1.8 epsilon_range_min = 155.0 epsilon_range_max = 310.0 delta_range_2p_min = 129.0 delta_range_2p_max = 162.0 delta_range_3p_min = 65.0 delta_range_3p_max = 104.0 p_distance_c1p_outbound_line_2p_max = 2.9 o3p_distance_c1p_outbound_line_2p_max = 2.4 bond_detection_distance_tolerance = 0.5 enable = True
- bond_min_distance = 1.2
- bond_max_distance = 1.8
- epsilon_range_min = 155.0
- epsilon_range_max = 310.0
- delta_range_2p_min = 129.0
- delta_range_2p_max = 162.0
- delta_range_3p_min = 65.0
- delta_range_3p_max = 104.0
- p_distance_c1p_outbound_line_2p_max = 2.9
- o3p_distance_c1p_outbound_line_2p_max = 2.4
- bond_detection_distance_tolerance = 0.5
- enable = True
- show_histogram_slotsbond_lengths = 5 nonbonded_interaction_distances = 5 bond_angle_deviations_from_ideal = 5 dihedral_angle_deviations_from_ideal = 5 chiral_volume_deviations_from_ideal = 5
- bond_lengths = 5
- nonbonded_interaction_distances = 5

- bond_angle_deviations_from_ideal = 5
- dihedral_angle_deviations_from_ideal = 5
- chiral_volume_deviations_from_ideal = 5
- show_max_itemsnot_linked = 5 bond_restraints_sorted_by_residual = 5 nonbonded_interactions_sorted_by_model_distance = 5 bond_angle_restraints_sorted_by_residual = 5 dihedral_angle_restraints_sorted_by_residual = 3 chirality_restraints_sorted_by_residual = 3 planarity_restraints_sorted_by_residual = 3 residues_with_excluded_nonbonded_symmetry_interactions = 12 fatal_problem_max_lines = 10
- not_linked = 5
- bond_restraints_sorted_by_residual = 5
- nonbonded_interactions_sorted_by_model_distance = 5
- bond_angle_restraints_sorted_by_residual = 5
- dihedral_angle_restraints_sorted_by_residual = 3
- chirality_restraints_sorted_by_residual = 3
- planarity_restraints_sorted_by_residual = 3
- residues_with_excluded_nonbonded_symmetry_interactions = 12
- fatal_problem_max_lines = 10
- ncs_groupThe definition of one NCS group. Note, that almost always in refinement programs they will be checked and filtered if needed.reference = None Residue selection string for the complete master NCS copyselection = None Residue selection string for each NCS copy location in ASU
- reference = None Residue selection string for the complete master NCS copy
- selection = None Residue selection string for each NCS copy location in ASU
- ncs_searchSet of parameters for NCS search procedure. Some of them also used for filtering user-supplied ncs_group.enabled = False Enable NCS restraints or constraints in refinement (in some cases may be switched on inside refinement program).exclude_selection = "element H or element D or water" Atoms selected by this selection will be excluded from the model before any NCS search and/or filtering procedures. There is no way atoms defined by this selection will be in NCS.chain_similarity_threshold = 0.85 Threshold for sequence similarity between matching chains. A smaller value may cause more chains to be grouped together and can lower the number of common residueschain_max_rmsd = 2. limit of rms difference between chains to be considered as copiesresidue_match_radius = 4.0 Maximum allowed distance difference between pairs of matching atoms of two residuestry_shortcuts = False Try very quick check to speed up the search when chains are identical. If failed, regular search will be performed automatically.minimum_number_of_atoms_in_copy = 3 Do not create ncs groups where master and copies would contain less than specified amount of atomsvalidate_user_supplied_groups = True Enable validation of user-supplied ncs_group. Need to exercise a lot of caution turning this off. This option is for developers only.
- enabled = False Enable NCS restraints or constraints in refinement (in some cases may be switched on inside refinement program).
- exclude_selection = "element H or element D or water" Atoms selected by this selection will be excluded from the model before any NCS search and/or filtering procedures. There is no way atoms defined by this selection will be in NCS.
- chain_similarity_threshold = 0.85 Threshold for sequence similarity between matching chains. A smaller value may cause more chains to be grouped together and can lower the number of common residues

- chain_max_rmsd = 2. limit of rms difference between chains to be considered as copies
- residue_match_radius = 4.0 Maximum allowed distance difference between pairs of matching atoms of two residues
- try_shortcuts = False Try very quick check to speed up the search when chains are identical. If failed, regular search will be performed automatically.
- minimum_number_of_atoms_in_copy = 3 Do not create ncs groups where master and copies would contain less than specified amount of atoms
- validate_user_supplied_groups = True Enable validation of user-supplied ncs_group. Need to exercise a lot of caution turning this off. This option is for developers only.
- clash_guardnonbonded_distance_threshold = 0.5 max_number_of_distances_below_threshold = 100 max_fraction_of_distances_below_threshold = 0.1
- nonbonded_distance_threshold = 0.5
- max_number_of_distances_below_threshold = 100
- max_fraction_of_distances_below_threshold = 0.1
- geometry_restraintseditsexcessive_bond_distance_limit = 10 bondaction = *add delete change atom_selection_1 = None atom_selection_2 = None symmetry_operation = None The bond is between atom_1 and symmetry_operation * atom_2, with atom_1 and atom_2 given in fractional coordinates. Example: symmetry_operation = -x-1,-y,zdistance_ideal = None sigma = None slack = None limit = -1.0 top_out = False angleaction = *add delete change atom_selection_1 = None atom_selection_2 = None atom_selection_3 = None angle_ideal = None sigma = None dihedralaction = *add delete change atom_selection_1 = None atom_selection_2 = None atom_selection_3 = None atom_selection_4 = None angle_ideal = None alt_angle_ideals = None sigma = None periodicity = 1 planarityaction = *add delete change atom_selection = None sigma = None parallelityaction = *add delete change atom_selection_1 = None atom_selection_2 = None sigma = 0.027 target_angle_deg = 0
- editsexcessive_bond_distance_limit = 10 bondaction = *add delete change atom_selection_1 = None atom_selection_2 = None symmetry_operation = None The bond is between atom_1 and symmetry_operation * atom_2, with atom_1 and atom_2 given in fractional coordinates. Example: symmetry_operation = -x-1,-y,zdistance_ideal = None sigma = None slack = None limit = -1.0 top_out = False angleaction = *add delete change atom_selection_1 = None atom_selection_2 = None atom_selection_3 = None angle_ideal = None sigma = None dihedralaction = *add delete change atom_selection_1 = None atom_selection_2 = None atom_selection_3 = None atom_selection_4 = None angle_ideal = None alt_angle_ideals = None sigma = None periodicity = 1 planarityaction = *add delete change atom_selection = None sigma = None parallelityaction = *add delete change atom_selection_1 = None atom_selection_2 = None sigma = 0.027 target_angle_deg = 0
- excessive_bond_distance_limit = 10
- bondaction = *add delete change atom_selection_1 = None atom_selection_2 = None symmetry_operation = None The bond is between atom_1 and symmetry_operation * atom_2, with atom_1 and atom_2 given in fractional coordinates. Example: symmetry_operation = -x-1,-y,zdistance_ideal = None sigma = None slack = None limit = -1.0 top_out = False
- action = *add delete change
- atom_selection_1 = None
- atom_selection_2 = None

- symmetry_operation = None The bond is between atom_1 and symmetry_operation * atom_2, with atom_1 and atom_2 given in fractional coordinates. Example: symmetry_operation = -x-1,-y,z
- distance_ideal = None
- sigma = None
- slack = None
- limit = -1.0
- top_out = False
- angleaction = *add delete change atom_selection_1 = None atom_selection_2 = None atom_selection_3 = None angle_ideal = None sigma = None
- action = *add delete change
- atom_selection_1 = None
- atom_selection_2 = None
- atom_selection_3 = None
- angle_ideal = None
- sigma = None
- dihedralaction = *add delete change atom_selection_1 = None atom_selection_2 = None atom_selection_3 = None atom_selection_4 = None angle_ideal = None alt_angle_ideals = None sigma = None periodicity = 1
- action = *add delete change
- atom_selection_1 = None
- atom_selection_2 = None
- atom_selection_3 = None
- atom_selection_4 = None
- angle_ideal = None
- alt_angle_ideals = None
- sigma = None
- periodicity = 1
- planarityaction = *add delete change atom_selection = None sigma = None
- action = *add delete change
- atom_selection = None
- sigma = None
- parallelityaction = *add delete change atom_selection_1 = None atom_selection_2 = None sigma = 0.027 target_angle_deg = 0
- action = *add delete change
- atom_selection_1 = None
- atom_selection_2 = None
- sigma = 0.027
- target_angle_deg = 0

- geometry_restraintsremoveangles = None dihedrals = None chiralities = None planarities = None parallelities = None
- removeangles = None dihedrals = None chiralities = None planarities = None parallelities = None
- angles = None
- dihedrals = None
- chiralities = None
- planarities = None
- parallelities = None
- ordered_solventlow_resolution = 2.8 Low resolution limit for water picking (at lower resolution water will not be picked even if requessted)mode = *second_half filter_only every_macro_cycle every_macro_cycle_after_first Choices for water picking strategy: auto - start water picking after ferst few macro-cycles, filter_only - remove water only, every_macro_cycle - do water update every macro-cyclemask_atoms_selection = protein and (name CA or name CB or name N or name C or name O) Mask macromolecule atoms in peak picking mapinclude_altlocs = False Search for water with altlocsn_cycles = 1 output_residue_name = HOH output_chain_id = S output_atom_name = O scattering_type = O Defines scattering factors for newly added watersprimary_map_type = mFobs-DFmodel Map used to identify candidate water sites - by default this is the standard difference map.primary_map_cutoff = 3. dist_min = 1.8 dist_max = 3.2 dist_min_altloc = 0.5 refine_oat = False Q & B coarse grid search.refine_adp = True Refine ADP for newly placed solvent.refine_occupancies = True Refine solvent occupancies.new_solvent = *isotropic anisotropic Based on the choice, added solvent will have isotropic or anisotropic b-factorsb_iso_min = 1.0 Minimum B-factor value, waters with smaller value will be rejectedb_iso_max = 80.0 Maximum B-factor value, waters with bigger value will be rejectedanisotropy_min = 0.1 For solvent refined as anisotropic: remove is less than this valueoccupancy_min = 0.2 Minimum occupancy value, waters with smaller value will be rejectedoccupancy_max = 1.0 Maximum occupancy value, waters with bigger value will be rejectedoccupancy = 1.0 Initial occupancy value for newly added waterfilter_at_start = True ignore_final_filtering_step = False correct_drifted_waters = True secondary_map_and_map_cc_filtercc_map_1_type = "Fmodel" cc_map_2_type = 2mFobs-DFmodel poor_cc_threshold = 0.70 poor_map_value_threshold = 1.0
- low_resolution = 2.8 Low resolution limit for water picking (at lower resolution water will not be picked even if requessted)
- mode = *second_half filter_only every_macro_cycle every_macro_cycle_after_first Choices for water picking strategy: auto - start water picking after ferst few macro-cycles, filter_only - remove water only, every_macro_cycle - do water update every macro-cycle
- mask_atoms_selection = protein and (name CA or name CB or name N or name C or name O) Mask macromolecule atoms in peak picking map
- include_altlocs = False Search for water with altlocs
- n_cycles = 1
- output_residue_name = HOH
- output_chain_id = S
- output_atom_name = O
- scattering_type = O Defines scattering factors for newly added waters

- primary_map_type = mFobs-DFmodel Map used to identify candidate water sites - by default this is the standard difference map.
- primary_map_cutoff = 3.
- dist_min = 1.8
- dist_max = 3.2
- dist_min_altloc = 0.5
- refine_oat = False Q & B coarse grid search.
- refine_adp = True Refine ADP for newly placed solvent.
- refine_occupancies = True Refine solvent occupancies.
- new_solvent = *isotropic anisotropic Based on the choice, added solvent will have isotropic or anisotropic b-factors
- b_iso_min = 1.0 Minimum B-factor value, waters with smaller value will be rejected
- b_iso_max = 80.0 Maximum B-factor value, waters with bigger value will be rejected
- anisotropy_min = 0.1 For solvent refined as anisotropic: remove is less than this value
- occupancy_min = 0.2 Minimum occupancy value, waters with smaller value will be rejected
- occupancy_max = 1.0 Maximum occupancy value, waters with bigger value will be rejected
- occupancy = 1.0 Initial occupancy value for newly added water
- filter_at_start = True
- ignore_final_filtering_step = False
- correct_drifted_waters = True
- secondary_map_and_map_cc_filtercc_map_1_type = "Fmodel" cc_map_2_type = 2mFobs-DFmodel poor_cc_threshold = 0.70 poor_map_value_threshold = 1.0
- cc_map_1_type = "Fmodel"
- cc_map_2_type = 2mFobs-DFmodel
- poor_cc_threshold = 0.70
- poor_map_value_threshold = 1.0
- cablam_idealizationrotation_angle = 30 angle to rotate the oxygen while searching for better conformationrequire_h_bond = True require presence of (new) h-bond after rotationdo_gm = False Run geometry minimization after rotationverbose = 0 Verbosity levels: 0: Absolutely no output 1: Essential info and errors 2: Everythingfind_ss_after_fixes = False re-evaluate SS after fixing Cablam outliers. May be helpful to identify new or extend previous SS elementssave_intermediates = False Save all cablam rotation for particular residue in separate file
- rotation_angle = 30 angle to rotate the oxygen while searching for better conformation
- require_h_bond = True require presence of (new) h-bond after rotation
- do_gm = False Run geometry minimization after rotation
- verbose = 0 Verbosity levels: 0: Absolutely no output 1: Essential info and errors 2: Everything
- find_ss_after_fixes = False re-evaluate SS after fixing Cablam outliers. May be helpful to identify new or extend previous SS elements
- save_intermediates = False Save all cablam rotation for particular residue in separate file

- peak_searchuse_sigma_scaled_maps = True Default is sigma scaled map, map in absolute scale is used otherwise.grid_step = 0.6 Grid step for map samplingmax_number_of_peaks = None map_next_to_modelmin_model_peak_dist = 1.8 max_model_peak_dist = 3.2 min_peak_peak_dist = 1.8 use_hydrogens = False peak_searchpeak_search_level = 1 max_peaks = 0 interpolate = True min_distance_sym_equiv = None general_positions_only = False min_cross_distance = 1.8 min_cubicle_edge = 5
- use_sigma_scaled_maps = True Default is sigma scaled map, map in absolute scale is used otherwise.
- grid_step = 0.6 Grid step for map sampling
- max_number_of_peaks = None
- map_next_to_modelmin_model_peak_dist = 1.8 max_model_peak_dist = 3.2 min_peak_peak_dist = 1.8 use_hydrogens = False
- min_model_peak_dist = 1.8
- max_model_peak_dist = 3.2
- min_peak_peak_dist = 1.8
- use_hydrogens = False
- peak_searchpeak_search_level = 1 max_peaks = 0 interpolate = True min_distance_sym_equiv = None general_positions_only = False min_cross_distance = 1.8 min_cubicle_edge = 5
- peak_search_level = 1
- max_peaks = 0
- interpolate = True
- min_distance_sym_equiv = None
- general_positions_only = False
- min_cross_distance = 1.8
- min_cubicle_edge = 5
- bulk_solvent_and_scalemode = slow *fast apply_back_trace = True bulk_solvent = True anisotropic_scaling = True k_sol_b_sol_grid_search = True minimization_k_sol_b_sol = True minimization_b_cart = True target = ls_wunit_k1 *ml symmetry_constraints_on_b_cart = True k_sol_max = 0.6 k_sol_min = 0.0 b_sol_max = 300.0 b_sol_min = 0.0 k_sol_grid_search_max = 0.6 k_sol_grid_search_min = 0.0 b_sol_grid_search_max = 80.0 b_sol_grid_search_min = 20.0 k_sol_step = 0.2 b_sol_step = 20.0 number_of_macro_cycles = 1 max_iterations = 25 min_iterations = 25 fix_k_sol = None fix_b_sol = None fix_b_cartb11 = None b22 = None b33 = None b12 = None b13 = None b23 = None
- mode = slow *fast
- apply_back_trace = True
- bulk_solvent = True
- anisotropic_scaling = True
- k_sol_b_sol_grid_search = True
- minimization_k_sol_b_sol = True
- minimization_b_cart = True
- target = ls_wunit_k1 *ml
- symmetry_constraints_on_b_cart = True

- k_sol_max = 0.6
- k_sol_min = 0.0
- b_sol_max = 300.0
- b_sol_min = 0.0
- k_sol_grid_search_max = 0.6
- k_sol_grid_search_min = 0.0
- b_sol_grid_search_max = 80.0
- b_sol_grid_search_min = 20.0
- k_sol_step = 0.2
- b_sol_step = 20.0
- number_of_macro_cycles = 1
- max_iterations = 25
- min_iterations = 25
- fix_k_sol = None
- fix_b_sol = None
- fix_b_cartb11 = None b22 = None b33 = None b12 = None b13 = None b23 = None
- b11 = None
- b22 = None
- b33 = None
- b12 = None
- b13 = None
- b23 = None
- alpha_betafree_reflections_per_bin = 140 number_of_macromolecule_atoms_absent = 225 n_atoms_included = 0 bf_atoms_absent = 15.0 final_error = 0.0 absent_atom_type = "O" method = *est calc estimation_algorithm = *analytical iterative verbose = -1 interpolation = True number_of_waters_absent = 613 sigmaa_estimatorkernel_width_free_reflections = 100 kernel_on_chebyshev_nodes = True number_of_sampling_points = 20 number_of_chebyshev_terms = 10 use_sampling_sum_weights = True
- free_reflections_per_bin = 140
- number_of_macromolecule_atoms_absent = 225
- n_atoms_included = 0
- bf_atoms_absent = 15.0
- final_error = 0.0
- absent_atom_type = "O"
- method = *est calc
- estimation_algorithm = *analytical iterative
- verbose = -1
- interpolation = True
- number_of_waters_absent = 613

• sigmaa_estimatorkernel_width_free_reflections = 100 kernel_on_chebyshev_nodes = True number_of_sampling_points = 20 number_of_chebyshev_terms = 10 use_sampling_sum_weights = True

• kernel_width_free_reflections = 100

• kernel_on_chebyshev_nodes = True

• number_of_sampling_points = 20

• number_of_chebyshev_terms = 10

• use_sampling_sum_weights = True

• maskuse_asu_masks = True solvent_radius = 1.1 shrink_truncation_radius = 0.9 use_resolution_based_gridding = False If enabled, mask grid step = d_min / grid_step_factorstep = 0.6 Grid step (in A) for solvent mask calculationn_real = None Gridding for mask calculationgrid_step_factor = 4.0 Grid step is resolution divided by grid_step_factorverbose = 1 mean_shift_for_mask_update = 0.1 Value of overall model shift in refinement to updates the mask.ignore_zero_occupancy_atoms = True Include atoms with zero occupancy into mask calculationignore_hydrogens = True Ignore H or D atoms in mask calculationn_radial_shells = 1 Number of shells in a radial shell bulk solvent modelradial_shell_width = 1.3 Radial shell widthFmask_res_high = 3.0 Assume Fmask=0 for reflections outside the range inf>d>=Fmask_res_high

• use_asu_masks = True

• solvent_radius = 1.1

• shrink_truncation_radius = 0.9

• use_resolution_based_gridding = False If enabled, mask grid step = d_min / grid_step_factor

• step = 0.6 Grid step (in A) for solvent mask calculation

• n_real = None Gridding for mask calculation

• grid_step_factor = 4.0 Grid step is resolution divided by grid_step_factor

• verbose = 1

• mean_shift_for_mask_update = 0.1 Value of overall model shift in refinement to updates the mask.

• ignore_zero_occupancy_atoms = True Include atoms with zero occupancy into mask calculation

• ignore_hydrogens = True Ignore H or D atoms in mask calculation

• n_radial_shells = 1 Number of shells in a radial shell bulk solvent model

• radial_shell_width = 1.3 Radial shell width

• Fmask_res_high = 3.0 Assume Fmask=0 for reflections outside the range inf>d>=Fmask_res_high

• tardyUnder developmentmode = every_macro_cycle *second_and_before_last once first first_half xray_weight_factor = 10 start_temperature_kelvin = 2500 final_temperature_kelvin = 300 velocity_scaling = True temperature_cap_factor = 1.5 excessive_temperature_factor = 5 number_of_cooling_steps = 500 number_of_time_steps = 1 time_step_pico_seconds = 0.001 temperature_degrees_of_freedom = *cartesian constrained minimization_max_iterations = 0 omit_bonds_with_slack_greater_than = 0 constrain_dihedrals_with_sigma_less_than = 10 near_singular_hinges_angular_tolerance_deg = 5 emulate_cartesian = False trajectory_directory = None prolsq_repulsion_function_changesenergy(delta) = c_rep*(max(0,(k_rep*vdw_distance)**irexp-delta**irexp))**rexpc_rep = None Usual value: 16k_rep = 0.75 Usual value: 1. Smaller values reduce the distance threshold at which the repulsive force becomes active.irexp = None Usual value: 1rexp = None Usual value: 4

• mode = every_macro_cycle *second_and_before_last once first first_half

- xray_weight_factor = 10
- start_temperature_kelvin = 2500
- final_temperature_kelvin = 300
- velocity_scaling = True
- temperature_cap_factor = 1.5
- excessive_temperature_factor = 5
- number_of_cooling_steps = 500
- number_of_time_steps = 1
- time_step_pico_seconds = 0.001
- temperature_degrees_of_freedom = *cartesian constrained
- minimization_max_iterations = 0
- omit_bonds_with_slack_greater_than = 0
- constrain_dihedrals_with_sigma_less_than = 10
- near_singular_hinges_angular_tolerance_deg = 5
- emulate_cartesian = False
- trajectory_directory = None
- prolsq_repulsion_function_changesenergy(delta) = c_rep*(max(0,(k_rep*vdw_distance)**irexp-delta**irexp))**rexpc_rep = None Usual value: 16k_rep = 0.75 Usual value: 1. Smaller values reduce the distance threshold at which the repulsive force becomes active.irexp = None Usual value: 1rexp = None Usual value: 4
- c_rep = None Usual value: 16
- k_rep = 0.75 Usual value: 1. Smaller values reduce the distance threshold at which the repulsive force becomes active.
- irexp = None Usual value: 1
- rexp = None Usual value: 4
- cartesian_dynamicstemperature = 300 number_of_steps = 200 time_step = 0.0005 initial_velocities_zero_fraction = 0 n_print = 100 verbose = -1 random_seed = None n_collect = 10 stop_cm_motion = True
- temperature = 300
- number_of_steps = 200
- time_step = 0.0005
- initial_velocities_zero_fraction = 0
- n_print = 100
- verbose = -1
- random_seed = None
- n_collect = 10
- stop_cm_motion = True
- simulated_annealingstart_temperature = 5000 final_temperature = 300 cool_rate = 100 number_of_steps = 50 time_step = 0.0005 initial_velocities_zero_fraction = 0 interleave_minimization =

False verbose = -1 n_print = 100 update_grads_shift = 0.3 random_seed = None refine_sites = True refine_adp = False max_number_of_iterations = 25 mode = every_macro_cycle *second_and_before_last once first first_half

- start_temperature = 5000
- final_temperature = 300
- cool_rate = 100
- number_of_steps = 50
- time_step = 0.0005
- initial_velocities_zero_fraction = 0
- interleave_minimization = False
- verbose = -1
- n_print = 100
- update_grads_shift = 0.3
- random_seed = None
- refine_sites = True
- refine_adp = False
- max_number_of_iterations = 25
- mode = every_macro_cycle *second_and_before_last once first first_half
- target_weightsoptimize_xyz_weight = False optimize_adp_weight = False r_free_only = False Use only R-free to choose the best weightwxc_scale = 0.5 wxu_scale = 1.0 wc = 1.0 wu = 1.0 fix_wxc = None fix_wxu = None shake_sites = True shake_adp = 10.0 regularize_ncycles = 50 verbose = 1 wnc_scale = 0.5 wnu_scale = 1.0 rmsd_cutoff_for_gradient_filtering = 3.0 force_optimize_weights = False weight_selection_criteriabonds_rmsd = None angles_rmsd = None r_free_minus_r_work = None r_free_range_width = None mean_diff_b_iso_bonded_fraction = None min_diff_b_iso_bonded = None
- optimize_xyz_weight = False
- optimize_adp_weight = False
- r_free_only = False Use only R-free to choose the best weight
- wxc_scale = 0.5
- wxu_scale = 1.0
- wc = 1.0
- wu = 1.0
- fix_wxc = None
- fix_wxu = None
- shake_sites = True
- shake_adp = 10.0
- regularize_ncycles = 50
- verbose = 1
- wnc_scale = 0.5
- wnu_scale = 1.0

- rmsd_cutoff_for_gradient_filtering = 3.0
- force_optimize_weights = False
- weight_selection_criteriabonds_rmsd = None angles_rmsd = None r_free_minus_r_work = None r_free_range_width = None mean_diff_b_iso_bonded_fraction = None min_diff_b_iso_bonded = None
- bonds_rmsd = None
- angles_rmsd = None
- r_free_minus_r_work = None
- r_free_range_width = None
- mean_diff_b_iso_bonded_fraction = None
- min_diff_b_iso_bonded = None
- iasb_iso_max = 100.0 occupancy_min = -1.0 occupancy_max = 1.5 ias_b_iso_max = 100.0 ias_b_iso_min = 0.0 ias_occupancy_min = 0.01 ias_occupancy_max = 3.0 initial_ias_occupancy = 1.0 build_ias_types = L R B BH ring_atoms = None use_map = True build_only = False file_prefix = None lone_pairatom_x = CA atom_xo = C atom_o = O peak_search_mapmap_type = *Fobs-Fmodel mFobs-DFmodel grid_step = 0.25 scaling = *volume sigma
- b_iso_max = 100.0
- occupancy_min = -1.0
- occupancy_max = 1.5
- ias_b_iso_max = 100.0
- ias_b_iso_min = 0.0
- ias_occupancy_min = 0.01
- ias_occupancy_max = 3.0
- initial_ias_occupancy = 1.0
- build_ias_types = L R B BH
- ring_atoms = None
- use_map = True
- build_only = False
- file_prefix = None
- lone_pairatom_x = CA atom_xo = C atom_o = O
- atom_x = CA
- atom_xo = C
- atom_o = O
- peak_search_mapmap_type = *Fobs-Fmodel mFobs-DFmodel grid_step = 0.25 scaling = *volume sigma
- map_type = *Fobs-Fmodel mFobs-DFmodel
- grid_step = 0.25
- scaling = *volume sigma
- ls_target_namestarget_name = *ls_wunit_k1 ls_wunit_k2 ls_wunit_kunit ls_wunit_k1_fixed ls_wunit_k1ask3_fixed ls_wexp_k1 ls_wexp_k2 ls_wexp_kunit ls_wff_k1 ls_wff_k2 ls_wff_kunit

ls_wff_k1_fixed ls_wff_k1ask3_fixed lsm_kunit lsm_k1 lsm_k2 lsm_k1_fixed lsm_k1ask3_fixed

- target_name = *ls_wunit_k1 ls_wunit_k2 ls_wunit_kunit ls_wunit_k1_fixed ls_wunit_k1ask3_fixed ls_wexp_k1 ls_wexp_k2 ls_wexp_kunit ls_wff_k1 ls_wff_k2 ls_wff_kunit ls_wff_k1_fixed ls_wff_k1ask3_fixed lsm_kunit lsm_k1 lsm_k2 lsm_k1_fixed lsm_k1ask3_fixed
- twinningtwin_target = *twin_lsq_f detwinmode = algebraic proportional *auto map_typestwofofc = *two_m_dtfo_d_fc two_dtfo_fc fofc = *m_dtfo_d_fc gradient m_gradient aniso_correct = False
- twin_target = *twin_lsq_f
- detwinmode = algebraic proportional *auto map_typestwofofc = *two_m_dtfo_d_fc two_dtfo_fc fofc = *m_dtfo_d_fc gradient m_gradient aniso_correct = False
- mode = algebraic proportional *auto
- map_typestwofofc = *two_m_dtfo_d_fc two_dtfo_fc fofc = *m_dtfo_d_fc gradient m_gradient aniso_correct = False
- twofofc = *two_m_dtfo_d_fc two_dtfo_fc
- fofc = *m_dtfo_d_fc gradient m_gradient
- aniso_correct = False
- structure_factors_and_gradients_accuracyalgorithm = *fft direct taam cos_sin_table = False grid_resolution_factor = 1/3. quality_factor = None u_base = None b_base = None wing_cutoff = None exp_table_one_over_step_size = None extraExtra parameters for additional algorithms, e.g. TAAM using DiSCaMB
- algorithm = *fft direct taam
- cos_sin_table = False
- grid_resolution_factor = 1/3.
- quality_factor = None
- u_base = None
- b_base = None
- wing_cutoff = None
- exp_table_one_over_step_size = None
- extraExtra parameters for additional algorithms, e.g. TAAM using DiSCaMB
- r_free_flagsfraction = 0.1 max_free = 2000 lattice_symmetry_max_delta = 5.0 Tolerance used in the determination of the highest lattice symmetry. Can be thought of as angle between lattice vectors that should line up perfectly if the symmetry is ideal. A typical value is 3 degrees.use_lattice_symmetry = True When generating Rfree flags, do so in the asymmetric unit of the highest lattice symmetry. The result is an Rfree set suitable for twin refinement.
- fraction = 0.1
- max_free = 2000
- lattice_symmetry_max_delta = 5.0 Tolerance used in the determination of the highest lattice symmetry. Can be thought of as angle between lattice vectors that should line up perfectly if the symmetry is ideal. A typical value is 3 degrees.
- use_lattice_symmetry = True When generating Rfree flags, do so in the asymmetric unit of the highest lattice symmetry. The result is an Rfree set suitable for twin refinement.

• reference_modelenabled = False Restrains the dihedral angles to a high-resolution reference structure to reduce overfitting at low resolution. You will need to specify a reference PDB file (in the input list in the main window) to use this option.file = None use_starting_model_as_reference = False sigma = 1.0 limit = 15.0 hydrogens = False Include dihedrals with hydrogen atomsmain_chain = True Include dihedrals formed by main chain atomsside_chain = True Include dihedrals formed by side chain atomsfix_outliers = True Try to fix rotamer outliers in refined modelstrict_rotamer_matching = False Make sure that rotamers in refinement model matches those in reference model even when they are not outliersauto_shutoff_for_ncs = False Do not apply to parts of structure covered by NCS restraintssecondary_structure_only = False Only apply reference model restraints to secondary structure elements (helices and sheets)reference_groupreference = None selection = None file_name = None this is to used internally to disambiguate cases where multiple reference models contain the same chain ID. This normally does not need to be set by the usersearch_optionsSet of parameters for NCS search procedure. Some of them also used for filtering user-supplied ncs_group.exclude_selection = "element H or element D or water" Atoms selected by this selection will be excluded from the model before any NCS search and/or filtering procedures. There is no way atoms defined by this selection will be in NCS.chain_similarity_threshold = 0.85 Threshold for sequence similarity between matching chains. A smaller value may cause more chains to be grouped together and can lower the number of common residueschain_max_rmsd = 100. limit of rms difference between chains to be considered as copiesresidue_match_radius = 1000 Maximum allowed distance difference between pairs of matching atoms of two residuestry_shortcuts = False Try very quick check to speed up the search when chains are identical. If failed, regular search will be performed automatically.minimum_number_of_atoms_in_copy = 3 Do not create ncs groups where master and copies would contain less than specified amount of atomsvalidate_user_supplied_groups = True Enable validation of user-supplied ncs_group. Need to exercise a lot of caution turning this off. This option is for developers only.

• enabled = False Restrains the dihedral angles to a high-resolution reference structure to reduce overfitting at low resolution. You will need to specify a reference PDB file (in the input list in the main window) to use this option.

• file = None

• use_starting_model_as_reference = False

• sigma = 1.0

• limit = 15.0

• hydrogens = False Include dihedrals with hydrogen atoms

• main_chain = True Include dihedrals formed by main chain atoms

• side_chain = True Include dihedrals formed by side chain atoms

• fix_outliers = True Try to fix rotamer outliers in refined model

• strict_rotamer_matching = False Make sure that rotamers in refinement model matches those in reference model even when they are not outliers

• auto_shutoff_for_ncs = False Do not apply to parts of structure covered by NCS restraints

• secondary_structure_only = False Only apply reference model restraints to secondary structure elements (helices and sheets)

• reference_groupreference = None selection = None file_name = None this is to used internally to disambiguate cases where multiple reference models contain the same chain ID. This normally does not need to be set by the user

• reference = None

- selection = None
- file_name = None this is to used internally to disambiguate cases where multiple reference models contain the same chain ID. This normally does not need to be set by the user
- search_optionsSet of parameters for NCS search procedure. Some of them also used for filtering user-supplied ncs_group.exclude_selection = "element H or element D or water" Atoms selected by this selection will be excluded from the model before any NCS search and/or filtering procedures. There is no way atoms defined by this selection will be in NCS.chain_similarity_threshold = 0.85 Threshold for sequence similarity between matching chains. A smaller value may cause more chains to be grouped together and can lower the number of common residueschain_max_rmsd = 100. limit of rms difference between chains to be considered as copiesresidue_match_radius = 1000 Maximum allowed distance difference between pairs of matching atoms of two residuestry_shortcuts = False Try very quick check to speed up the search when chains are identical. If failed, regular search will be performed automatically.minimum_number_of_atoms_in_copy = 3 Do not create ncs groups where master and copies would contain less than specified amount of atomsvalidate_user_supplied_groups = True Enable validation of user-supplied ncs_group. Need to exercise a lot of caution turning this off. This option is for developers only.
- exclude_selection = "element H or element D or water" Atoms selected by this selection will be excluded from the model before any NCS search and/or filtering procedures. There is no way atoms defined by this selection will be in NCS.
- chain_similarity_threshold = 0.85 Threshold for sequence similarity between matching chains. A smaller value may cause more chains to be grouped together and can lower the number of common residues
- chain_max_rmsd = 100. limit of rms difference between chains to be considered as copies
- residue_match_radius = 1000 Maximum allowed distance difference between pairs of matching atoms of two residues
- try_shortcuts = False Try very quick check to speed up the search when chains are identical. If failed, regular search will be performed automatically.
- minimum_number_of_atoms_in_copy = 3 Do not create ncs groups where master and copies would contain less than specified amount of atoms
- validate_user_supplied_groups = True Enable validation of user-supplied ncs_group. Need to exercise a lot of caution turning this off. This option is for developers only.
- ion_placementdebug = False ion_chain_id = X initial_occupancy = 1.0 Occupancy for newly placed ions - if less than 1.0, the occupancy may be refined automatically in future runs of phenix.refine.initial_b_iso = Auto refine_ion_occupancies = True Toggles refinement of occupancies for newly placed ions. This will only happen if the occupancy refinement strategy is selected.refine_ion_adp = *Auto isotropic anisotropic none B-factor refinement type for newly placed ions. At medium-to-high resolution, anisotropic refinement may be preferrable for the heavier elements.refine_anomalous = True If True and the wavelength is specified, any newly placed ions will have anomalous scattering factors refined. This is unlikely to affect R-factors but should flatten the anomalous LLG map.max_distance_between_like_charges = 3.5 use_svm = True require_valence = False Toggles the use of valence calculationsambiguous_valence_cutoff = 0.5 Cutoff to uniquely select one of many metals by comparing its observed and expected valenced_min_strict_valence = 1.5 anom_map_type = *residual simple llg Type of anomalous difference map to use. Default is a residual map, showing only unmodeled scattering.find_anomalous_substructure = Auto use_phaser = True Toggles the use of Phaser for calculation of f-prime values.aggressive = False Toggles more permissive settings for flagging waters as heavier elements. Not recommended for automated use.map_sampling_radius = 2.0 svmsvm_name =

*Auto heavy merged_high_res Name of SVM classifier to usefiltered_outputs = True Apply an filter to the outputs of the SVM, ensuring they are chemically sanemin_score = 0.2 min_score_above = 0.1 min_fraction_of_next = 2.0 watermin_2fofc_level = 1.8 Minimum water 2mFo-DFc map value. Waters below this cutoff will not be analyzed further.max_fofc_level = 3.0 Maximum water mFo-DFc map valuemax_anom_level = 3.0 Maximum water anomalous map valuemax_occ = 1.0 Maximum water occupancymin_b_iso = 1.0 Minimum water isotropic B-factorfp_max = 1.0 fpp_max = 0 max_stddev_b_iso = 5 min_frac_b_iso = 0.2 min_frac_calpha_b_iso = 0.75 max_frac_calpha_2fofc = 1.2 min_2fofc_coordinating = 0.9 Minimum 2mFo-DFc for waters involved in coordination shells.phaserllgc_ncycles = None distance_cutoff = 1.5 distance_cutoff_same_site = 0.7 fpp_ratio_min = 0.2 Minimum ratio of refined/theoretical f-double-prime.fpp_ratio_max = 1.1 Maximum ratio of refined/theoretical f-double-prime.chloridemax_distance_to_amide_n = 3.5 Max distance to amide N atommax_distance_to_cation = 3.5 min_distance_to_anion = 3.5 min_distance_to_other_sites = 2.1 max_distance_to_hydroxyl = 3.5 delta_amide_h_angle = 20 Allowed deviation (in degrees) from ideal N-H-X angledelta_planar_angle = 10 max_deviation_from_plane = 0.8

• debug = False

• ion_chain_id = X

• initial_occupancy = 1.0 Occupancy for newly placed ions - if less than 1.0, the occupancy may be refined automatically in future runs of phenix.refine.

• initial_b_iso = Auto

• refine_ion_occupancies = True Toggles refinement of occupancies for newly placed ions. This will only happen if the occupancy refinement strategy is selected.

• refine_ion_adp = *Auto isotropic anisotropic none B-factor refinement type for newly placed ions. At medium-to-high resolution, anisotropic refinement may be preferrable for the heavier elements.

• refine_anomalous = True If True and the wavelength is specified, any newly placed ions will have anomalous scattering factors refined. This is unlikely to affect R-factors but should flatten the anomalous LLG map.

• max_distance_between_like_charges = 3.5

• use_svm = True

• require_valence = False Toggles the use of valence calculations

• ambiguous_valence_cutoff = 0.5 Cutoff to uniquely select one of many metals by comparing its observed and expected valence

• d_min_strict_valence = 1.5

• anom_map_type = *residual simple llg Type of anomalous difference map to use. Default is a residual map, showing only unmodeled scattering.

• find_anomalous_substructure = Auto

• use_phaser = True Toggles the use of Phaser for calculation of f-prime values.

• aggressive = False Toggles more permissive settings for flagging waters as heavier elements. Not recommended for automated use.

• map_sampling_radius = 2.0

• svmsvm_name = *Auto heavy merged_high_res Name of SVM classifier to usefiltered_outputs = True Apply an filter to the outputs of the SVM, ensuring they are chemically sanemin_score = 0.2 min_score_above = 0.1 min_fraction_of_next = 2.0

• svm_name = *Auto heavy merged_high_res Name of SVM classifier to use

- filtered_outputs = True Apply an filter to the outputs of the SVM, ensuring they are chemically sane
- min_score = 0.2
- min_score_above = 0.1
- min_fraction_of_next = 2.0
- watermin_2fofc_level = 1.8 Minimum water 2mFo-DFc map value. Waters below this cutoff will not be analyzed further.max_fofc_level = 3.0 Maximum water mFo-DFc map valuemax_anom_level = 3.0 Maximum water anomalous map valuemax_occ = 1.0 Maximum water occupancymin_b_iso = 1.0 Minimum water isotropic B-factorfp_max = 1.0 fpp_max = 0 max_stddev_b_iso = 5 min_frac_b_iso = 0.2 min_frac_calpha_b_iso = 0.75 max_frac_calpha_2fofc = 1.2 min_2fofc_coordinating = 0.9 Minimum 2mFo-DFc for waters involved in coordination shells.
- min_2fofc_level = 1.8 Minimum water 2mFo-DFc map value. Waters below this cutoff will not be analyzed further.
- max_fofc_level = 3.0 Maximum water mFo-DFc map value
- max_anom_level = 3.0 Maximum water anomalous map value
- max_occ = 1.0 Maximum water occupancy
- min_b_iso = 1.0 Minimum water isotropic B-factor
- fp_max = 1.0
- fpp_max = 0
- max_stddev_b_iso = 5
- min_frac_b_iso = 0.2
- min_frac_calpha_b_iso = 0.75
- max_frac_calpha_2fofc = 1.2
- min_2fofc_coordinating = 0.9 Minimum 2mFo-DFc for waters involved in coordination shells.
- phaserllgc_ncycles = None distance_cutoff = 1.5 distance_cutoff_same_site = 0.7 fpp_ratio_min = 0.2 Minimum ratio of refined/theoretical f-double-prime.fpp_ratio_max = 1.1 Maximum ratio of refined/theoretical f-double-prime.
- llgc_ncycles = None
- distance_cutoff = 1.5
- distance_cutoff_same_site = 0.7
- fpp_ratio_min = 0.2 Minimum ratio of refined/theoretical f-double-prime.
- fpp_ratio_max = 1.1 Maximum ratio of refined/theoretical f-double-prime.
- chloridemax_distance_to_amide_n = 3.5 Max distance to amide N atommax_distance_to_cation = 3.5 min_distance_to_anion = 3.5 min_distance_to_other_sites = 2.1 max_distance_to_hydroxyl = 3.5 delta_amide_h_angle = 20 Allowed deviation (in degrees) from ideal N-H-X angledelta_planar_angle = 10 max_deviation_from_plane = 0.8
- max_distance_to_amide_n = 3.5 Max distance to amide N atom
- max_distance_to_cation = 3.5
- min_distance_to_anion = 3.5
- min_distance_to_other_sites = 2.1
- max_distance_to_hydroxyl = 3.5

- delta_amide_h_angle = 20 Allowed deviation (in degrees) from ideal N-H-X angle
- delta_planar_angle = 10
- max_deviation_from_plane = 0.8
- guiMiscellaneous parameters for phenix.refine GUIbase_output_dir = None tmp_dir = None send_notification = False notify_email = None add_hydrogens = False Runs phenix.ready_set to add hydrogens prior to refinement.skip_rsr = False skip_kinemage = False phil_file = None ready_set_hydrogensneutron_option = *all_h all_d hd_and_h hd_and_d all_hd add_h_to_water = False add_d_to_water = False neutron_exchange_hydrogens = False Add deuteriums in exchangeable sitesperdeuterate = False Add deuteriums in all possible sitesmigrationPHIL scopes for helping with migrationsdata_managerfmodelmnopScope of X-ray data and free-R flagshigh_resolution = None low_resolution = None twin_law = None outliers_rejection = True Remove basic wilson outliers , extreme wilson outliers , and beamstop shadow outliersfrench_wilson_scale = True sigma_fobs_rejection_criterion = None sigma_iobs_rejection_criterion = None ignore_all_zeros = True force_anomalous_flag_to_be_equal_to = None convert_to_non_anomalous_if_ratio_pairs_lone_less_than_threshold = 0.5 french_wilsonmax_bins = 60 Maximum number of resolution binsmin_bin_size = 40 Minimum number of reflections per binr_free_flagsrequired = True Specify if free-r flags are must be present (or else generated)test_flag_value = None This value is usually selected automatically - do not change unless you really know what you're doing!ignore_r_free_flags = False Use all reflections in refinement (work and test)disable_suitability_test = False ignore_pdb_hexdigest = False If True, disables safety check based on MD5 hexdigests stored in PDB files produced by previous runs.generate = False Generate R-free flags (if not available in input files)fraction = 0.1 max_free = 2000 lattice_symmetry_max_delta = 5 use_lattice_symmetry = True use_dataman_shells = False Used to avoid biasing of the test set by certain types of non-crystallographic symmetry.n_shells = 20 electron_dataScope of X-ray data and free-R flagshigh_resolution = None low_resolution = None twin_law = None outliers_rejection = True Remove basic wilson outliers , extreme wilson outliers , and beamstop shadow outliersfrench_wilson_scale = True sigma_fobs_rejection_criterion = None sigma_iobs_rejection_criterion = None ignore_all_zeros = True force_anomalous_flag_to_be_equal_to = None convert_to_non_anomalous_if_ratio_pairs_lone_less_than_threshold = 0.5 french_wilsonmax_bins = 60 Maximum number of resolution binsmin_bin_size = 40 Minimum number of reflections per binr_free_flagsrequired = True Specify if free-r flags are must be present (or else generated)test_flag_value = None This value is usually selected automatically - do not change unless you really know what you're doing!ignore_r_free_flags = False Use all reflections in refinement (work and test)disable_suitability_test = False ignore_pdb_hexdigest = False If True, disables safety check based on MD5 hexdigests stored in PDB files produced by previous runs.generate = False Generate R-free flags (if not available in input files)fraction = 0.1 max_free = 2000 lattice_symmetry_max_delta = 5 use_lattice_symmetry = True use_dataman_shells = False Used to avoid biasing of the test set by certain types of non-crystallographic symmetry.n_shells = 20 refinementinputexperimental_phasesScope of experimental phase information (HL coefficients)file_name = None labels = None monomersScope of monomers information (CIF files)file_name = None Monomer file(s) name (CIF)electron_datafile_name = None labels = None r_free_flagsfile_name = None This is normally the same as the file containing Fobs and is usually selected automatically.label = None neutron_datafile_name = None labels = None r_free_flagsfile_name = None This is normally the same as the file containing Fobs and is usually selected automatically.label = None pdbfile_name = None Model file(s) name (PDB)neutron_file_name = None Model file(s) name (PDB)electron_file_name = None Model file(s) name (PDB)xray_datafile_name = None labels = None r_free_flagsfile_name = N...
- base_output_dir = None

- tmp_dir = None
- send_notification = False
- notify_email = None
- add_hydrogens = False Runs phenix.ready_set to add hydrogens prior to refinement.
- skip_rsr = False
- skip_kinemage = False
- phil_file = None
- ready_set_hydrogensneutron_option = *all_h all_d hd_and_h hd_and_d all_hd add_h_to_water = False add_d_to_water = False neutron_exchange_hydrogens = False Add deuteriums in exchangeable sitesperdeuterate = False Add deuteriums in all possible sites
- neutron_option = *all_h all_d hd_and_h hd_and_d all_hd
- add_h_to_water = False
- add_d_to_water = False
- neutron_exchange_hydrogens = False Add deuteriums in exchangeable sites
- perdeuterate = False Add deuteriums in all possible sites
- migrationPHIL scopes for helping with migrationsdata_managerfmodelmnopScope of X-ray data and free-R flagshigh_resolution = None low_resolution = None twin_law = None outliers_rejection = True Remove basic wilson outliers , extreme wilson outliers , and beamstop shadow outliersfrench_wilson_scale = True sigma_fobs_rejection_criterion = None sigma_iobs_rejection_criterion = None ignore_all_zeros = True force_anomalous_flag_to_be_equal_to = None convert_to_non_anomalous_if_ratio_pairs_lone_less_than_threshold = 0.5 french_wilsonmax_bins = 60 Maximum number of resolution binsmin_bin_size = 40 Minimum number of reflections per binr_free_flagsrequired = True Specify if free-r flags are must be present (or else generated)test_flag_value = None This value is usually selected automatically - do not change unless you really know what you're doing!ignore_r_free_flags = False Use all reflections in refinement (work and test)disable_suitability_test = False ignore_pdb_hexdigest = False If True, disables safety check based on MD5 hexdigests stored in PDB files produced by previous runs.generate = False Generate R-free flags (if not available in input files)fraction = 0.1 max_free = 2000 lattice_symmetry_max_delta = 5 use_lattice_symmetry = True use_dataman_shells = False Used to avoid biasing of the test set by certain types of non-crystallographic symmetry.n_shells = 20 electron_dataScope of X-ray data and free-R flagshigh_resolution = None low_resolution = None twin_law = None outliers_rejection = True Remove basic wilson outliers , extreme wilson outliers , and beamstop shadow outliersfrench_wilson_scale = True sigma_fobs_rejection_criterion = None sigma_iobs_rejection_criterion = None ignore_all_zeros = True force_anomalous_flag_to_be_equal_to = None convert_to_non_anomalous_if_ratio_pairs_lone_less_than_threshold = 0.5 french_wilsonmax_bins = 60 Maximum number of resolution binsmin_bin_size = 40 Minimum number of reflections per binr_free_flagsrequired = True Specify if free-r flags are must be present (or else generated)test_flag_value = None This value is usually selected automatically - do not change unless you really know what you're doing!ignore_r_free_flags = False Use all reflections in refinement (work and test)disable_suitability_test = False ignore_pdb_hexdigest = False If True, disables safety check based on MD5 hexdigests stored in PDB files produced by previous runs.generate = False Generate R-free flags (if not available in input files)fraction = 0.1 max_free = 2000 lattice_symmetry_max_delta = 5 use_lattice_symmetry = True use_dataman_shells = False Used to avoid biasing of the test set by certain types of non-crystallographic symmetry.n_shells = 20 refinementinputexperimental_phasesScope of

experimental phase information (HL coefficients)file_name = None labels = None monomersScope of monomers information (CIF files)file_name = None Monomer file(s) name (CIF)electron_datafile_name = None labels = None r_free_flagsfile_name = None This is normally the same as the file containing Fobs and is usually selected automatically.label = None neutron_datafile_name = None labels = None r_free_flagsfile_name = None This is normally the same as the file containing Fobs and is usually selected automatically.label = None pdbfile_name = None Model file(s) name (PDB)neutron_file_name = None Model file(s) name (PDB)electron_file_name = None Model file(s) name (PDB)xray_datafile_name = None labels = None r_free_flagsfile_name = None This is normally the same as the file containing Fobs and is usually selected automatically.label = None sequencefile_name = None Sequence data in a text file (supported formats include FASTA, PIR, and raw text). Currently this is only used by the PHENIX GUI for validation.

• data_managerfmodelmnopScope of X-ray data and free-R flagshigh_resolution = None low_resolution = None twin_law = None outliers_rejection = True Remove basic wilson outliers , extreme wilson outliers , and beamstop shadow outliersfrench_wilson_scale = True sigma_fobs_rejection_criterion = None sigma_iobs_rejection_criterion = None ignore_all_zeros = True force_anomalous_flag_to_be_equal_to = None convert_to_non_anomalous_if_ratio_pairs_lone_less_than_threshold = 0.5 french_wilsonmax_bins = 60 Maximum number of resolution binsmin_bin_size = 40 Minimum number of reflections per binr_free_flagsrequired = True Specify if free-r flags are must be present (or else generated)test_flag_value = None This value is usually selected automatically - do not change unless you really know what you're doing!ignore_r_free_flags = False Use all reflections in refinement (work and test)disable_suitability_test = False ignore_pdb_hexdigest = False If True, disables safety check based on MD5 hexdigests stored in PDB files produced by previous runs.generate = False Generate R-free flags (if not available in input files)fraction = 0.1 max_free = 2000 lattice_symmetry_max_delta = 5 use_lattice_symmetry = True use_dataman_shells = False Used to avoid biasing of the test set by certain types of non-crystallographic symmetry.n_shells = 20 electron_dataScope of X-ray data and free-R flagshigh_resolution = None low_resolution = None twin_law = None outliers_rejection = True Remove basic wilson outliers , extreme wilson outliers , and beamstop shadow outliersfrench_wilson_scale = True sigma_fobs_rejection_criterion = None sigma_iobs_rejection_criterion = None ignore_all_zeros = True force_anomalous_flag_to_be_equal_to = None convert_to_non_anomalous_if_ratio_pairs_lone_less_than_threshold = 0.5 french_wilsonmax_bins = 60 Maximum number of resolution binsmin_bin_size = 40 Minimum number of reflections per binr_free_flagsrequired = True Specify if free-r flags are must be present (or else generated)test_flag_value = None This value is usually selected automatically - do not change unless you really know what you're doing!ignore_r_free_flags = False Use all reflections in refinement (work and test)disable_suitability_test = False ignore_pdb_hexdigest = False If True, disables safety check based on MD5 hexdigests stored in PDB files produced by previous runs.generate = False Generate R-free flags (if not available in input files)fraction = 0.1 max_free = 2000 lattice_symmetry_max_delta = 5 use_lattice_symmetry = True use_dataman_shells = False Used to avoid biasing of the test set by certain types of non-crystallographic symmetry.n_shells = 20

• fmodelmnopScope of X-ray data and free-R flagshigh_resolution = None low_resolution = None twin_law = None outliers_rejection = True Remove basic wilson outliers , extreme wilson outliers , and beamstop shadow outliersfrench_wilson_scale = True sigma_fobs_rejection_criterion = None sigma_iobs_rejection_criterion = None ignore_all_zeros = True force_anomalous_flag_to_be_equal_to = None convert_to_non_anomalous_if_ratio_pairs_lone_less_than_threshold = 0.5 french_wilsonmax_bins = 60 Maximum number of resolution binsmin_bin_size = 40 Minimum number of reflections per binr_free_flagsrequired = True Specify if free-r flags are must be present (or else generated)test_flag_value = None This value is usually selected automatically - do not change unless you really know what you're doing!ignore_r_free_flags = False Use all reflections in refinement (work and test)disable_suitability_test = False ignore_pdb_hexdigest = False If True, disables safety check based on MD5 hexdigests stored in PDB files produced by previous runs.generate = False Generate R-free flags (if not available in input files)fraction = 0.1 max_free = 2000 lattice_symmetry_max_delta = 5 use_lattice_symmetry = True use_dataman_shells = False Used to avoid biasing of the test set by certain types of non-crystallographic symmetry.n_shells = 20 electron_dataScope of X-ray data and free-R flagshigh_resolution = None low_resolution = None twin_law = None outliers_rejection = True Remove basic wilson outliers , extreme wilson outliers , and beamstop shadow outliersfrench_wilson_scale = True sigma_fobs_rejection_criterion = None sigma_iobs_rejection_criterion = None ignore_all_zeros = True force_anomalous_flag_to_be_equal_to = None convert_to_non_anomalous_if_ratio_pairs_lone_less_than_threshold = 0.5 french_wilsonmax_bins = 60 Maximum number of resolution binsmin_bin_size = 40 Minimum number of reflections per binr_free_flagsrequired = True Specify if free-r flags are must be present (or else generated)test_flag_value = None This value is usually selected automatically - do not change unless you really know what you're doing!ignore_r_free_flags = False Use all reflections in refinement (work and test)disable_suitability_test = False ignore_pdb_hexdigest = False If True, disables safety check based on MD5 hexdigests stored in PDB files produced by previous runs.generate = False Generate R-free flags (if not available in input files)fraction = 0.1 max_free = 2000 lattice_symmetry_max_delta = 5 use_lattice_symmetry = True use_dataman_shells = False Used to avoid biasing of the test set by certain types of non-crystallographic symmetry.n_shells = 20

- mnopScope of X-ray data and free-R flagshigh_resolution = None low_resolution = None twin_law = None outliers_rejection = True Remove basic wilson outliers , extreme wilson outliers , and beamstop shadow outliersfrench_wilson_scale = True sigma_fobs_rejection_criterion = None sigma_iobs_rejection_criterion = None ignore_all_zeros = True force_anomalous_flag_to_be_equal_to = None convert_to_non_anomalous_if_ratio_pairs_lone_less_than_threshold = 0.5 french_wilsonmax_bins = 60 Maximum number of resolution binsmin_bin_size = 40 Minimum number of reflections per binr_free_flagsrequired = True Specify if free-r flags are must be present (or else generated)test_flag_value = None This value is usually selected automatically - do not change unless you really know what you're doing!ignore_r_free_flags = False Use all reflections in refinement (work and test)disable_suitability_test = False ignore_pdb_hexdigest = False If True, disables safety check based on MD5 hexdigests stored in PDB files produced by previous runs.generate = False Generate R-free flags (if not available in input files)fraction = 0.1 max_free = 2000 lattice_symmetry_max_delta = 5 use_lattice_symmetry = True use_dataman_shells = False Used to avoid biasing of the test set by certain types of non-crystallographic symmetry.n_shells = 20
- high_resolution = None
- low_resolution = None
- twin_law = None
- outliers_rejection = True Remove basic wilson outliers , extreme wilson outliers , and beamstop shadow outliers
- french_wilson_scale = True
- sigma_fobs_rejection_criterion = None
- sigma_iobs_rejection_criterion = None
- ignore_all_zeros = True
- force_anomalous_flag_to_be_equal_to = None
- convert_to_non_anomalous_if_ratio_pairs_lone_less_than_threshold = 0.5

- french_wilsonmax_bins = 60 Maximum number of resolution binsmin_bin_size = 40 Minimum number of reflections per bin
- max_bins = 60 Maximum number of resolution bins
- min_bin_size = 40 Minimum number of reflections per bin
- r_free_flagsrequired = True Specify if free-r flags are must be present (or else generated)test_flag_value = None This value is usually selected automatically - do not change unless you really know what you're doing!ignore_r_free_flags = False Use all reflections in refinement (work and test)disable_suitability_test = False ignore_pdb_hexdigest = False If True, disables safety check based on MD5 hexdigests stored in PDB files produced by previous runs.generate = False Generate R-free flags (if not available in input files)fraction = 0.1 max_free = 2000 lattice_symmetry_max_delta = 5 use_lattice_symmetry = True use_dataman_shells = False Used to avoid biasing of the test set by certain types of non-crystallographic symmetry.n_shells = 20
- required = True Specify if free-r flags are must be present (or else generated)
- test_flag_value = None This value is usually selected automatically - do not change unless you really know what you're doing!
- ignore_r_free_flags = False Use all reflections in refinement (work and test)
- disable_suitability_test = False
- ignore_pdb_hexdigest = False If True, disables safety check based on MD5 hexdigests stored in PDB files produced by previous runs.
- generate = False Generate R-free flags (if not available in input files)
- fraction = 0.1
- max_free = 2000
- lattice_symmetry_max_delta = 5
- use_lattice_symmetry = True
- use_dataman_shells = False Used to avoid biasing of the test set by certain types of non-crystallographic symmetry.
- n_shells = 20
- electron_dataScope of X-ray data and free-R flagshigh_resolution = None low_resolution = None twin_law = None outliers_rejection = True Remove basic wilson outliers , extreme wilson outliers , and beamstop shadow outliersfrench_wilson_scale = True sigma_fobs_rejection_criterion = None sigma_iobs_rejection_criterion = None ignore_all_zeros = True force_anomalous_flag_to_be_equal_to = None convert_to_non_anomalous_if_ratio_pairs_lone_less_than_threshold = 0.5 french_wilsonmax_bins = 60 Maximum number of resolution binsmin_bin_size = 40 Minimum number of reflections per binr_free_flagsrequired = True Specify if free-r flags are must be present (or else generated)test_flag_value = None This value is usually selected automatically - do not change unless you really know what you're doing!ignore_r_free_flags = False Use all reflections in refinement (work and test)disable_suitability_test = False ignore_pdb_hexdigest = False If True, disables safety check based on MD5 hexdigests stored in PDB files produced by previous runs.generate = False Generate R-free flags (if not available in input files)fraction = 0.1 max_free = 2000 lattice_symmetry_max_delta = 5 use_lattice_symmetry = True use_dataman_shells = False Used to avoid biasing of the test set by certain types of non-crystallographic symmetry.n_shells = 20
- high_resolution = None
- low_resolution = None

• twin_law = None

• outliers_rejection = True Remove basic wilson outliers , extreme wilson outliers , and beamstop shadow outliers

• french_wilson_scale = True

• sigma_fobs_rejection_criterion = None

• sigma_iobs_rejection_criterion = None

• ignore_all_zeros = True

• force_anomalous_flag_to_be_equal_to = None

• convert_to_non_anomalous_if_ratio_pairs_lone_less_than_threshold = 0.5

• french_wilsonmax_bins = 60 Maximum number of resolution binsmin_bin_size = 40 Minimum number of reflections per bin

• max_bins = 60 Maximum number of resolution bins

• min_bin_size = 40 Minimum number of reflections per bin

• r_free_flagsrequired = True Specify if free-r flags are must be present (or else generated)test_flag_value = None This value is usually selected automatically - do not change unless you really know what you're doing!ignore_r_free_flags = False Use all reflections in refinement (work and test)disable_suitability_test = False ignore_pdb_hexdigest = False If True, disables safety check based on MD5 hexdigests stored in PDB files produced by previous runs.generate = False Generate R-free flags (if not available in input files)fraction = 0.1 max_free = 2000 lattice_symmetry_max_delta = 5 use_lattice_symmetry = True use_dataman_shells = False Used to avoid biasing of the test set by certain types of non-crystallographic symmetry.n_shells = 20

• required = True Specify if free-r flags are must be present (or else generated)

• test_flag_value = None This value is usually selected automatically - do not change unless you really know what you're doing!

• ignore_r_free_flags = False Use all reflections in refinement (work and test)

• disable_suitability_test = False

• ignore_pdb_hexdigest = False If True, disables safety check based on MD5 hexdigests stored in PDB files produced by previous runs.

• generate = False Generate R-free flags (if not available in input files)

• fraction = 0.1

• max_free = 2000

• lattice_symmetry_max_delta = 5

• use_lattice_symmetry = True

• use_dataman_shells = False Used to avoid biasing of the test set by certain types of non-crystallographic symmetry.

• n_shells = 20

• refinementinputexperimental_phasesScope of experimental phase information (HL coefficients)file_name = None labels = None monomersScope of monomers information (CIF files)file_name = None Monomer file(s) name (CIF)electron_datafile_name = None labels = None r_free_flagsfile_name = None This is normally the same as the file containing Fobs and is usually selected automatically.label = None neutron_datafile_name = None labels = None r_free_flagsfile_name

= None This is normally the same as the file containing Fobs and is usually selected automatically.label = None pdbfile_name = None Model file(s) name (PDB)neutron_file_name = None Model file(s) name (PDB)electron_file_name = None Model file(s) name (PDB)xray_datafile_name = None labels = None r_free_flagsfile_name = None This is normally the same as the file containing Fobs and is usually selected automatically.label = None sequencefile_name = None Sequence data in a text file (supported formats include FASTA, PIR, and raw text). Currently this is only used by the PHENIX GUI for validation.

• inputexperimental_phasesScope of experimental phase information (HL coefficients)file_name = None labels = None monomersScope of monomers information (CIF files)file_name = None Monomer file(s) name (CIF)electron_datafile_name = None labels = None r_free_flagsfile_name = None This is normally the same as the file containing Fobs and is usually selected automatically.label = None neutron_datafile_name = None labels = None r_free_flagsfile_name = None This is normally the same as the file containing Fobs and is usually selected automatically.label = None pdbfile_name = None Model file(s) name (PDB)neutron_file_name = None Model file(s) name (PDB)electron_file_name = None Model file(s) name (PDB)xray_datafile_name = None labels = None r_free_flagsfile_name = None This is normally the same as the file containing Fobs and is usually selected automatically.label = None sequencefile_name = None Sequence data in a text file (supported formats include FASTA, PIR, and raw text). Currently this is only used by the PHENIX GUI for validation.

• experimental_phasesScope of experimental phase information (HL coefficients)file_name = None labels = None

• file_name = None

• labels = None

• monomersScope of monomers information (CIF files)file_name = None Monomer file(s) name (CIF)

• file_name = None Monomer file(s) name (CIF)

• electron_datafile_name = None labels = None r_free_flagsfile_name = None This is normally the same as the file containing Fobs and is usually selected automatically.label = None

• file_name = None

• labels = None

• r_free_flagsfile_name = None This is normally the same as the file containing Fobs and is usually selected automatically.label = None

• file_name = None This is normally the same as the file containing Fobs and is usually selected automatically.

• label = None

• neutron_datafile_name = None labels = None r_free_flagsfile_name = None This is normally the same as the file containing Fobs and is usually selected automatically.label = None

• file_name = None

• labels = None

• r_free_flagsfile_name = None This is normally the same as the file containing Fobs and is usually selected automatically.label = None

• file_name = None This is normally the same as the file containing Fobs and is usually selected automatically.

• label = None

• pdbfile_name = None Model file(s) name (PDB)neutron_file_name = None Model file(s) name (PDB)electron_file_name = None Model file(s) name (PDB)

• file_name = None Model file(s) name (PDB)

• neutron_file_name = None Model file(s) name (PDB)

• electron_file_name = None Model file(s) name (PDB)

• xray_datafile_name = None labels = None r_free_flagsfile_name = None This is normally the same as the file containing Fobs and is usually selected automatically.label = None

• file_name = None

• labels = None

• r_free_flagsfile_name = None This is normally the same as the file containing Fobs and is usually selected automatically.label = None

• file_name = None This is normally the same as the file containing Fobs and is usually selected automatically.

• label = None

• sequencefile_name = None Sequence data in a text file (supported formats include FASTA, PIR, and raw text). Currently this is only used by the PHENIX GUI for validation.

• file_name = None Sequence data in a text file (supported formats include FASTA, PIR, and raw text). Currently this is only used by the PHENIX GUI for validation.

• amberParameters for using Amber in refinement.use_amber = False Use Amber for all the gradients in refinementtopology_file_name = None A topology file needed by Amber. Can be generated using phenix.AmberPrep.coordinate_file_name = None A coordinate file needed by Amber. Can be generated using phenix.AmberPrep.order_file_name = None A file that maps amber atom numbers to phenix atom numbers.wxc_factor = 1. restraint_wt = 0. restraintmask = '' reference_file_name = '' bellymask = '' If given, turn on belly in sanderqmmask = '' If given, turn on QM/MM with the given maskqmcharge = 0 Charge of the QM/MM regionnetcdf_trajectory_file_name = '' If given, turn on writing netcdf trajectoryprint_amber_energies = False Print details of Amber energies during refinement

• use_amber = False Use Amber for all the gradients in refinement

• topology_file_name = None A topology file needed by Amber. Can be generated using phenix.AmberPrep.

• coordinate_file_name = None A coordinate file needed by Amber. Can be generated using phenix.AmberPrep.

• order_file_name = None A file that maps amber atom numbers to phenix atom numbers.

• wxc_factor = 1.

• restraint_wt = 0.

• restraintmask = ''

• reference_file_name = ''

• bellymask = '' If given, turn on belly in sander

• qmmask = '' If given, turn on QM/MM with the given mask

• qmcharge = 0 Charge of the QM/MM region

• netcdf_trajectory_file_name = '' If given, turn on writing netcdf trajectory

• print_amber_energies = False Print details of Amber energies during refinement

• qiQMworking_directory = None qm_restraintsselection = None selection for core of atoms to calculate new restraints via a QM geometry minimisationrun_in_macro_cycles = *first_only first_and_last all last_only test the steps of the refinement that the restraints generation is runbuffer = 3.5 distance to include entire residues into the enviroment of the coreignore_lack_of_h_on_ligand = False skip check on protonation of ligand for entities such as MgF3capping_groups = True cleanup = all *most None calculate = *in_situ_opt starting_energy final_energy starting_strain final_strain starting_bound final_bound starting_binding final_binding gradients Choose QM calculations to runwrite_files = *restraints pdb_core pdb_buffer pdb_final_core pdb_final_buffer which ligand or cluster files to writeprotein_optimisation_freeze = *all None main_chain main_chain_to_beta main_chain_to_delta torsions the parts of protein residues that are frozen when an amino acid is the main selectionremove_water = False restraints_filename = Auto restraints filename is based on model name if not specifiedignore_x_h_distance_protein = False skip check on transfer of proton during QM optimisationinclude_nearest_neighbours_in_optimisation = False include the side chains of protein in the QM optimisationinclude_inter_residue_restraints = False do_not_update_restraints = False For testing and maybe getting strain energy of standard restraintsbuffer_selection = None use this instead of distance from selectionspecific_atom_chargesatom_selection = None charge = None specific_atom_multiplicitiesatom_selection = None multiplicity = None freeze_specific_atomsatom_selection = None packageprogram = *mopac test charge = Auto multiplicity = Auto method = Auto basis_set = Auto solvent_model = None nproc = 1 read_output_to_skip_opt_if_available = False ignore_input_differences = False view_output = None qm_gradientsselection = None selection for core of atoms to calculate new restraints via a QM geometry minimisationrun_in_macro_cycles = first_only first_and_last *all last_only test the steps of the refinement that the restraints generation is runbuffer = 3.5 distance to include entire residues into the enviroment of the coreignore_lack_of_h_on_ligand = False skip check on protonation of ligand for entities such as MgF3capping_groups = True cleanup = all *most None specific_atom_chargesatom_selection = None charge = None specific_atom_multiplicitiesatom_selection = None multiplicity = None packageprogram = *mopac test charge = Auto multiplicity = Auto method = Auto basis_set = Auto solvent_model = None nproc = 1 read_output_to_skip_opt_if_available = False ignore_input_differences = False view_output = None

- working_directory = None

- qm_restraintsselection = None selection for core of atoms to calculate new restraints via a QM geometry minimisationrun_in_macro_cycles = *first_only first_and_last all last_only test the steps of the refinement that the restraints generation is runbuffer = 3.5 distance to include entire residues into the enviroment of the coreignore_lack_of_h_on_ligand = False skip check on protonation of ligand for entities such as MgF3capping_groups = True cleanup = all *most None calculate = *in_situ_opt starting_energy final_energy starting_strain final_strain starting_bound final_bound starting_binding final_binding gradients Choose QM calculations to runwrite_files = *restraints pdb_core pdb_buffer pdb_final_core pdb_final_buffer which ligand or cluster files to writeprotein_optimisation_freeze = *all None main_chain main_chain_to_beta main_chain_to_delta torsions the parts of protein residues that are frozen when an amino acid is the main selectionremove_water = False restraints_filename = Auto restraints filename is based on model name if not specifiedignore_x_h_distance_protein = False skip check on transfer of proton during QM optimisationinclude_nearest_neighbours_in_optimisation = False include the side chains of protein in the QM optimisationinclude_inter_residue_restraints = False do_not_update_restraints = False For testing and maybe getting strain energy of standard restraintsbuffer_selection = None use this instead of distance from selectionspecific_atom_chargesatom_selection = None charge = None specific_atom_multiplicitiesatom_selection = None multiplicity = None freeze_specific_atomsatom_selection = None packageprogram = *mopac test charge = Auto multiplicity = Auto method = Auto basis_set = Auto solvent_model = None nproc = 1 read_output_to_skip_opt_if_available = False ignore_input_differences = False view_output = None

- selection = None selection for core of atoms to calculate new restraints via a QM geometry minimisation
- run_in_macro_cycles = *first_only first_and_last all last_only test the steps of the refinement that the restraints generation is run
- buffer = 3.5 distance to include entire residues into the enviroment of the core
- ignore_lack_of_h_on_ligand = False skip check on protonation of ligand for entities such as MgF3
- capping_groups = True
- cleanup = all *most None
- calculate = *in_situ_opt starting_energy final_energy starting_strain final_strain starting_bound final_bound starting_binding final_binding gradients Choose QM calculations to run
- write_files = *restraints pdb_core pdb_buffer pdb_final_core pdb_final_buffer which ligand or cluster files to write
- protein_optimisation_freeze = *all None main_chain main_chain_to_beta main_chain_to_delta torsions the parts of protein residues that are frozen when an amino acid is the main selection
- remove_water = False
- restraints_filename = Auto restraints filename is based on model name if not specified
- ignore_x_h_distance_protein = False skip check on transfer of proton during QM optimisation
- include_nearest_neighbours_in_optimisation = False include the side chains of protein in the QM optimisation
- include_inter_residue_restraints = False
- do_not_update_restraints = False For testing and maybe getting strain energy of standard restraints
- buffer_selection = None use this instead of distance from selection
- specific_atom_chargesatom_selection = None charge = None
- atom_selection = None
- charge = None
- specific_atom_multiplicitiesatom_selection = None multiplicity = None
- atom_selection = None
- multiplicity = None
- freeze_specific_atomsatom_selection = None
- atom_selection = None
- packageprogram = *mopac test charge = Auto multiplicity = Auto method = Auto basis_set = Auto solvent_model = None nproc = 1 read_output_to_skip_opt_if_available = False ignore_input_differences = False view_output = None
- program = *mopac test
- charge = Auto
- multiplicity = Auto
- method = Auto
- basis_set = Auto
- solvent_model = None

- nproc = 1
- read_output_to_skip_opt_if_available = False
- ignore_input_differences = False
- view_output = None
- qm_gradientsselection = None selection for core of atoms to calculate new restraints via a QM geometry minimisationrun_in_macro_cycles = first_only first_and_last *all last_only test the steps of the refinement that the restraints generation is runbuffer = 3.5 distance to include entire residues into the enviroment of the coreignore_lack_of_h_on_ligand = False skip check on protonation of ligand for entities such as MgF3capping_groups = True cleanup = all *most None specific_atom_chargesatom_selection = None charge = None specific_atom_multiplicitiesatom_selection = None multiplicity = None packageprogram = *mopac test charge = Auto multiplicity = Auto method = Auto basis_set = Auto solvent_model = None nproc = 1 read_output_to_skip_opt_if_available = False ignore_input_differences = False view_output = None
- selection = None selection for core of atoms to calculate new restraints via a QM geometry minimisation
- run_in_macro_cycles = first_only first_and_last *all last_only test the steps of the refinement that the restraints generation is run
- buffer = 3.5 distance to include entire residues into the enviroment of the core
- ignore_lack_of_h_on_ligand = False skip check on protonation of ligand for entities such as MgF3
- capping_groups = True
- cleanup = all *most None
- specific_atom_chargesatom_selection = None charge = None
- atom_selection = None
- charge = None
- specific_atom_multiplicitiesatom_selection = None multiplicity = None
- atom_selection = None
- multiplicity = None
- packageprogram = *mopac test charge = Auto multiplicity = Auto method = Auto basis_set = Auto solvent_model = None nproc = 1 read_output_to_skip_opt_if_available = False ignore_input_differences = False view_output = None
- program = *mopac test
- charge = Auto
- multiplicity = Auto
- method = Auto
- basis_set = Auto
- solvent_model = None
- nproc = 1
- read_output_to_skip_opt_if_available = False
- ignore_input_differences = False
- view_output = None

- denreference_file = None gamma = 0.5 kappa = 0.1 weight = 30.0 sigma = 0.44 optimize = True If selected, will run a grid search to determine optimal values of the weight and gamma parameters for DEN refinement. This is very slow, but highly recommended, and can be parallelized across multiple CPU cores.opt_gamma_values = 0.0, 0.2, 0.4, 0.6, 0.8, 1.0 opt_weight_values = 3.0, 10.0, 30.0, 100.0, 300.0 num_cycles = 10 kappa_burn_in_cycles = 2 bulk_solvent_and_scale = True refine_adp = True final_refinement_cycle = False verbose = False annealing_type = *torsion cartesian select strategy to apply DEN restraintsminimize_c_alpha_only = False output_kinemage = False output kinemage representation of starting DEN restraintsrestraint_networklower_distance_cutoff = 3.0 upper_distance_cutoff = 15.0 sequence_separation_low = 0 sequence_separation_limit = 10 exclude_hydrogens = True ndistance_ratio = 1.0 export_den_pairs = False den_network_file = None
- reference_file = None
- gamma = 0.5
- kappa = 0.1
- weight = 30.0
- sigma = 0.44
- optimize = True If selected, will run a grid search to determine optimal values of the weight and gamma parameters for DEN refinement. This is very slow, but highly recommended, and can be parallelized across multiple CPU cores.
- opt_gamma_values = 0.0, 0.2, 0.4, 0.6, 0.8, 1.0
- opt_weight_values = 3.0, 10.0, 30.0, 100.0, 300.0
- num_cycles = 10
- kappa_burn_in_cycles = 2
- bulk_solvent_and_scale = True
- refine_adp = True
- final_refinement_cycle = False
- verbose = False
- annealing_type = *torsion cartesian select strategy to apply DEN restraints
- minimize_c_alpha_only = False
- output_kinemage = False output kinemage representation of starting DEN restraints
- restraint_networklower_distance_cutoff = 3.0 upper_distance_cutoff = 15.0 sequence_separation_low = 0 sequence_separation_limit = 10 exclude_hydrogens = True ndistance_ratio = 1.0 export_den_pairs = False den_network_file = None
- lower_distance_cutoff = 3.0
- upper_distance_cutoff = 15.0
- sequence_separation_low = 0
- sequence_separation_limit = 10
- exclude_hydrogens = True
- ndistance_ratio = 1.0
- export_den_pairs = False
- den_network_file = None